

Share Buyback Valuation & Simulation

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Abstract

This paper debunks the prevalent belief that share buybacks and dividends are equivalent methods of returning capital to shareholders. In reality, a share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders.

Whether a share buyback causes a gain or loss for the remaining shareholders depends on the price paid relative to the present value of the subsequent dividends and share-prices.

Formulas are presented for properly measuring the effect of a share buyback on intrinsic shareholder value. Simulations are used to give a complete overview of possible outcomes.

Formulas are also provided for measuring the actual gain/loss of historical share buybacks. These formulas are used on thousands of the largest U.S. stocks since the late 1990s, which shows that the actual effect of the many trillions of dollars spent on share buybacks has been an even split between gains and losses for the remaining shareholders. But shockingly 10% of U.S. companies have at some point in time destroyed more than 40% of their shareholder value from making share buybacks instead of dividend payouts. Some companies have destroyed their entire shareholder value because of share buybacks.

Several case studies of individual companies show the actual effect of their historical share buybacks on the shareholder value. Simulations also demonstrate how financial managers could have assessed the effect of share buybacks at different points in time.

This paper should be understandable by anyone who invests a little effort. It mostly uses simple math and basic financial concepts like Present Value, and half the pages are plots.

“When the world is sliding into a decline, to excel is easy.” – Hagakure

A Shocking Revelation

The first plot below shows the cumulative share buybacks and dividends for the iconic U.S. motorcycle company Harley Davidson. Between 1999 and 2026, the company spent about \$34b on share buybacks and only about \$12b on dividends. These amounts have been adjusted for the time-value of money by compounding each year's amount forward at an 8% annual rate.

The second plot shows the cumulative gains and losses caused by all these share buybacks. This is measured by the so-called Return on Intrinsic Value (ROIV). When ROIV is zero it means that share buybacks and dividends would have been equally valuable to the remaining shareholders. When ROIV is positive it means the remaining shares gained that much value as a result of the share buybacks. When ROIV is negative the buybacks caused a loss for the remaining shareholders.

This company's share buybacks destroyed almost half of the shareholder value, compared to having just paid out the same amount of money as dividends instead. This is revealed by the cumulative ROIV being almost -50% towards the end. This gigantic loss was caused by the share-price generally declining over the years, and the time-value of money made the decline even worse. This ROIV was calculated using a modest 8% discount-rate. If you think the discount-rate should have been higher for this particular company, then ROIV would have shown an even greater loss.

Share buybacks are real investments a company is making in itself on behalf of the remaining shareholders. Buybacks are not magically protected from declining share-prices and the time-value of money. **It is delusional to keep believing that share buybacks and dividends are equivalent!**

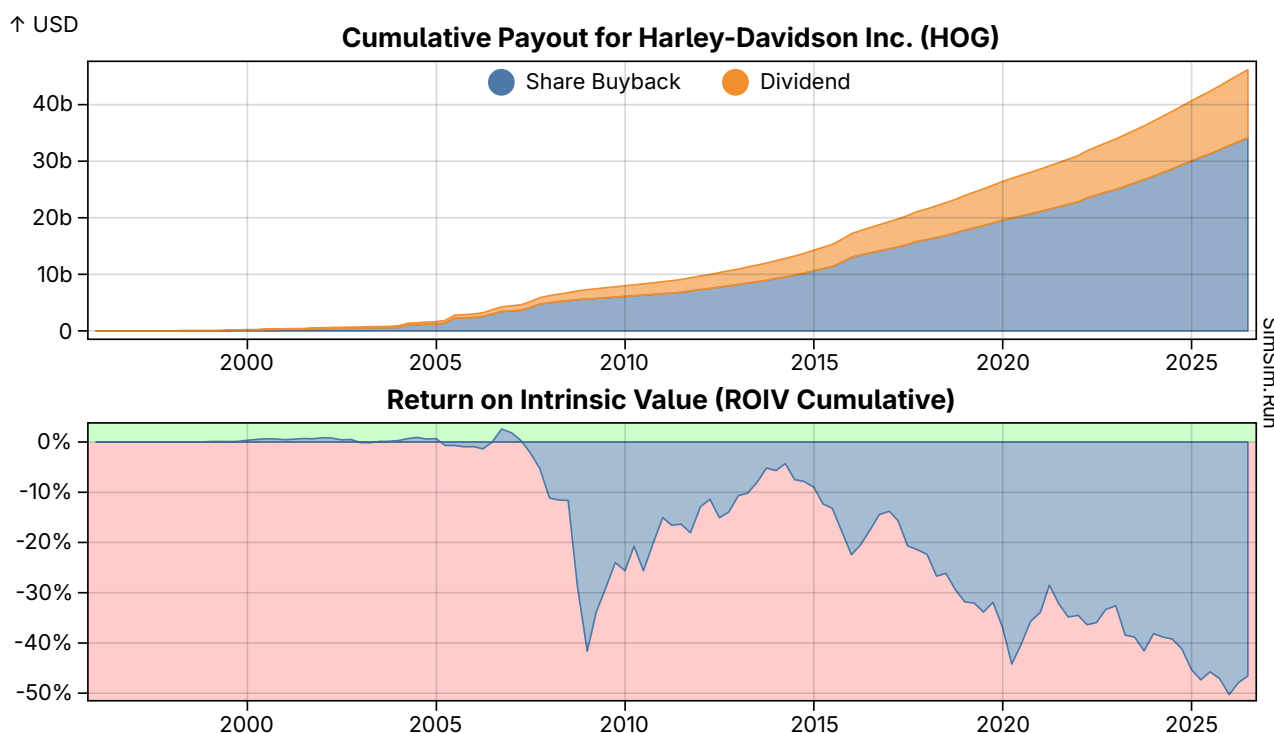


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AI Disclosure

Artificial Intelligence (AI) was not used in the research of the first edition of this paper. The Google Gemini 3.1 Pro AI chat-bot was first used to write section 12 for fun, where AI explains the basics of share buyback valuation. When another AI session later reviewed the paper, it amusingly said this was the best layman's explanation it had ever read. So you may want to read section 12 first.

AI was then used to carefully review the paper in steps. AI checked all formulas and found an error in Eq.(4) for calculating the excise tax of a share buyback. AI also found a few cascading rounding errors in numerical examples. AI also made a few suggestions for clarifications of the text, and checked the spelling and grammar (but sometimes I refused to comply). AI also carefully analyzed the case studies and automatically made some test calculations to verify all the plots looked correct.

AI's review was so impressive that I started using AI for quickly testing ideas and proving difficult math formulas, that I would then spend days proving myself to make sure they were indeed correct. I had long discussions with AI on section 5 and AI made me realize my mistakes, but then AI sent me down many deep rabbit-holes to test AI's claims that also turned out to be wrong. Neither of us got section 5 completely right on our own. It was only through collaboration that we finally got it.

It is truly stunning what AI is capable of today compared to just two years ago. It will be interesting to see where this ends. Soon, AI will probably be able to innovate entirely on its own initiative.

Preface

This is the story about how this paper is the culmination of 15 years of trials and tribulations.

I am originally a computer scientist. Investing was a complete mystery to me until I stumbled over a book about some old geezer named Warren Buffett, whom I had never heard of before. His investment philosophy made sense to me, namely that there is a difference between the market price of a stock and its intrinsic value to long-term investors. If you can estimate the intrinsic value and it is very different from the share-price, then you can do well at investing. That made sense to me, but I later found out how difficult it is to do in practice.

After I finished my formal education, I read the books by Warren Buffett's teacher Ben Graham, who was obviously a genius, but I learned through hard experience that his valuation methods were not really suitable for modern times, because Graham's methods were developed during a time when industry was very different. I also studied finance and accounting textbooks by myself, which is why I sometimes reinvented concepts and gave them new names when I was unaware that a concept already existed under another name, and there were no AI chat-bots to explain it back then.

I also made some software to help discover and analyze stocks. I came across a company that had good profits for years, but it spent most or all of the profits on share buybacks instead of dividends, and then suddenly the company went bankrupt. So I wondered what the intrinsic value was for the shareholders who held onto their shares for that whole time, and who had forgone dividends to increase their ownership of the company through share buybacks, but ended up with nothing. They would obviously have been much better off with dividends instead of all those share buybacks.

There were only a few books on share buybacks. As I recall, I went to a business school library in Berlin and another library in Copenhagen to read those few books. One book was just a compilation of academic papers on the subject. All the theories were based on the so-called Efficient Market Hypothesis and the Modigliani-Miller theorem. They always assumed that the share-price was exactly equal to the intrinsic value, so there was no difference between share buybacks and dividends; which I knew was false from the company that went bankrupt after making buybacks.

Share buybacks were becoming the main way for corporations to "return cash to shareholders" (a phrase I deplore because it is so misleading), and I realized how important it was and that it was my chance of doing something very important in life. So I started working on the formulas for valuing share buybacks. I thought I could write a short paper in maybe two months. Although the core ideas and formulas are fairly simple, the rabbit hole turned out to be much deeper than I expected.

I used axiomatic proof techniques that I had learned as a student of mathematics and computer science. This is similar to reasoning from "first principles" in physics. You start by only assuming what is obviously true, i.e. a share buyback reduces the company's cash holdings and the number of shares outstanding. You must NOT assume that the intrinsic value is exactly equal to the share-price, which is what all the finance professors assumed, but that is not obviously true in the real world, so you would need to make a big leap of faith to assume that, and that is pseudo-science.

Once you make the distinction between the intrinsic value and share-price, it then becomes a matter of modeling and studying the value transformation caused by a share buyback in different scenarios.

Altogether I probably spent about two years researching and writing what eventually became a 200-page treatise on share buyback valuation [Pedersen 2012]. I knew the treatise was a very important contribution to finance, because it debunked many of the core hypotheses of academic finance, and it also provided valuation formulas that were derived using proper axiomatic reasoning, and which could be used in real life to improve the capital allocation for the majority of corporate profits.

I was naturally excited about my important discovery, so I told a family relative who was a CFO. He exploded in anger, yelling and screaming at me completely out of control. I later learned this is known as a narcissistic rage, because his whole identity was based on him being a finance expert and he saw me as inferior, so when I excitedly told him something very important about finance that he did not know, he felt it was a mortal attack on his ego, so he exploded in a narcissistic rage.

I also wrote to any professor and researcher who was even remotely interested in share buybacks. I think I suggested they use a “sledge hammer and chainsaw” to see if they could make my theory fall apart. All of the main researchers in the field just ignored me, except for one who immediately dismissed the entire treatise after a quick glance, and he was also very condescending. This is even though it is their paid job to engage in discussions with people whose theories oppose their own beliefs, for the sake of scientific progress. Nobody ever refuted my work. They just ignored it.

I also wrote a shorter paper with the main parts of the treatise, which I tried to get published in nearly ten of the leading finance and business journals. I sent letters directly to the editors asking if it would be possible to publish the paper as “free access” if it was accepted, because I had paid for all the research myself and I wanted it to benefit as many people as possible. The editors themselves responded very quickly that they were not interested in publishing the paper at all, but they refused to give me any reason why.

I also contacted financial consultancy agencies because they could use my work to help their clients make better share buyback decisions. Nobody ever responded. I later realized that they might be held legally liable if they suddenly admitted that they had advised their clients to buyback shares for trillions of dollars, if many of those share buybacks had likely destroyed shareholder value.

Since the first edition of my treatise was released almost 14 years ago, I am only aware of a single professor who used my treatise to teach his students how to properly value share buybacks. But he never spread the knowledge beyond his own classroom, and sadly he passed away a few years ago.

I continued work on other projects in computer science and investment finance e.g. developing new and vastly superior algorithms for portfolio optimization (like [this paper](#)). But I got tired of writing research papers with highly original and important contributions that were just being ignored.

So I started developing the [SimSim.Run](#) web-site, originally just to make it easy for people to value share buybacks using Monte Carlo simulation with probability distributions from historical data. During the development of the web-site I also extended the formulas from the original treatise, and that gradually evolved into this paper, which is now a major extension of the original treatise.

This is surely one of the most important and useful finance papers ever written. It could change the world if people would just read it and use it. Hopefully it won't be ignored for another 15 years.

I would like to [thank everyone](#) who helped throughout this ordeal.

1 Introduction

A share buyback is when a company repurchases some of its outstanding shares to increase the remaining shareholders' ownership of the company. In USA the majority of corporate profits are being used for share buybacks instead of dividend payouts.

Finance professors used to say that a share buyback signaled the company's management had inside information that the stock was undervalued. But there were countless examples of share buybacks done at peak share-prices just before the company and its stock started to decline. In some cases the companies even went bankrupt just a few years after they had made massive share buybacks.

Finance professors then changed their story so that share buybacks were now a perfect substitute for dividend payouts, possibly even with a tax advantage depending on the tax-rates for dividends and capital gains. This is known as the "substitution hypothesis" which is commonly believed to be true, and is probably the reason that the majority of profits are used for buybacks instead of dividends.

The "proof" of this claim usually takes one of two forms: Either the share-price is assumed to be exactly equal to the so-called intrinsic value to shareholders; or the valuation ratio such as P/E or P/B is assumed to remain the exact same in the future. Neither of these is a valid assumption in reality, but finance professors then utter the phrase "*all else being equal*" as some kind of magic incantation that they believe turns absurd claims into solid science.

Anyone else can see the absurdity of these claims. For example, a company that buys back shares at price \$100 after which the share-price drops to \$50, could obviously have bought back twice as many shares if they had just waited. So timing matters greatly for the efficacy of share buybacks. In the extreme case where a company goes bankrupt after it has made large share buybacks, the shareholders would obviously have been infinitely better off with dividend payouts instead of share buybacks. So share buybacks are clearly not a perfect substitute for dividend payouts.

The reality is that a share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders. Whether this results in a gain or loss for the remaining shareholders depends on how long they hold their shares and what the future dividends and share-prices will be.

Sections 2 and 3 show how to properly value share buybacks using the future earnings and share-prices. As the future is uncertain, we use simulations with probability distributions for the input values, so we can see the entire range of possible outcomes and their probabilities.

Sections 4 and 5 show how to calculate the returns on historical share buybacks using the actual dividends and share-prices. These methods are vastly superior to how academia only considers the correlation between share buybacks and subsequent abnormal stock-returns, as our new formulas let us see the actual effect of historical share buybacks and issuances on the intrinsic shareholder value.

Section 7 uses these valuation formulas on historical data for thousands of the largest U.S. stocks since the late 1990s. This shows that the actual effect of the many trillions of dollars spent on share buybacks has probably just been an even split between gains and losses for the shareholders.

Sections 8 to 11 are case studies for individual companies that show the actual effect of their historical share buybacks. Simulations are also used to demonstrate how the financial managers of those companies could have assessed the effect of share buybacks at different points in time.

1.1 Other Resources

This paper is a major extension to the original treatise on share buyback valuation [Pedersen 2012]. Section 2 of that treatise contains an extensive literature review which still seems current for today even though it was written almost 14 years ago, because academia is still stuck in their old dogma.

Another recent paper considers the valuation and simulation of share issuances [Pedersen 2024].

The [SimSim.Run](#) web-site makes it easy to run the simulations. [Video tutorials](#) are also available.

1.2 Advice

For financial managers trying to decide whether to make a share buyback:

- A share buyback destroys shareholder value when the price paid is above the present value of the subsequent dividends and share-prices. It is similar to a regular investment in the stock.
- Don't buyback shares if you don't really understand their effect. You cannot destroy shareholder value if you just make a dividend payout instead.
- Don't buyback shares to try and boost Earnings Per Share (EPS) or offset dilution from stock options. It still destroys shareholder value if the share-price is higher than the intrinsic value.
- You can usually only estimate the intrinsic shareholder value without knowing the exact value. So it is essential to make simulations to get an overview of the possible outcomes, and then have a margin of safety to lower the risk of destroying shareholder value from a share buyback.
- Try different inputs for your simulations to better understand how share buybacks would affect the shareholder value in different scenarios for the future of your company. If you need to make wildly optimistic assumptions about the future, in order for the share buyback to result in a gain to shareholders, then you should probably refrain from making the share buyback. Conversely, if your assumptions about the future are very conservative and the simulations still show a significant gain in shareholder value, then you may want to make a large share buyback.
- Avoid making share buybacks when the stock seems to be over-valued because it will likely destroy shareholder value. It would be much better to make a special dividend payout, or retain the cash to improve the financial strength of the company, so you can take full advantage the next time the share-price declines and then buyback a much greater number of shares.
- Once you properly understand the effect of share buybacks, you can use them more strategically to increase shareholder value whenever a good opportunity arises, without having to grow the company's sales and earnings, cut expenses, overwork your employees, etc. All you have to do is buyback shares when the price is significantly below the intrinsic value.
- Inform your shareholders about your share buyback plans for full transparency: How do you value share buybacks to assess whether they will likely cause a gain or loss for shareholders? Do you fund the share buybacks using excess cash, profits or borrowings? You should regularly inform the shareholders about recent share buybacks and your future plans. The shareholders are your business partners so they should be properly informed about these very important issues.

2 Basic Formulas

To properly understand the effect of a share buyback on long-term shareholder value, you should always keep in mind that a share buyback has two direct consequences (these are the two axioms):

1. A share buyback reduces the amount of cash the company has. This is a cost.
2. A share buyback reduces the number of shares outstanding. This is a benefit.

A share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders. The question is whether this benefits or harms the remaining shareholders. The answer is that it depends on the share-price relative to the so-called intrinsic value to shareholders.

2.1 Notation

Throughout this paper, we use the letter V to indicate the scenario WITHOUT a share buyback, and the letter W indicates the scenario WITH a share buyback. The letter V is meant as an abbreviation for the word “value”. It also indicates that the value is typically unknown, similar to the letter X in mathematics. The letter W was chosen as an abbreviation for the phrase “value WITH a buyback”.

When written as individual upper-case letters they denote the per-share values for the two scenarios. Lower-case v and w indicate the corresponding total values. When used as a super-script for another variable, it indicates that variable is either considered with (W) or without (V) a share buyback.

2.2 Shares

First let $Shares^V$ (note the indicator V) denote the number of shares WITHOUT the buyback, and let $Shares^W$ (note the indicator W) denote the number of shares WITH the buyback, which just subtracts the number of shares repurchased given the buyback amount and share-price:

$$Shares^W = Shares^V - \frac{Buyback}{SharePrice} \quad (1)$$

The market-value of all the company’s shares before the share buyback is defined as (note the V):

$$MarketCap^V = Shares^V \cdot SharePrice \quad (2)$$

This can be used in Eq.(1) to write the share-count after the buyback as (note the W and V):

$$Shares^W = Shares^V \cdot \left(1 - \frac{Buyback}{MarketCap^V} \right) \quad (3)$$

If there is an excise tax on share buybacks, which is calculated as a percentage of the market value of the repurchased shares, then it effectively decreases the buyback amount as follows:

$$Shares^W = Shares^V \cdot \left(1 - \frac{Buyback / (1 + TaxRateBuyback)}{MarketCap^V} \right) \quad (4)$$

2.3 Intrinsic Value

To measure the gain/loss in shareholder value that results from a share buyback, we also need to define the intrinsic value v (lower-case) of a company to its shareholders. This is the total amount and not per-share. It can be defined in different ways. The simplest is to consider the intrinsic value to be the company's excess cash that could either be used for dividends or share buybacks now, plus the present value of all future earnings for eternity that could also be paid out as dividends:

$$v = \text{Excess Cash} + \sum_{t=1}^{\infty} \frac{\text{Earnings}_t}{(1+d)^t} \quad (5)$$

Let TaxRateDividend be the tax-rate for dividends, and recall from above that Shares^V is the number of shares WITHOUT the buyback.

The per-share, after-tax intrinsic value WITHOUT a share buyback is denoted V (upper-case):

$$V = v \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^V} \quad (6)$$

A share buyback reduces the total intrinsic value by the buyback amount so the value becomes:

$$w = v - \text{Buyback} \quad (7)$$

The per-share, after-tax intrinsic value WITH the share buyback is denoted W (upper-case) and is calculated using the reduced Shares^W from Eq.(4):

$$W = (v - \text{Buyback}) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^W} \quad (8)$$

2.4 Alternative Definitions of Intrinsic Value

The intrinsic value v can be defined in different ways and it will result in the same formulas for valuing a share buyback, if two requirements are satisfied: The intrinsic value v is decreased with the buyback amount as in Eq.(7), and the number of shares is decreased as in Eqs.(1) and (4), so the combined effect is that a share buyback transforms the per-share intrinsic value from V to W as in Eq.(8). Section 3 also considers share-price changes, so the buyback amount is subtracted from only the excess cash and earnings part of the intrinsic value, and the formula for W is therefore different.

2.5 Effect on Market Capitalization

Ignoring other factors that may affect the share-price, the isolated effect of a share buyback on the company's market-cap, is that the buyback amount is merely subtracted. To see this, hold the share-price constant and calculate the market-cap WITH the share buyback using Shares^W from Eq.(4):

$$\begin{aligned} \text{MarketCap}^W &= \text{Shares}^W \cdot \text{SharePrice} \\ &= \text{MarketCap}^V - \text{Buyback} / (1 + \text{TaxRateBuyback}) \end{aligned} \quad (9)$$

If the buyback amount was used for a dividend payout instead, then the number of shares would be unchanged, and the share-price and market-cap would decrease by that amount instead. The effect on the market-cap is identical in the two scenarios (except for the excise tax on share buybacks). But the effect on per-share intrinsic value depends on the share-price relative to the intrinsic value.

2.6 Equilibrium

The so-called equilibrium is where the per-share intrinsic value is the same with and without a share buyback, that is, $W = V$. If the buyback tax is zero then the equilibrium is where the market-cap is exactly equal to the intrinsic value, that is, $MarketCap^V = v$. When also having a buyback tax, the market-cap must be proportionately smaller than the intrinsic value. The equilibrium is written as an inequality so it is easy to see when the buyback causes a gain or loss to the per-share intrinsic value:

$$W > V \Leftrightarrow MarketCap^V < \frac{v}{1 + TaxRateBuyback} \quad (10)$$

2.7 Net Value Added

We define the Net Value Added (NVA) as the monetary amount (e.g. dollar amount) of intrinsic value that was either gained or lost from making a share buyback instead of a dividend payout.

The monetary value gained or lost per share that remains after the share buyback is:

$$NVA \text{ Per Share} = W - V \quad (11)$$

The total monetary value gained or lost on ALL shares that remain after the share buyback is:

$$NVA = (W - V) \cdot Shares^W \quad (12)$$

2.8 Relative Value

The relative value of a share buyback versus a dividend payout for the same amount, is the ratio between the value WITH the share buyback W from Eq.(8), and the value WITHOUT the share buyback V from Eq.(6). The formula can be reduced to this:

$$\frac{W}{V} = \frac{1 - \frac{Buyback}{v}}{1 - \frac{Buyback / (1 + TaxRateBuyback)}{MarketCap^V}} \quad (13)$$

Note that the overall numerator is how much of the intrinsic value remains after the share buyback, and the overall denominator is how much of the market-cap remains after the share buyback.

Also note that both the number of shares and the dividend tax-rate cancel out, so if we are considering the value to “eternal” shareholders who will never sell their shares, then the dividend tax-rate is irrelevant when valuing a share buyback. Further below we extend the formulas to also consider capital gains, so the tax-rates for both capital gains and dividends become relevant.

2.9 Return on Intrinsic Value (ROIV)

Subtracting 1 from the relative value makes it easier to see when the share buyback results in a gain (+) or loss (-) relative to a dividend payout. We call this the Return on Intrinsic Value (ROIV):

$$ROIV = \frac{W}{V} - 1 \quad (14)$$

This can also be written as:

$$ROIV = \frac{W - V}{V} \quad (15)$$

Note that ROIV is ill-defined when the intrinsic value is zero or negative. And because the formula is non-linear, the losses are greater than gains for similar degrees of mispricing, which has profound implications for many aspects of finance, as explained in [Pedersen 2012].

2.10 Return on Buyback (ROB)

The ROIV ratio in Eq.(14) tells us how a share buyback impacts the per-share intrinsic value for the remaining shareholders. But it is also useful to consider the gain/loss of shareholder value relative to the amount that was spent on the share buyback. This makes it easy to compare the gain/loss to other investment returns. This is called the Return on Buyback (ROB) ratio which is defined as:

$$ROB = \frac{W - V}{(Buyback / Shares^V) \cdot (1 - TaxRateDividend)} \quad (16)$$

The numerator is the difference between the intrinsic value of the company with (W) and without (V) the share buyback. The denominator is the buyback amount adjusted for the number of shares and dividend taxes, so it is a fair comparison to the numerator which is also per-share and after-tax.

It is important that we use the number of shares BEFORE the buyback $Shares^V$ in the denominator in Eq.(16). This means the denominator equals the per-share and after-tax dividend that each share would otherwise have received if there were no share buyback. The ROB ratio is essentially a cost-benefit ratio for a share buyback versus a dividend payout, where the numerator is the per-share benefit and the denominator is the per-share cost. Furthermore, if we had instead used $Shares^W$ in the denominator then Eq.(16) would collapse into just measuring the stock's mispricing without the buyback's impact on the per-share intrinsic value.

The ROB ratio was not used in [Pedersen 2012] and was first used on the SimSim.Run web-site.

2.11 Comparing Return Ratios

It is important you understand the difference between the ROIV and ROB ratios: ROIV measures the change in per-share intrinsic value for the remaining shareholders, while ROB measures that change relative to the buyback amount. This will hopefully be made clear from the examples below.

Note that the ROIV and ROB ratios in Eqs.(15) and (16) have the same numerator, which is the difference $W - V$ in per-share intrinsic value with (W) and without (V) the share buyback. This gain/loss is then compared to either the intrinsic value in the ROIV ratio, or the buyback amount in the ROB ratio, to show the magnitude of the per-share gain/loss relative to these two base-lines.

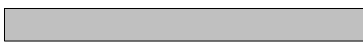
Both the ROIV and ROB ratios will have the same sign and show either gain or loss for the same intrinsic values, because the sign of both ratios are determined by the numerator $W - V$.

The magnitudes of the ROIV and ROB ratios are different because of their different denominators. For example, the ROB ratio can be very high which shows a share buyback would result in a very high return on the amount spent on the buyback. But if the buyback amount is very small, then the actual impact on the per-share intrinsic value is also very small, as measured by the ROIV ratio.

The ROIV ratio is ill-defined for zero or negative intrinsic values V , but the ROB ratio can also be calculated for these, because the ROB ratio does not divide by the intrinsic value.

2.12 Visual Examples

Let us now consider a few visual examples. The figure below demonstrates the effect of a share buyback when the intrinsic value v is 95% of $MarketCap^V$ and the share buyback is 20% of $MarketCap^V$. If the share-price remains unchanged after the buyback then the mispricing is magnified slightly as the intrinsic value decreases to become 93.75% of $MarketCap^W$, down from 95% prior to the share buyback. This means the per-share intrinsic value has decreased about -1.3% as a result of the share buyback, which is what the ROIV ratio in Eq.(14) calculates. If instead v is 105% of $MarketCap^V$ then the per-share intrinsic value would have increased by about +1.2% as a result of the share buyback (note that the ROIV ratio is not symmetric). The equilibrium is when $MarketCap^V = v$ so the share buyback has equal effect on the $MarketCap$ and intrinsic value.

Before Share Buyback	
$MarketCap^V$	
$v = 95 \% \cdot MarketCap^V$	
Share Buyback is The Portion in Dark Grey	
$Buyback = 20 \% \cdot MarketCap^V$	
$Buyback \simeq 21 \% \cdot v$	
After Share Buyback	
$MarketCap^W$	
$w = 93.75 \% \cdot MarketCap^W$	

The figure below demonstrates a much greater mispricing where the intrinsic value v is only 25% of $MarketCap^V$ and the share buyback is again 20% of $MarketCap^V$. If the share-price remains unchanged after the buyback then the mispricing is greatly magnified as the intrinsic value becomes only 6.25% of $MarketCap^W$, down from 25% prior to the share buyback. So the per-share intrinsic value has decreased 75% as a result of the share buyback, which is the ROIV ratio from Eq.(14).

Before Share Buyback	
$MarketCap^V$	
$v = 25 \% \cdot MarketCap^V$	
Share Buyback is The Portion in Dark Grey	
$Buyback = 20 \% \cdot MarketCap^V$	
$Buyback = 80 \% \cdot v$	
After Share Buyback	
$MarketCap^W$	
$w = 6.25 \% \cdot MarketCap^W$	

2.13 Numerical Examples

Let us now do the math for an example where the stock is slightly over-valued. Say $MarketCap^V$ of all the company's shares before the buyback is \$10b. The company has \$1b in excess cash they want to either use for a share buyback or dividend payout. If the present value of all future earnings is \$7b then the total intrinsic value is $v = \$8b$ because it also includes the excess cash. Assume the dividend and buyback tax-rates are zero in these examples.

The per-share intrinsic value WITHOUT the share buyback is calculated using Eq.(6):

$$V = \frac{v}{Shares^V} = \frac{\$8b}{1b} = \$8 \quad (17)$$

The number of shares after the buyback is calculated using Eq.(3):

$$Shares^W = Shares^V \cdot \left(1 - \frac{Buyback}{MarketCap^V}\right) = 1b \cdot \left(1 - \frac{\$1b}{\$10b}\right) = 0.9b \quad (18)$$

The per-share intrinsic value WITH the share buyback is calculated using Eq.(8):

$$W = \frac{v - Buyback}{Shares^W} = \frac{\$8b - \$1b}{0.9b} \simeq \$7.78 \quad (19)$$

The gain/loss of per-share intrinsic value from making a share buyback versus a dividend payout is the ROIV ratio calculated using Eq.(14):

$$ROIV = \frac{W}{V} - 1 \simeq \frac{\$7.78}{\$8} - 1 \simeq -2.8\% \quad (20)$$

The ROIV ratio can also be calculated using the formula for the relative value in Eq.(13):

$$ROIV = \frac{W}{V} - 1 = \frac{1 - \frac{Buyback}{v}}{1 - \frac{Buyback}{MarketCap^V}} - 1 = \frac{1 - \frac{\$1b}{\$8b}}{1 - \frac{\$1b}{\$10b}} - 1 \simeq -2.8\% \quad (21)$$

So the share buyback would incur a loss to the remaining shareholders of about -2.8% relative to a dividend payout. This may seem like a minor loss, but remember that this is a loss of intrinsic value for all remaining shares, so it may actually be a substantial loss compared to the buyback amount.

To better understand the magnitude of the loss we may calculate the ROB ratio using Eq.(16), which measures the net effect of the share buyback on the per-share intrinsic value, relative to the amount that was spent on the buyback:

$$ROB = \frac{W - V}{Buyback / Shares^V} \simeq \frac{\$7.78 - \$8}{\$1b / 1b} \simeq -22\% \quad (22)$$

The share buyback can be considered a type of investment that the company is making in itself on behalf of its remaining shareholders, and the ROB ratio measures the return on that investment. In this example, it would result in a loss of about -22% on the invested amount. No financial manager would make such a stupid investment, if they knew the math!

As shown in some of the case studies further below, when financial managers consistently buyback shares that are overvalued, the value destruction is compounded and becomes massive over time.

Let us consider another example where the stock is under-valued by 50%, so $MarketCap^V$ is still \$10b but now the intrinsic value is actually $v = \$20b$, which again contains excess cash of \$1b that we will either use on a share buyback or a dividend.

The per-share intrinsic value WITHOUT the share buyback is calculated using Eq.(6):

$$V = \frac{v}{Shares^V} = \frac{\$20b}{1b} = \$20 \quad (23)$$

The per-share intrinsic value WITH the share buyback is calculated using Eq.(8) and Eq.(18):

$$W = \frac{v - Buyback}{Shares^W} = \frac{\$20b - \$1b}{0.9b} \simeq \$21.11 \quad (24)$$

The gain/loss of per-share intrinsic value is the ROIV ratio calculated using Eq.(14):

$$ROIV = \frac{W}{V} - 1 \simeq \frac{\$21.11}{\$20} - 1 \simeq 5.6\% \quad (25)$$

So the share buyback would result in a gain to the remaining shareholders of about 5.6% relative to a dividend payout. This may seem like a small gain, but remember that the ROIV ratio measures the gain/loss relative to the per-share intrinsic value. So let us calculate the ROB ratio using Eq.(16) to measure the gain/loss relative to the amount used for the share buyback:

$$ROB = \frac{W - V}{Buyback / Shares^V} \simeq \frac{\$21.11 - \$20}{\$1b / 1b} \simeq 111\% \quad (26)$$

So the effect of the share buyback would be similar to the company making an investment for \$1b whose present value was about \$2.11b for a 111% gain. The return is more than twice the invested amount, simply because the repurchased shares were under-valued by 50%. Every financial manager would make such an investment, if they knew the math!

These examples show that share buybacks may have a significant impact on long-term shareholder value when the share-price does not exactly equal the intrinsic value, and the impact can be very large in case of severe over- and under-valuation of the stock.

You should try and experiment with different inputs in the formulas above, to see how a share buyback would impact the long-term shareholder value.

2.14 Simulation

We typically don't know the exact intrinsic value of a company, because it is based on estimates of the future earnings. So we should repeat the calculations above with many different assumptions. You could enter the valuation formulas in a computer spreadsheet and run some simulations, but it is much easier to use the [SimSim.Run](#) web-site that was made specifically for running this kind of simulation (click the link to try it). We use the same input assumptions as above, except for the intrinsic value which is now a normal-distributed (bell-shaped) random variable with mean \$10b and standard deviation \$1b.

Figure 1 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. Note the ROIV distribution is skewed so the losses are greater than the gains, even though the intrinsic value v is normal-distributed. The asymmetry of the ROIV ratio is because Eq.(14) is non-linear with respect to the intrinsic value v .

Figure 2 again shows the simulated ROIV ratio from Eq.(14), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the current share-price changes, while the other variables remain the same.

Figure 3 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. Note the ROB distribution is symmetric and bell-shaped like the normal-distributed intrinsic value v (but with sampling noise), which is because Eq.(16) is linear with respect to the intrinsic value v .

Figure 4 again shows the simulated ROB ratio from Eq.(16), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROB ratio would change when the current share-price changes, while the other variables remain the same.

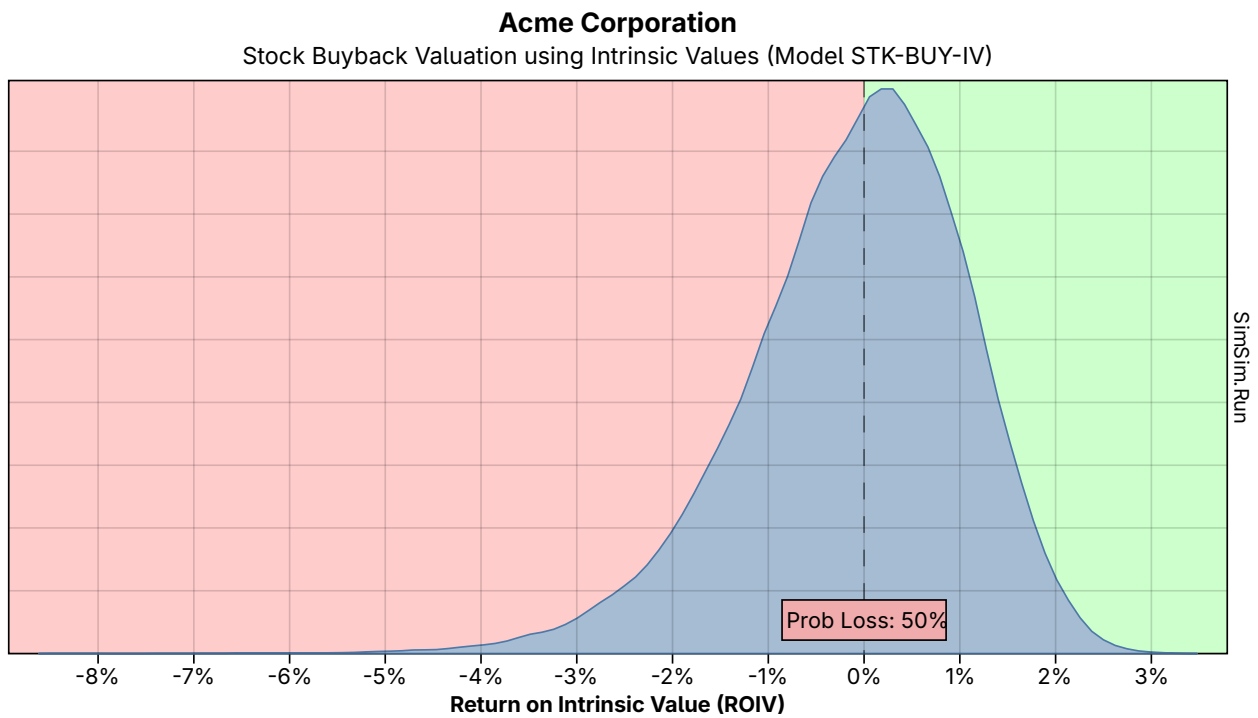


Figure 1: Simulation of $ROIV$ in Eq.(14) using a random variable for the intrinsic value v .

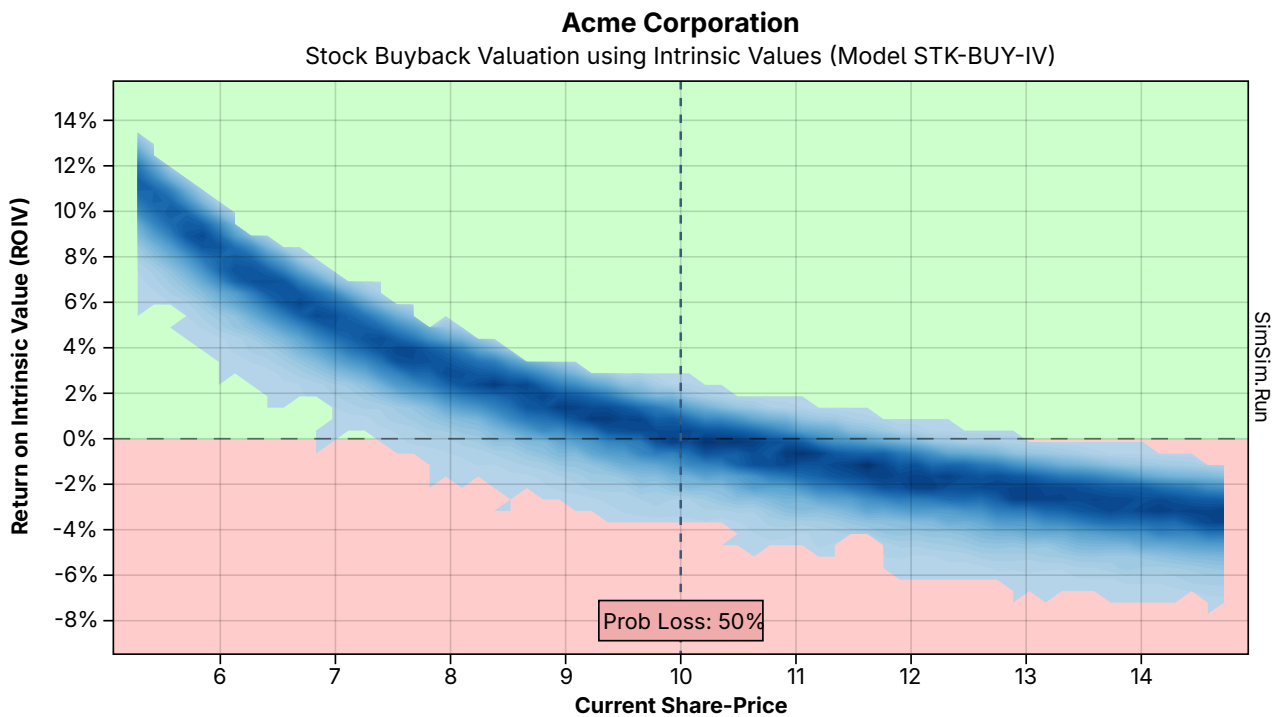


Figure 2: Simulation of $ROIV$ in Eq.(14) using a random variable for the intrinsic value v , and varying the current share-price (or MarketCap) on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.
The distribution in Figure 1 is the vertical dashed blue line here.

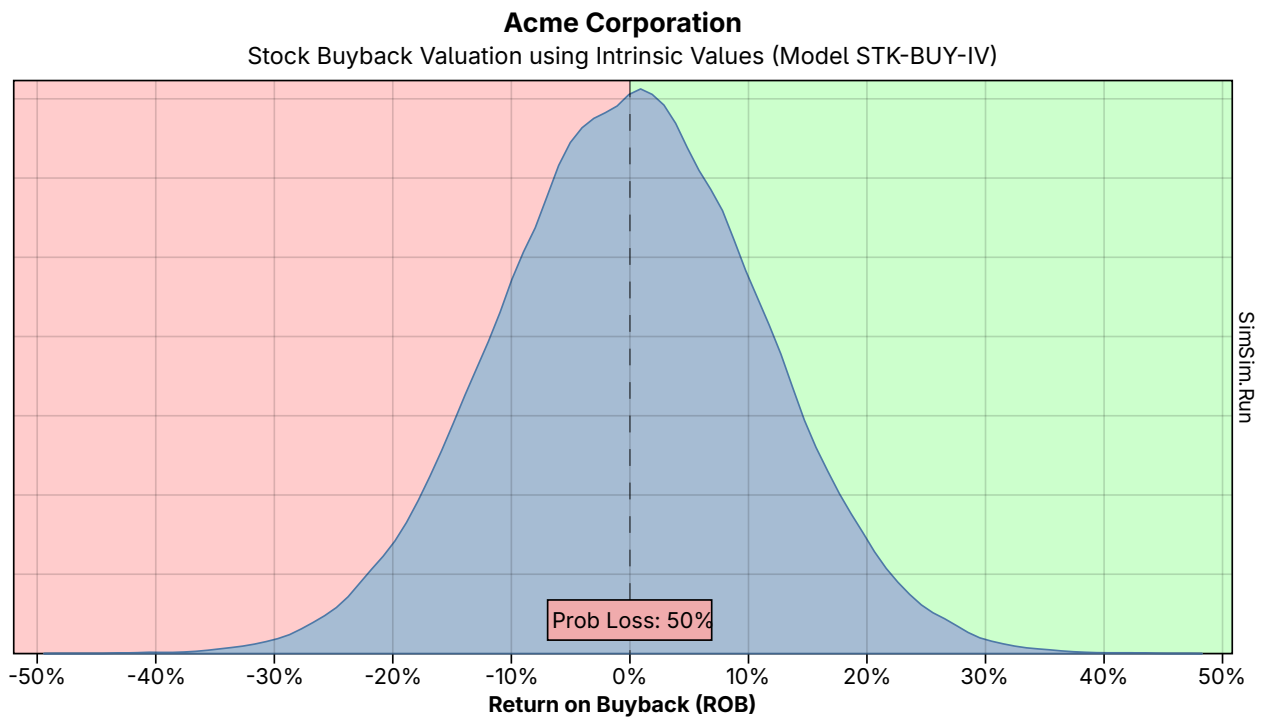


Figure 3: Simulation of **ROB** in Eq.(16) using a random variable for the intrinsic value v .

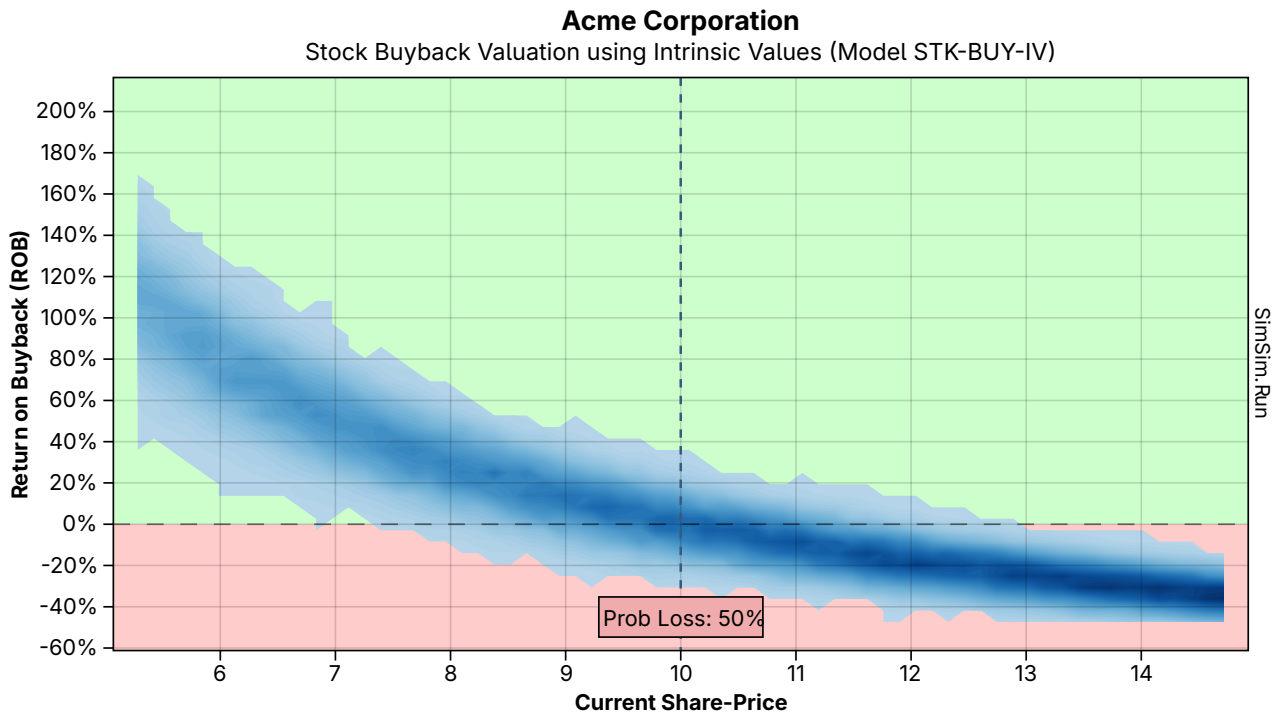


Figure 4: Simulation of **ROB** in Eq.(16) using a random variable for the intrinsic value v , and varying the current share-price (or MarketCap) on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 3 is the vertical dashed blue line here.

3 Capital Gains

In [Pedersen 2012] it was argued that share buybacks should always be made for the sake of the remaining shareholders, which ultimately means that the valuation should only consider the so-called “eternal” shareholders who will never sell their shares, so there is no need to consider future share-prices when valuing share buybacks.

But one of the main arguments in favor of share buybacks over dividends is that if capital gains taxes are lower than dividend taxes, then share buybacks are claimed to be a more tax-efficient way of “returning” capital to shareholders. The “proof” of this claim typically assumes that the valuation ratio remains constant in the future, but that is obviously not a valid assumption in the real world.

We can extend the formulas for share buyback valuation to also include capital gains/losses from future changes in share-prices. The intrinsic value for shareholders is now calculated from the excess cash, the present value of the earnings until the future year N which are assumed to be paid out as dividends, and the present value of $SharePrice_N$ for the future year N .

We use the same future $MarketCap_N$ in year N for both scenarios without and with a share buyback. We typically estimate it from the product of the future earnings (or sales) and the P/E (or P/S) ratio, which are usually much easier to estimate than trying to estimate the future $MarketCap_N$ directly:

$$MarketCap_N = Earnings_N \cdot P/E_N \quad (27)$$

The share-price in future year N is different for the two scenarios without and with a share buyback, because $MarketCap_N$ is divided by $Shares^V$ without, and $Shares^W$ with a buyback from Eq.(4):

$$\begin{aligned} SharePrice_N^V &= MarketCap_N / Shares^V && \text{WITHOUT Share Buyback} \\ SharePrice_N^W &= MarketCap_N / Shares^W && \text{WITH Share Buyback} \end{aligned} \quad (28)$$

The capital gains depend on each investor's purchase price for the shares, so they may be different for all investors and we therefore have to make a reasonable compromise and calculate the capital gains between the current $SharePrice$ and $SharePrice_N^V$ for future year N (and for $SharePrice_N^W$):

$$Capital\ Gains_N^V = SharePrice_N^V - SharePrice \quad (29)$$

The tax on only positive capital gains can be calculated as follows (and similarly for $SharePrice_N^W$):

$$TaxCapGains_N^V = TaxRateCapGains \cdot \max(Capital\ Gains_N^V, 0) \quad (30)$$

The after-tax, present value of $SharePrice_N^V$ is then as follows (and similarly for $SharePrice_N^W$):

$$PV\ SharePrice_N^V = \frac{SharePrice_N^V - TaxCapGains_N^V}{(1+d)^N} \quad (31)$$

We can now calculate the after-tax, per-share intrinsic value WITHOUT the share buyback:

$$V_N = \left(\text{Excess Cash} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^V} + \text{PV SharePrice}_N^V \quad (32)$$

And the after-tax, per-share intrinsic value WITH the share buyback:

$$W_N = \left(\text{Excess Cash} - \text{Buyback} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^W} + \text{PV SharePrice}_N^W \quad (33)$$

3.1 Equilibrium

The equilibrium is where a share buyback does not change the after-tax, per-share intrinsic value. It is written as an inequality $W_N > V_N$ so it is easy to see when the share buyback causes a gain or loss. Using the definitions from Eqs.(32) and (33) above, the equilibrium can be reduced to:¹

$$W_N > V_N \Leftrightarrow$$

$$\text{MarketCap}^V < \frac{\text{Excess Cash} + \sum_{t=1}^N \frac{\text{Earnings}_t}{(1+d)^t} + \frac{\text{MarketCap}_N}{(1+d)^N} \cdot \frac{(1 - \text{TaxRateCapGains})}{(1 - \text{TaxRateDividend})}}{1 + \text{TaxRateBuyback}} \quad (34)$$

Note that the tax-rates are just some of the factors that determine whether a share buyback causes a gain or loss to the remaining shareholders. The main factor is still the stock's overall mispricing.

3.2 Return Ratios & Relative Value

In practice, we calculate V_N and W_N using Eqs.(32) and (33) and use the results in Eqs.(14) and (16) to calculate the ROIV and ROB ratios for assessing the gain/loss caused by the share buyback.

It is also interesting to study the formula for the relative value of the share buyback W_N/V_N , which can be reduced to the following when using the definitions of V_N and W_N from Eqs.(32) and (33):

$$\frac{W_N}{V_N} = \frac{1 - \frac{\text{Buyback}}{V_N \cdot \text{Shares}^V} \cdot \left((1 - \text{TaxRateDividend}) + \frac{\text{TaxRateCapGains}}{(1 + \text{TaxRateBuyback}) \cdot (1+d)^N} \right)}{1 - \frac{\text{Buyback} / (1 + \text{TaxRateBuyback})}{\text{MarketCap}^V}} \quad (35)$$

Note that the overall numerator is how much of the intrinsic value remains after the share buyback, and the overall denominator is how much of the market-cap remains after the share buyback. This is similar to Eq.(13) for an “eternal” shareholder where there are no capital gains, except in Eq.(35) the overall numerator is also adjusted for all the different tax-rates in a complex manner.

¹ Eqs.(34) and (35) are some of the only formulas in this paper that are challenging to derive because there are so many variables. We assume there are capital gains both WITH and WITHOUT a share buyback. If both cases have losses instead then the capital gains tax-rate can be set to zero. The formulas are more complicated when the future share-price WITH a buyback has a capital gain, and the future share-price WITHOUT a buyback has a capital loss.

3.3 Example

Now consider an example that uses the formulas above with the following assumptions:

$$\begin{aligned} \text{Shares}^V &= 1 \text{ b} & \text{SharePrice} &= \$ 10 & \text{MarketCap}^V &= \text{Shares}^V \cdot \text{SharePrice} = \$ 10 \text{ b} \\ \text{Buyback} &= \text{Excess Cash} = \$ 1 \text{ b} \\ \text{Earnings}_t &= \$ 500 \text{ m for all future years } t \end{aligned} \quad (36)$$

$$\text{MarketCap}_N = P/E_N \cdot \text{Earnings}_N = 20 \cdot \$ 500 \text{ m} = \$ 10 \text{ b}$$

$$d = 10 \% \quad \text{TaxRateDividend} = 30 \% \quad \text{TaxRateCapGains} = 10 \%$$

The number of shares after the buyback is calculated using Eq.(1), assuming zero excise tax:

$$\text{Shares}^W = \text{Shares}^V - \frac{\text{Buyback}}{\text{SharePrice}} = 1 \text{ b} - \frac{\$ 1 \text{ b}}{\$ 10} = 0.9 \text{ b} \quad (37)$$

The future share-price in year N without and with a share buyback is calculated using Eq.(28):

$$\begin{aligned} \text{SharePrice}_N^V &= \text{MarketCap}_N / \text{Shares}^V = \$ 10 \text{ b} / 1 \text{ b} = \$ 10 \\ \text{SharePrice}_N^W &= \text{MarketCap}_N / \text{Shares}^W = \$ 10 \text{ b} / 0.9 \text{ b} \simeq \$ 11.11 \end{aligned} \quad (38)$$

The capital gains tax in the two scenarios are:

$$\begin{aligned} \text{TaxCapGains}_N^V &= 10 \% \cdot (\text{SharePrice}_N^V - \text{SharePrice}) = 10 \% \cdot (\$ 10 - \$ 10) = \$ 0 \\ \text{TaxCapGains}_N^W &= 10 \% \cdot (\text{SharePrice}_N^W - \text{SharePrice}) = 10 \% \cdot (\$ 11.11 - \$ 10) \simeq \$ 0.11 \end{aligned} \quad (39)$$

The present value of the future share-prices net of capital gains tax in the two scenarios are:

$$\begin{aligned} \text{PV SharePrice}_N^V &= (\text{SharePrice}_N^V - \text{TaxCapGains}_N^V) / 1.1^N = (\$ 10 - \$ 0) / 1.1^N \\ \text{PV SharePrice}_N^W &= (\text{SharePrice}_N^W - \text{TaxCapGains}_N^W) / 1.1^N = (\$ 11.11 - \$ 0.11) / 1.1^N \end{aligned} \quad (40)$$

For example, if we use $N = 5$ years for the investment duration then the present values are:

$$\begin{aligned} \text{PV SharePrice}_5^V &= (\$ 10 - \$ 0) / 1.1^5 \simeq \$ 6.21 \\ \text{PV SharePrice}_5^W &= (\$ 11.11 - \$ 0.11) / 1.1^5 \simeq \$ 6.83 \end{aligned} \quad (41)$$

The after-tax, per-share intrinsic value WITHOUT the share buyback is calculated using Eq.(32):

$$\begin{aligned}
 V_5 &= \left(\text{Excess Cash} + \sum_{t=1}^5 \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^V} + \text{PV SharePrice}_5^V \\
 &= \left(\$1\text{ b} + \sum_{t=1}^5 \frac{\$500\text{ m}}{1.1^t} \right) \cdot \frac{1 - 30\%}{1\text{ b}} + \$6.21 \simeq \$8.236
 \end{aligned} \tag{42}$$

The after-tax, per-share intrinsic value WITH the share buyback is calculated using Eq.(33):

$$\begin{aligned}
 W_5 &= \left(\text{Excess Cash} - \text{Buyback} + \sum_{t=1}^5 \frac{\text{Earnings}_t}{(1+d)^t} \right) \cdot \frac{1 - \text{TaxRateDividend}}{\text{Shares}^W} + \text{PV SharePrice}_5^W \\
 &= \left(\$1\text{ b} - \$1\text{ b} + \sum_{t=1}^5 \frac{\$500\text{ m}}{1.1^t} \right) \cdot \frac{1 - 30\%}{0.9\text{ b}} + \$6.83 \simeq \$8.304
 \end{aligned} \tag{43}$$

The gain/loss of per-share intrinsic value from the share buyback versus a dividend payout is the ROIV ratio calculated using Eq.(14) with the values W_5 and V_5 above for $N = 5$ years:

$$\text{ROIV}_5 = \frac{W_5}{V_5} - 1 \simeq \frac{\$8.304}{\$8.236} - 1 \simeq 0.8\% \tag{44}$$

So the share buyback would increase the per-share intrinsic value to the remaining shareholders by about +0.8% relative to a dividend payout.

We may also calculate the ROB ratio, which measures the net effect of the share buyback relative to the amount that was spent on the buyback. Because the tax-rates are different for dividends and capital gains, we must use the original formula for the ROB ratio in Eq.(16) as follows:

$$\begin{aligned}
 \text{ROB}_5 &= \frac{W_5 - V_5}{(\text{Buyback} / \text{Shares}^V) \cdot (1 - \text{TaxRateDividend})} \\
 &\simeq \frac{\$8.304 - \$8.236}{(\$1\text{ b} / 1\text{ b}) \cdot (1 - 30\%)} \simeq 9.7\%
 \end{aligned} \tag{45}$$

A share buyback is a kind of investment the company is making in itself on behalf of the remaining shareholders, and the ROB ratio measures the return on that investment. In this example it would be a gain of about +9.7% on the invested amount.

When using these formulas by hand, you should be careful with rounding errors in the intermediate calculations, which can significantly affect the final result. This is not a problem on a computer.

You should try and experiment with different inputs in the formulas above, to see how a share buyback would impact the intrinsic shareholder value. For example, if you just set the investment duration to $N = 10$ years instead of $N = 5$ as above, then the share buyback actually causes a loss in shareholder value. Why is that? Try and ponder this question before continuing to the next section.

3.4 Simulation

We obviously don't know the exact future earnings and share-prices, so we should repeat the calculations above with many different assumptions to assess the possible effect of a share buyback. You could enter the formulas and run simulations in a computer spreadsheet, but it is much easier to use the [SimSim.Run](#) web-site that was specifically made for this purpose (click the link to try it).

We use the same assumptions as Eq.(36) in the example above, except that some of the variables are now normal-distributed (bell-shaped) random variables with mean μ and standard deviation σ :

$$\begin{aligned} \text{Earnings}_t &\sim N(\mu = \$500\text{ m}, \sigma = \$100\text{ m}) \text{ for all future years } t \\ P/E_N &\sim N(\mu = 20, \sigma = 3) \\ d &\sim N(\mu = 10\%, \sigma = 1\%) \end{aligned} \tag{46}$$

Figure 5 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. This is a violin-plot that shows the ROIV distributions vertically for all investment periods N between 1 and 10 years. The ROIV ratios are calculated from V_N and W_N in Eqs.(32) and (33) and thousands of random values from Eq.(46).

Figure 6 again shows the simulated ROIV ratio from Eq.(14), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis, and all investment periods N from 1 to 10 years are also combined. This makes it easy to see how the ROIV ratio would change for all investment periods overall when the current share-price changes, while all other variables remain the same.

Figure 7 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's original shareholders, but relative to the buyback amount.

Figure 8 again shows the simulated ROB ratio from Eq.(16), but here the current share-price (or equivalently the *MarketCap*) is varied on the x-axis, and all investment periods N from 1 to 10 years are also combined. This makes it easy to see how the ROB ratio would change for all investment periods overall when the current share-price changes, while all other variables remain the same.

Note in Figure 5 and Figure 7 that the ROIV and ROB distributions are drifting downwards for longer investment periods N , even though the distributions in Eq.(46) are the same for all years. If you try running this simulation yourself, you will see that the drift gets worse for larger discount-rates. It is also possible for the distributions to drift upwards instead of downwards. It all depends on the choice of discount-rate, earnings, valuation ratios, etc.

In this example, the distributions are drifting downwards because the future earnings and P/E ratios don't grow in successive years. The present value of the share-price decreases from one more year of discounting, but adding one more year of discounted earnings is not sufficient to make up for that decrease, so the total present value drifts downwards.

What this means in practice is that you should always use probability distributions when valuing share buybacks, and consider many investment periods, as there is rarely just a single intrinsic value of a stock. Even if the future earnings, share-prices, and discount-rates are all held constant, the present value of a stock may vary simply from changing the investment period.

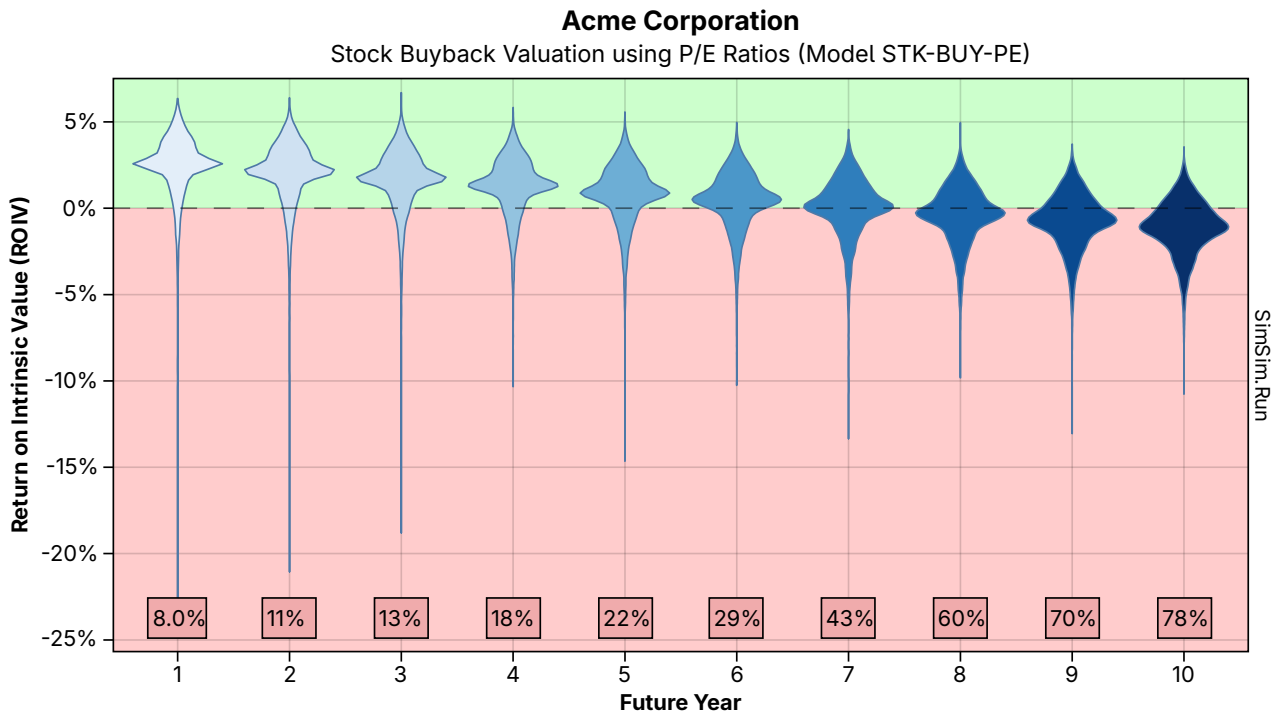


Figure 5: Simulation of **ROIV** in Eq.(14) using random variables for V_N and W_N in Eqs.(32) and (33).

This violin-plot shows the ROIV distributions for investment periods between 1-10 years.

Red boxes show the probability of loss for each investment period when making a share buyback.

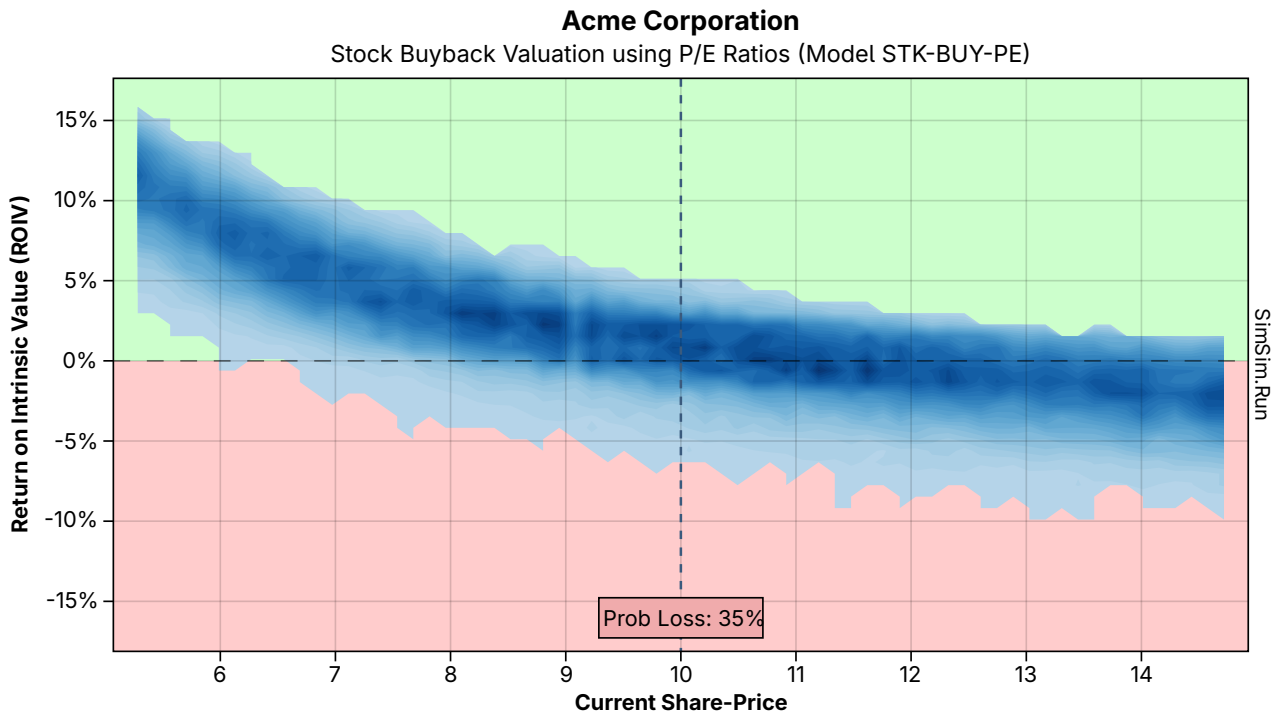


Figure 6: Simulation of **ROIV** in Eq.(14) using random variables for V_N and W_N in Eqs.(32) and (33),

and varying the current share-price (or MarketCap) on the x-axis.

This shows the ROIV ratio for all investment periods between 1-10 years combined.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The red box shows the overall probability of loss for all investment periods when making a share buyback.

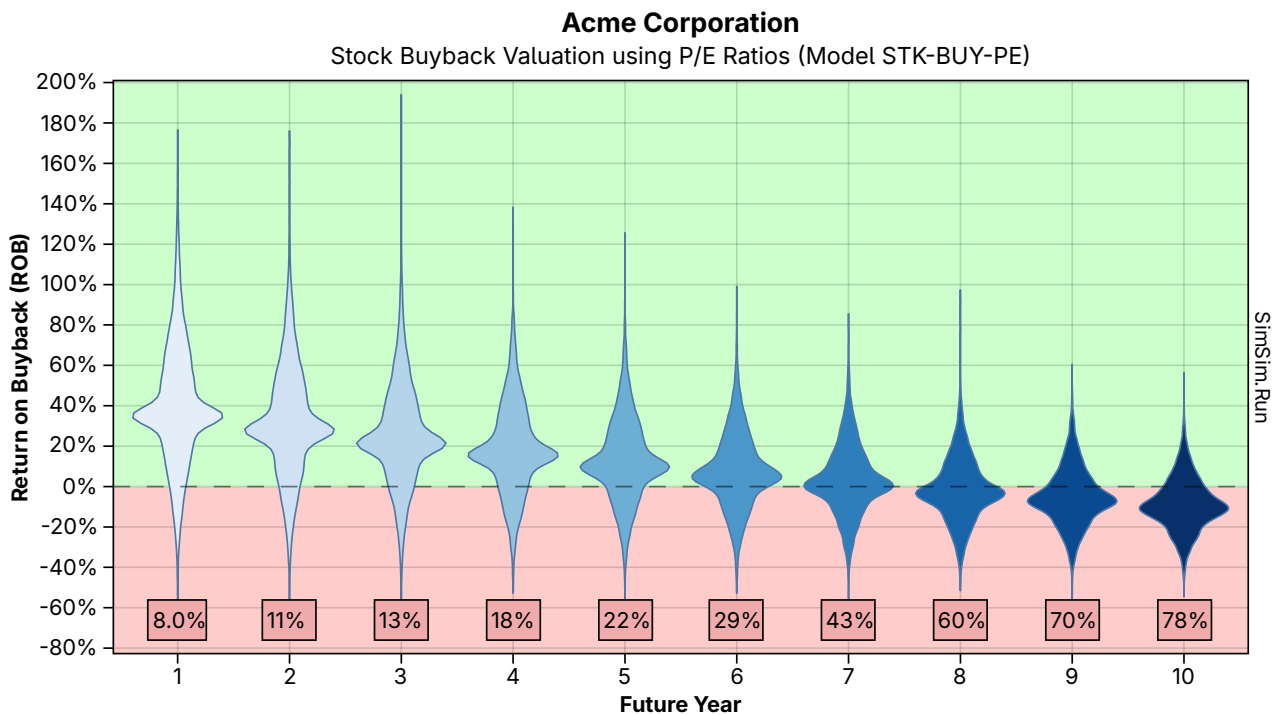


Figure 7: Simulation of **ROB** in Eq.(16) using random variables for V_N and W_N in Eqs.(32) and (33).

This violin-plot shows the ROB distributions for investment periods between 1-10 years.

Red boxes show the probability of loss for each investment period when making a share buyback.

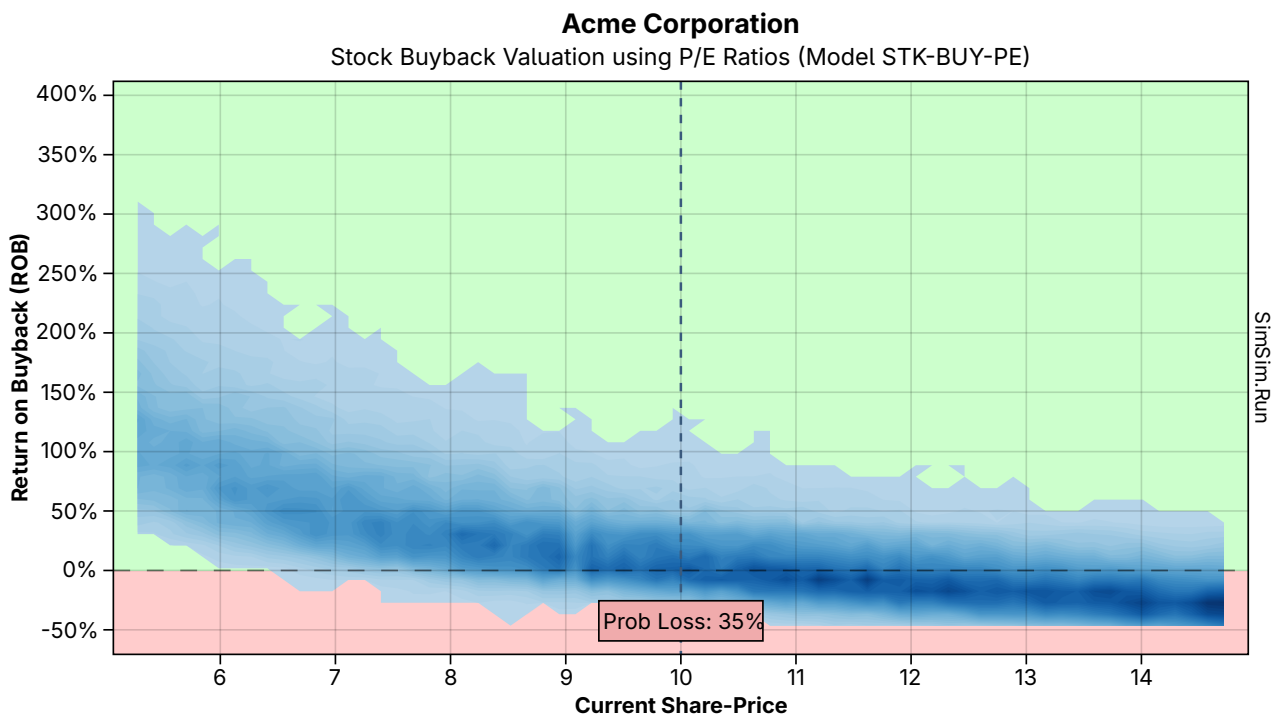


Figure 8: Simulation of **ROB** in Eq.(16) using random variables for V_N and W_N in Eqs.(32) and (33),

and varying the current share-price (or MarketCap) on the x-axis.

This shows the ROB ratio for all investment periods between 1-10 years combined.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The red box shows the overall probability of loss for all investment periods when making a share buyback.

4 Historical Returns – Forward Looking

In the previous sections 2 and 3, we derived formulas for valuing share buybacks using estimates of the company's future share-prices and earnings that could be paid out as dividends.

In this section, we present valuation formulas using historical data for share-prices and dividends to assess the actual gain/loss of past share buybacks and issuances for the shareholders at that time.

You can think of the previous sections as valuing share buybacks looking into an uncertain future, while this section is valuing past share buybacks and issuances using actual historical data.

This section is still “forward looking” because we calculate the effect of past share buybacks for the shareholders who sold their shares in one of the following years. This lets us see exactly which quarters with share buybacks and issuances caused gains or losses for the remaining shareholders.

In the next section 5, we present formulas for valuing the cumulative returns of past share buybacks, which lets us see the compounded effect of many share buybacks over long time horizons.

A company's financial cash-flow statements should have the share buyback and issuance amounts reported separately so we can value them separately. But in some financial databases like the one we use in the case studies further below, the amounts are combined into a *Net Buyback* amount that is positive for a share buyback and negative for an issuance. You can still use the formulas below if you have a financial database with separate amounts for the share buybacks and issuances, by simply using the same signs + or - for those amounts, as we have defined for *Net Buyback* here.

4.1 Shares

To calculate the effect of a past share buyback or issuance, we must know the number of shares that were repurchased or issued. If you have access to a detailed financial database, it might have the actual number of shares that you can use directly. But the low-cost database we use here does not have this data, so instead we have to try and estimate the number of shares repurchased or issued.

We cannot simply calculate the difference in shares outstanding at the end of the current and previous time periods (typically financial quarters), because that would conflate share buybacks during a period with the exercise of stock options that may have been granted several years prior at much lower share-prices. Some companies make very large share buybacks just to offset the dilution from exercised stock options. This would greatly distort the valuation of the share buyback that happened during a period. So we must estimate the number of shares repurchased or issued.

Let $Shares_{Start}^W$ denote the number of historical shares outstanding at the end of some *Start* period WITH the share buyback or issuance that occurred during that *Start* period (typically a financial quarter). This is the historical number of shares that we obtain from the financial database. It is assumed to be adjusted for stock-splits.

Let $Shares_{Start}^{NBB}$ denote the actual net number of shares bought back (positive) or issued (negative).

Let $Shares_{Start}^V$ denote the number of shares WITHOUT the share buyback or issuance during the *Start* period. This is the number that we have to calculate by adding back the repurchased shares:

$$Shares_{Start}^V = Shares_{Start}^W + Shares_{Start}^{NBB} \quad (47)$$

If the financial database does not have the actual number of shares bought back or issued, then we can estimate the number using the net buyback amount and the average share-price for the period:

$$Shares_{Start}^{NBB} \simeq \frac{Net\ Buyback_{Start}}{\text{mean}[SharePrice_{Start}]} \quad (48)$$

For simplicity, we ignore excise tax on the net buyback amount in the formulas below, but it could be taken into account by deducting the excise tax only from the positive net buyback amounts:

$$Shares_{Start}^{NBB} \simeq \frac{Net\ Buyback_{Start} / (1 + TaxRateBuyback_{Start})}{\text{mean}[SharePrice_{Start}]} \quad \text{if } Net\ Buyback_{Start} > 0 \quad (49)$$

Inserting Eq.(48) into Eq.(47) gives us an estimate for the number of shares WITHOUT the share buyback or issuance during the *Start* period. This is basically the reverse of Eq.(1):

$$Shares_{Start}^V \simeq Shares_{Start}^W + \frac{Net\ Buyback_{Start}}{\text{mean}[SharePrice_{Start}]} \quad (50)$$

The average market-value of the shares WITH the net share buyback during the *Start* period is:

$$\text{mean}[MarketCap_{Start}^W] = \text{mean}[SharePrice_{Start}] \cdot Shares_{Start}^W \quad (51)$$

Using this in Eq.(50), we can estimate the number of shares WITHOUT the net share buyback as:

$$Shares_{Start}^V \simeq Shares_{Start}^W \cdot \left(1 + \frac{Net\ Buyback_{Start}}{\text{mean}[MarketCap_{Start}^W]} \right) \quad (52)$$

We typically don't know the exact dates of share buybacks and issuances, so we use the average $SharePrice_{Start}$ for the entire quarterly *Start* period, which causes estimation errors in the formulas. But we must use the number of shares outstanding at the end of the *Start* period, $Shares_{Start}^W$, in order for Eq.(50) to be equivalent to Eq.(52). It may seem counter-intuitive, but we should not use the weighted average number of shares in the *Start* period, because that causes larger estimation errors.

4.2 Intrinsic Value

In the previous sections 2 and 3, we first defined the per-share intrinsic value WITHOUT the share buyback, which was denoted V , and from this we subtracted the buyback amount, and adjusted the number of shares to get the intrinsic value WITH the share buyback, which was denoted W .

Here we do the opposite: We start by defining the intrinsic value WITH the historical share buyback or issuance W , and from this we estimate the value WITHOUT the share buyback or issuance V .

We also want to assess the gain/loss of a share buyback or issuance for different investors who might have held the shares for 1 or 2 or 3 years, etc. To do this, we will calculate the intrinsic value for shareholders between all possible *Start* and *End* periods (typically financial quarters).

The per-share intrinsic value WITH the actual share buyback or issuance during the *Start* period, for a shareholder who owns the shares between the *Start* and *End* periods, is defined as the present value of the dividends during that time-range, and the average share-price for the *End* period:²

$$W_{[Start, End]} = \sum_{t=Start}^{End} \frac{Dividend\ Per\ Share_t}{(1+d)^{t-Start}} + \frac{mean[SharePrice_{End}]}{(1+d)^{End-Start}} \quad (53)$$

Note that we include the dividends in both the *Start* and *End* periods, and we use the average share-price for the *End* period to avoid using single-day outliers for the ending share-price. This gives the best estimate of the intrinsic value for shareholders between the *Start* and *End* periods. If we had excluded the dividend in the *Start* period then it would also artificially inflate the ROIV ratio.

If we had not included the dividend in the *Start* period, then we could have used the following formula to “undo” the share buyback in the *Start* period, by scaling the per-share intrinsic value W up to the total amount, then add the net buyback amount, and finally adjust the number of shares:

$$V_{[Start, End]} \stackrel{?}{=} \frac{W_{[Start, End]} \cdot Shares_{Start}^W + Net\ Buyback_{Start}}{Shares_{Start}^V} \quad (54)$$

We do include the dividend in the *Start* period because it gives a more accurate intrinsic value W , and this creates a problem when “undoing” the share buyback in Eq.(54), because we don’t know whether the dividend came before or after the buyback in the *Start* period. If the dividend came AFTER the buyback then Eq.(54) is correct. But if the dividend came BEFORE the buyback then Eq.(54) under-estimates V , because the repurchased shares also received a dividend before they left, so we would have to add $Shares_{Start}^{NBB} \cdot Dividend\ Per\ Share_{Start}$ to the numerator in Eq.(54).

We generally don’t know whether the dividend or buyback came first, and they may also have been interleaved during the financial quarter. Share issuances also create similar problems in Eq.(54). So we do not use Eq.(54) to estimate the per-share intrinsic value without the net buyback.

2 The best way of calculating the exponents of the discount-factors in Eq.(53) is to subtract the *Start* time-stamp from all the time-stamps, and then convert these time-differences into the number of years relative to the *Start* time. This also handles financial data with varying time-frequency.

We can avoid the timing problem of the dividend versus buyback in Eq.(54), by first subtracting the *Dividend Per Share* (abbreviated as *DPS*) from W , and then add the total dividend amount instead.

The per-share intrinsic value WITHOUT the share buyback or issuance in the *Start* period is then:

$$V_{[Start, End]} = \frac{(W_{[Start, End]} - DPS_{Start}) \cdot Shares_{Start}^W + Dividend_{Start} + Net\ Buyback_{Start}}{Shares_{Start}^V} \quad (55)$$

4.3 Dividend Per Share

Dividends and share buybacks, as well as share issuances e.g. from vested stock options, may all be interleaved within a single financial quarter. The exact timing of these events may cause a small gain or loss when calculating the difference in per-share intrinsic value $W - V$, which may cause extreme ROB ratios when divided by a very small buyback amount. To prevent this, we enforce the ordering of dividend payouts to always come before buybacks, by calculating the dividend per share from the total dividend amount in the cash-flow statement, divided by the number of shares WITHOUT the share buyback in the *Start* period. We cannot use the weighted average number of shares, as that may also cause extreme ROB ratios for the same reason. The correct conversion is:

$$Dividend\ Per\ Share_{Start} \simeq Dividend_{Start} / Shares_{Start}^V \quad (56)$$

4.4 Return Ratios

The Return on Intrinsic Value (ROIV) measures the gain/loss in per-share intrinsic value resulting from the share buyback or issuance during the *Start* period, for the shareholders who owned their shares until the *End* period. It is defined like Eq.(14) but using W from Eq.(53) and V from Eq.(55):

$$ROIV_{[Start, End]} = \frac{W_{[Start, End]}}{V_{[Start, End]}} - 1 \quad (57)$$

The Return on Buyback (ROB) measures the gain/loss in per-share intrinsic value, relative to the amount that was either spent on a share buyback or received in a share issuance, for an investor who owned the shares between the *Start* and *End* periods. The formula for the ROB ratio is similar to Eq.(16), but in order to handle both share buyback (positive) and issuance (negative) in a single formula, we need to take the absolute value of the net buyback amount, so the sign of the ROB ratio is only determined by the difference in the per-share intrinsic value:

$$ROB_{[Start, End]} = \frac{W_{[Start, End]} - V_{[Start, End]}}{|Net\ Buyback_{Start}| / Shares_{Start}^V} \quad (58)$$

We have experimented with various 2D plots to show the return ratios for all possible *Start* and *End* periods, but the plots were hard to understand. So we just show the mean and standard deviation for each *Start* period and all subsequent *End* periods. Several examples are shown in sections 8 to 11.

4.5 Example

Let us try and use the formulas above to value a share buyback for the remaining shareholders who owned their shares for several years more. The first 2 rows of the table below show the average share-price and dividend per share for each year, which is the data we would obtain from a financial database. The next two rows show the per-share intrinsic value with (W) and without (V) a share buyback in the first period. The final two rows measure the gain/loss to the remaining shareholders who owned their shares from the beginning until year t .

Year t	1	2	3	4	5
$\text{mean}[SharePrice_t]$	\$100	\$50	\$80	\$120	\$150
$Dividend\ Per\ Share_t$	\$5	\$5.25	\$5.50	\$5.75	\$6
$W_{[Start=1, End=t]}$	\$105	\$56.16	\$83.16	\$114.40	\$133.81
$V_{[Start=1, End=t]}$	\$105	\$61.04	\$85.35	\$113.46	\$130.93
$ROIV_{[Start=1, End=t]}$	0%	-8.0%	-2.6%	+0.8%	+2.2%
$ROB_{[Start=1, End=t]}$	0%	-49%	-22%	+9.4%	+29%

Whether the share buyback in the 1st period resulted in a gain or loss to the shareholders at the time, depends on how long they held on to their shares, as shown in the table's last two rows.

In this example, the ROIV and ROB ratios are both 0% in the same year as the buyback was made. When using quarterly financial data from the real world, the return ratios are usually close to zero but not exactly equal to zero, because that would require the share buyback had been made at the exact average share-price for the quarterly period, and the other estimates were also exactly correct.

For the shareholders who sold their shares in the 2nd year, the share buyback caused a loss in the per-share intrinsic value of $ROIV_{[1,2]} \simeq -8.0\%$, and when compared to the amount used for the share buyback the loss was $ROB_{[1,2]} \simeq -49\%$, because the share-price was only half of what was paid in the share buyback in the 1st year. So the intrinsic value would have been significantly greater if the amount used for the share buyback in the 1st year had instead been paid out as a dividend.

For all holding periods between 1 and 5 years, the mean ROIV was about -1.5% and the standard deviation was about 3.6%. The mean ROB was about -6.5% and the standard deviation was about 27%. Sometimes the ROIV and ROB means have opposite signs, which may happen because of the non-linearity of the ROIV formula, which makes losses have greater magnitude than gains. This issue was analyzed in great detail in [Pedersen 2012] because it has important implications.

4.5.1 Calculation Example

Let us demonstrate how the values in the table above are calculated. For this we need the buyback amount and the share-count after the buyback. This data would also come from a financial database:

$$Net\ Buyback_1 = \$10\ b \quad Shares_1^W = 0.9\ b \quad (59)$$

The number of shares WITHOUT the share buyback in the first period is estimated using Eq.(50):

$$Shares_1^V \simeq Shares_1^W + \frac{Net\ Buyback_1}{\text{mean}[SharePrice_1]} = 0.9\ b + \frac{\$10\ b}{\$100} = 1\ b \quad (60)$$

The total dividend amount is calculated from the per-share dividend using the reverse of Eq.(56):

$$Dividend_1 \simeq Dividend\ Per\ Share_1 \cdot Shares_1^V = \$5 \cdot 1\ b = \$5\ b \quad (61)$$

We then consider an investment between $Start = 1$ and $End = 3$ years. We calculate the per-share intrinsic value with (W) and without (V) the share buyback, so we can compare the two scenarios. We use a discount-rate of $d = 8\%$ in the present value calculations.

The per-share intrinsic value WITH the share buyback is calculated using Eq.(53):

$$\begin{aligned} W_{[Start=1, End=3]} &= \sum_{t=Start}^{End} \frac{Dividend\ Per\ Share_t}{(1+d)^{t-Start}} + \frac{SharePrice_{End}}{(1+d)^{End-Start}} \\ &= \frac{\$5}{1.08^0} + \frac{\$5.25}{1.08^1} + \frac{\$5.50}{1.08^2} + \frac{\$80}{1.08^2} \simeq \$83.16 \end{aligned} \quad (62)$$

The per-share intrinsic value WITHOUT the share buyback is calculated using Eq.(55):

$$\begin{aligned} V_{[Start=1, End=3]} &= \frac{(W_{[Start, End]} - DPS_{Start}) \cdot Shares_{Start}^W + Dividend_{Start} + Net\ Buyback_{Start}}{Shares_{Start}^V} \\ &\simeq \frac{(\$83.16 - \$5) \cdot 0.9\ b + \$5\ b + \$10\ b}{1\ b} \simeq \$85.34 \end{aligned} \quad (63)$$

The ROIV ratio is calculated using Eq.(57):

$$ROIV_{[Start=1, End=3]} = \frac{W_{[Start, End]}}{V_{[Start, End]}} - 1 = \frac{\$83.16}{\$85.34} - 1 \simeq -2.6\% \quad (64)$$

The ROB ratio is calculated using Eq.(58):

$$ROB_{[Start=1, End=3]} = \frac{W_{[Start, End]} - V_{[Start, End]}}{|Net\ Buyback_{Start}| / Shares_{Start}^V} \simeq \frac{\$83.16 - \$85.34}{\$10\ b / 1\ b} \simeq -22\% \quad (65)$$

In practice, we let a computer calculate the ROIV and ROB ratios like this for all $Start$ and End periods, and then it calculates the mean and standard deviation for each $Start$ period and all the subsequent End periods. This results in two time-series with the means and standard deviations which can then be plotted together, as demonstrated in the case studies in sections 8 to 11.

4.6 Caveats

- Taxes for dividends and capital gains are ignored here because they are different between investors and the taxes may also change over time. But Eqs.(53) and (55) could be extended similarly to section 3 to include taxes for dividends and capital gains.
- Taxes on share buybacks are calculated on the net buyback amount in Eq.(49), meaning that share issuances are a tax-shield. That is how the tax works in USA. But other countries may tax the buyback amount regardless of share issuances which requires changes to Eq.(49).
- Funding of share buybacks is ignored because it is too difficult to try and determine whether a past share buyback was funded from excess cash, operating cash-flows, or debt issuance.
- Historical share issuance valuation can be quite misleading, because a share issuance is normally used to fund growth, or pay down debt to avoid bankruptcy. So the intrinsic value without a share issuance usually cannot be calculated accurately by merely adjusting the intrinsic value with the share issuance amount and adjusting the number of shares, as we do here, because the future earnings might be very different without the share issuance. So the valuation formulas above only give a very rough estimate of the gain/loss resulting from a past share issuance.
- Share issuance to fund an acquisition or merger with another company does not appear on the cash-flow statement of the parent company because it is not a cash transaction. It is disclosed in the notes of the financial statements. This data is not available in the financial database we use here, so this kind of share issuance is not shown in the plots further below.
- A spin-off is when a parent company divests a subsidiary into a separate company, and gives the shares in the new company to the shareholders of the parent company. This should be included when valuing historical share buybacks and issuances for the parent company, by considering the market-value of the new shares as a special dividend payout in Eq.(53). This data is available in the financial database we use here, so it is included in our calculations.
- A split-off is when a parent company divests a subsidiary into a separate company, and the shareholders of the parent company have the choice of getting shares in the new company in exchange for some of their shares in the parent company, which are then cancelled / retired. This is a form of share buyback where the parent company pays for its own shares with a subsidiary company instead of cash. This is a non-cash transaction that is disclosed in the notes of the financial statements of the parent company. This data is not available in the financial database we use here, so split-offs and related buybacks are not shown in the plots further below. The subsequent dividends and share-prices for the parent company already reflect the decreased share count and divestiture of the subsidiary, so further adjustments for the split-off are not needed in Eq.(53). A future research project could study the valuation of split-offs to assess whether a long-term shareholder should choose the parent or subsidiary company, when the market-value of their shares may not equal their intrinsic values.

5 Historical Returns – Cumulative

In the previous section 4, we presented formulas for valuing historical share buybacks and issuances to assess which ones caused gains or losses for the remaining shareholders. Those formulas were forward looking, meaning they isolated and measured the effect of ONE period's share buyback or issuance on all future investment periods. That was used to assess which historical share buybacks and issuances caused gains or losses for the remaining shareholders.

In this section, we present valuation formulas for measuring the cumulative gains or losses resulting from ALL the share buybacks between some *Start* and *End* periods. When the results are plotted, it clearly shows the long-term compounding effect of share buybacks, which sometimes reveal shocking losses for companies that believed share buybacks were just more tax-efficient dividends.

These formulas only consider share buybacks and ignore issuances, because the cumulative sum of share buybacks (positive) and issuances (negative) may cross zero, which causes extreme values when used as a denominator in the ROB ratio. The low-cost financial database we use here only contains the net buyback amount (i.e. share buyback minus issuance amount), so we set the negative net buyback amounts to zero before inserting the amounts into the valuation formulas below.

This section was by far the most difficult to research of this entire paper, because there are different ways of defining the cumulative valuation formulas, and it is very easy to get confused and fail to support all scenarios. After much deliberation and trial and error, we have arrived at the following formulas which are fairly simple and should be meaningful and valid for all scenarios.

5.1 Shares

To “undo” all share buybacks between some *Start* and *End* periods, we cannot just use the original number of shares outstanding in the *Start* period for all subsequent periods, because there might be other changes to the number of shares, e.g. from issuing shares for employee stock-options or to pay for acquisitions, or share retirements following stock split-offs of subsidiaries. We must retain all these other changes to the number of shares while only “undoing” the share buybacks.

In section 4.1, we derived two kinds of formulas for “undoing” a single period's share buyback or issuance either by using addition in Eq.(50), or by using multiplication in Eq.(52). The result is the exact same when considering a single period's share buyback or issuance.

But when “undoing” multiple years of share buybacks while keeping other share changes, we can only use multiplication to adjust the number of shares, so it properly scales the shares in subsequent years. This is easier to understand with a small example: Imagine a company with 1 million shares bought back its shares until there was only 1000 shares left. Then the company issued 100 shares to acquire another company. This corresponded to 10% of the company's shares and hence 10% of its market-cap. Afterwards there were 1100 shares outstanding. If we then try and “undo” all the share buybacks using addition, the final number of shares would be 1,000,100, as if the acquisition had still only cost 100 shares, but that would only correspond to 0.01% of the company's adjusted shares instead of 10%. The number of shares issued should be adjusted so it is on the same scale. We do this by multiplying all cumulative changes in the number of shares instead of adding them.

Let $Shares_t^W$ denote the number of historical shares outstanding at the end of some period t WITH all the share buybacks and issuances that occurred until then. This is the historical number of shares that we obtain from a financial database. This is assumed to be adjusted for stock-splits.

Let $Shares_t^{BB}$ denote the actual number of shares bought back in period t . Share issuance is ignored.

Let $Shares_{[Start, End]}^{V Cum}$ denote the number of shares outstanding at the *End* period WITHOUT ALL the share buybacks between the *Start* and *End* periods. This is the number of shares we must estimate.

To “undo” the share buyback for ONE period t , we scale the number of shares with the following factor. The numerator is the number of shares WITHOUT the buyback in that one period, and the denominator is the number of shares WITH the buyback in that period. It can be reduced slightly:

$$\frac{Shares_t^W + Shares_t^{BB}}{Shares_t^W} = 1 + \frac{Shares_t^{BB}}{Shares_t^W} \quad (66)$$

Note that if we multiply this with $Shares_t^W$ it is the same as adding back the repurchased shares:

$$\frac{Shares_t^W + Shares_t^{BB}}{Shares_t^W} \cdot Shares_t^W = Shares_t^W + Shares_t^{BB} \quad (67)$$

We can then “undo” all the share buybacks between the *Start* and *End* periods by multiplying all the intermediate factors in Eq.(66) together, so the number of shares WITHOUT ALL the buybacks is:

$$Shares_{[Start, End]}^{V Cum} = Shares_{End}^W \cdot \prod_{t=Start}^{End} \left(1 + \frac{Shares_t^{BB}}{Shares_t^W} \right) \quad (68)$$

This formula is generally NOT the same as merely adding back all the repurchased shares. Over decades of both share buybacks and issuances, the compounded difference can be significant.

If the financial database does not have the actual number of shares bought back, then we can estimate it from the buyback amount (net of excise tax), and the average share-price for the period:

$$Shares_t^{BB} \simeq \frac{Buyback_t / (1 + TaxRateBuyback_t)}{\text{mean}[SharePrice_t]} \quad (69)$$

The average market-cap for period t can be calculated from the average $SharePrice_t$ for the entire period and $Shares_t^W$ for the end of the period (you should not use the weighted average shares):

$$\text{mean}[MarketCap_t^W] = \text{mean}[SharePrice_t] \cdot Shares_t^W \quad (70)$$

We can then rewrite Eq.(68) to estimate the number of shares WITHOUT all the share buybacks as:

$$Shares_{[Start, End]}^{V Cum} \simeq Shares_{End}^W \cdot \prod_{t=Start}^{End} \left(1 + \frac{Buyback_t / (1 + TaxRateBuyback_t)}{\text{mean}[MarketCap_t^W]} \right) \quad (71)$$

5.2 Intrinsic Value

The per-share intrinsic value WITH all the historical share buybacks and issuances between the *Start* and *End* periods, for a shareholder who owns the shares during this time-range, is the present value of all the dividends and the share-price at the end. The formula is exactly the same as Eq.(53) that was used when valuing a single net share buyback. It is reprinted here for easy reference:

$$W_{[Start, End]} = \sum_{t=Start}^{End} \frac{Dividend\ Per\ Share_t}{(1+d)^{t-Start}} + \frac{\text{mean}[SharePrice_{End}]}{(1+d)^{End-Start}} \quad (72)$$

The per-share intrinsic value WITHOUT all the share buybacks between the *Start* and *End* periods, assumes all the buyback amounts were instead paid out as dividends, and all the share buybacks were also “undone” to increase the number of shares outstanding. The present value is then:

$$\begin{aligned} V_{[Start, End]}^{Cum} &= \sum_{t=Start}^{End} \frac{(Dividend_t + Buyback_t) / Shares_{[Start, t]}^{V\ Cum}}{(1+d)^{t-Start}} \\ &+ \frac{\text{mean}[MarketCap_{End}^W] / Shares_{[Start, End]}^{V\ Cum}}{(1+d)^{End-Start}} \end{aligned} \quad (73)$$

5.3 Dividend Per Share

For the same reason as explained in section 4.3, we should always calculate the dividend per share used in Eq.(72), as the total dividend amount divided by the number of shares WITHOUT the share buyback in that one period:

$$Dividend\ Per\ Share_t \simeq Dividend_t / (Shares_t^W + Shares_t^{BB}) \quad (74)$$

5.4 Return Ratios

The cumulative Return on Intrinsic Value (ROIV) measures the gain/loss in per-share intrinsic value caused by ALL the share buybacks between the *Start* and *End* periods. It is defined similarly to the original Eq.(14) but using *W* and *V* from Eqs.(72) and (73):

$$ROIV_{[Start, End]}^{Cum} = \frac{W_{[Start, End]}}{V_{[Start, End]}^{Cum}} - 1 \quad (75)$$

The cumulative Return on Buyback (ROB) measures the gain/loss in intrinsic value, relative to the present value of ALL the buyback amounts between the *Start* and *End* periods:

$$ROB_{[Start, End]}^{Cum} = \frac{W_{[Start, End]} - V_{[Start, End]}^{Cum}}{PV\ Buyback\ Per\ Share_{[Start, End]}} \quad (76)$$

Similarly to the original ROB ratio in Eq.(16), it is important that we divide the buyback amounts by the number of shares WITHOUT all the buybacks, $Shares^{V Cum}_{[Start, t]}$, so the denominator of the ROB ratio equals the dividend that each share would otherwise have received if there were no buybacks:

$$PV \text{ Buyback Per Share}_{[Start, End]} = \sum_{t=Start}^{End} \frac{Buyback_t / Shares^{V Cum}_{[Start, t]}}{(1+d)^{t-Start}} \quad (77)$$

As previously discussed in section 2.11, the ROIV and ROB ratios have the same numerator $W - V$ which determines the sign of the ratios, and the different denominators determine their magnitude. We can see this connection clearly in the plots of the cumulative ROIV and ROB ratios in the case studies further below, where the plotted curves look very similar but have different magnitudes.

5.5 Monetary Amounts

One motivation for doing the cumulative valuation was to show the monetary gain or loss caused by decades of share buybacks. But it seems impossible to accurately measure this in all scenarios when there are both share buybacks and issuances. Let us explain why, and what we must show instead.

5.5.1 Cumulative Buyback Amounts

First consider the cumulative sum of all share buyback amounts between some *Start* and *End* periods. The buyback amounts are compounded forward to their future value at the *End* period to adjust for the time-value of money, because \$1 twenty years ago is not worth the same as \$1 today:

$$Cumulative \text{ Buyback}_{[Start, End]} = \sum_{t=Start}^{End} Buyback_t \cdot (1+d)^{End-t} \quad (78)$$

But this is NOT the amount that the remaining shareholders at the *End* period could have received as dividends instead. To see why, consider a simple example where a company starts with 100m shares, and after decades of share buybacks they end up with only 10m shares. Say the company spent a total of \$50b on all these share buybacks. For simplicity ignore the time-value of money. If all these buyback amounts had instead been paid out as dividends, the 10m shares that ultimately remained at the *End* period, would only have received $10 \text{ m} / 100 \text{ m} \cdot \$50 \text{ b} = \$5 \text{ b}$ as dividends, because the other 90m shares would not have been repurchased and should also receive dividends.

5.5.2 Using the End Shares

We can try and adjust the cumulative buyback amount in Eq.(78) by dividing each period's buyback amount by $Shares^{V Cum}_{[Start, t]}$ which is the period's number of shares WITHOUT all the share buybacks. This gives us the per-share buyback amount for each period as if there were no buybacks. We then scale the entire sum by $Shares^W_{End}$ which is the actual number of shares remaining in the *End* period:

$$Adj \text{ Cumulative Buyback}_{[Start, End]} \stackrel{?}{=} Shares^W_{End} \cdot \sum_{t=Start}^{End} \frac{Buyback_t}{Shares^{V Cum}_{[Start, t]}} \cdot (1+d)^{End-t} \quad (79)$$

But this formula only works if the company makes no share issuances. To see this, let us continue the small example from above, where a company originally had 100m shares, and after spending \$50b on buybacks over several decades, they end up with only 10m shares. This means each of the remaining shares have forgone $\$50b / 100m = \500 in dividends to make all those share buybacks. This \$500 is the value of the sum in Eq.(79) when ignoring the time-value of money for simplicity. Scaling this \$500 per-share amount by the number of shares actually outstanding at the *End* period, $Shares_{End}^W = 10m$, we get the adjusted cumulative buyback amount of \$5b as expected.

But now assume the company issues 100m new shares so $Shares_{End+1}^W = 110m$. Multiplying this by the \$500 per-share amount that was spent on all the previous share buybacks, the result is an adjusted cumulative buyback amount of \$55b instead of the correct \$5b. This shows how share issuances will cause Eq.(79) to create “phantom” buyback amounts that never actually existed.

5.5.3 Using the Period Shares

Let us now try and adjust the cumulative buyback amount in Eq.(78), by only summing the part of each period’s buyback amount that the shareholders at that time could have received as dividends. We do this by scaling each period’s buyback amount with $Shares_t^W / Shares_{[Start, t]}^{V Cum}$ which is the fraction of the shareholders still remaining in that period after all the prior share buybacks:

$$Adj\ Cumulative\ Buyback_{[Start, End]} \stackrel{?}{=} \sum_{t=Start}^{End} Buyback_t \cdot \frac{Shares_t^W}{Shares_{[Start, t]}^{V Cum}} \cdot (1+d)^{End-t} \quad (80)$$

But even if there were only share buybacks and no issuances, this formula has the same problem as the original Eq.(78). If the company continues to buyback shares, then the adjusted buyback amount that is added for each period is overstated, because some of those shareholders will end up selling their shares back to the company, but the buyback amounts attributed to them are still counted.

5.5.4 Net Value Added

The problem is essentially the same for the monetary Net Value Added (NVA) from all the share buybacks. For the shareholders who own the shares between the *Start* and *End* periods, we could try and calculate the total NVA by using the per-share intrinsic values with (W) and without (V) all the share buybacks from Eqs.(72) and (73), and scale their difference by the remaining number of shares at the *End* period. We also need to adjust for the time-value of money because W and V are present values at the *Start* period. So the formula for the total cumulative NVA would be:

$$Cumulative\ NVA_{[Start, End]} \stackrel{?}{=} \left(W_{[Start, End]} - V_{[Start, End]}^{Cum} \right) \cdot Shares_{End}^W \cdot (1+d)^{End-Start} \quad (81)$$

But similarly to the cumulative buyback amount in Eq.(79), this only works when there are only share buybacks, and it fails and creates “phantom” value when there are also share issuances.

We could instead try and sum each period's per-share gain/loss scaled by the number of shares actually outstanding in that period, and also adjust for the time-value of money. In this formula, $V_{[t, End]}$ should only “undo” the one share buyback in period t using Eq.(55) from section 4.2:

$$Cumulative\ NVA_{[Start, End]} \stackrel{?}{=} \sum_{t=Start}^{End} (W_{[t, End]} - V_{[t, End]}) \cdot Shares_t^W \cdot (1+d)^{End-t} \quad (82)$$

But similarly to the cumulative buyback amount in Eq.(80), this fails even if there are only share buybacks, because it overstates each period's total NVA when there are subsequent share buybacks.

5.5.5 Gordian Knot

After having tried even more approaches, it seems impossible to define meaningful and accurate formulas for the cumulative total amounts from the perspective of the remaining shareholders, if the company has also issued new shares along the way. It seems that the only valid solution is to only calculate and plot the cumulative buyback amounts using Eq.(78) and similarly for dividends, and the cumulative ROIV and ROB ratios using Eqs.(75) and (76) which are on a per-share basis.

5.6 Example

The following table shows an example of cumulative share buybacks over 5 years. The first 4 rows show the data we would get from a financial database: Share-price, dividend per share, buyback amount, and the number of shares outstanding at the end of each period. The next two rows show the per-share intrinsic value with (W) and without (V) all the share buybacks. The final two rows measure the cumulative gain/loss to the remaining shareholders caused by all the share buybacks.

Year t	1	2	3	4	5
mean $[SharePrice_t]$	\$100	\$50	\$80	\$120	\$150
$Dividend\ Per\ Share_t$	\$5	\$5.25	\$5.50	\$5.75	\$6
$Buyback_t$	\$10b	\$4.5b	\$6.48b	\$8.75b	\$9.84b
$Shares_t^W$	900m	810m	729m	656m	590m
$W_{[Start=1, End=t]}$	\$105	\$56.16	\$83.16	\$114.40	\$133.81
$V_{[Start=1, End=t]}^{Cum}$	\$105	\$61.04	\$82.92	\$105.69	\$118.42
$ROIV_{[Start=1, End=t]}^{Cum}$	0%	-8.0%	+0.3%	+8.2%	+13%
$ROB_{[Start=1, End=t]}^{Cum}$	0%	-34%	+1.3%	+33%	+45%

The first two rows are the same as in section 4.5, which showed forward-looking valuation for only a single share buyback for 10% of the market-cap in the 1st year, and measured whether that one share buyback caused a gain or loss for the shareholders who remained into the following years.

In this example, we make share buybacks in all years for 10% of the pre-buyback market-cap in each year, and we calculate the cumulative gain/loss for the remaining shareholders. The ROIV and ROB ratios are both 0% in the 1st year. When using financial data from the real world, the return ratios are usually close to zero but not exactly equal to zero, because that would require the share buyback had been made at the exact average share-price for the quarterly period.

For the shareholders who owned the shares from year 1 to 2, the share buybacks in those two years caused a loss of $ROIV_{[1,2]}^{Cum} \simeq -8.0\%$ to the per-share intrinsic value, because the buybacks in year 1 were made at a share-price of \$100 which then dropped to only \$50 in year 2, so the shareholders would have been better off with a dividend in year 1. But 10% of the shares were also bought back in year 2 and the share-price then increased from \$50 to \$80 in year 3, so the shareholders who held onto their shares until year 3 would have gained +0.3% on the intrinsic value of their shares.

More buybacks were made in years 3 to 5 while the share-price kept increasing, so the shareholders who held onto year 5 would have gained +13% on the per-share intrinsic value, as a result of the company repurchasing shares instead of paying dividends for the same amount of money in those 5 years. The gain was +45% when compared to the amount that was spent on all the share buybacks.

5.6.1 Calculation Example

Let us demonstrate how the values in the table above are calculated. These results may be slightly different from the table above due to small cascading rounding errors. We consider an investment duration of 3 years from $Start = 1$ to $End = 3$ and we are using a discount-rate of $d = 8\%$.

First we estimate the number of shares bought back in each of the 3 years using Eq.(69):

$$\begin{aligned} Shares_1^{BB} &\simeq \frac{Buyback_1}{\text{mean}[SharePrice_1]} = \frac{\$ 10 b}{\$ 100} = 100 m \\ Shares_2^{BB} &\simeq \frac{\$ 4.5 b}{\$ 50} = 90 m \quad Shares_3^{BB} \simeq \frac{\$ 6.48 b}{\$ 80} = 81 m \end{aligned} \quad (83)$$

These are used in Eq.(68) to estimate the number of shares WITHOUT all the cumulative buybacks. There are no share issuances, so all these equal the original 1b shares outstanding. For example:

$$\begin{aligned} Shares_{[Start=1, End=3]}^{V Cum} &= Shares_{End}^W \cdot \prod_{t=Start}^{End} \left(1 + \frac{Shares_t^{BB}}{Shares_t^W} \right) \\ &= 0.73 b \cdot \left(1 + \frac{100 m}{0.9 b} \right) \cdot \left(1 + \frac{90 m}{0.81 b} \right) \cdot \left(1 + \frac{81 m}{0.73 b} \right) = 1 b \end{aligned} \quad (84)$$

The per-share intrinsic value WITH all the share buybacks is calculated using Eq.(72):

$$\begin{aligned}
 W_{[Start=1, End=3]} &= \sum_{t=Start}^{End} \frac{Dividend\ Per\ Share_t}{(1+d)^{t-Start}} + \frac{SharePrice_{End}}{(1+d)^{End-Start}} \\
 &= \frac{\$ 5}{1.08^0} + \frac{\$ 5.25}{1.08^1} + \frac{\$ 5.5}{1.08^2} + \frac{\$ 80}{1.08^2} \simeq \$ 83.16
 \end{aligned} \tag{85}$$

In practice you should calculate the per-share dividends from the total cash-flow amounts, as explained in section 4.3. But in this example we do the reverse calculation using Eq.(74):

$$\begin{aligned}
 Dividend_t &= Dividend\ Per\ Share_t \cdot (Shares_t^W + Shares_t^{BB}) \\
 Dividend_1 &= \$ 5 \cdot (900\ m + 100\ m) = \$ 5\ b \\
 Dividend_2 &= \$ 5.25 \cdot (810\ m + 90\ m) = \$ 4.725\ b \\
 Dividend_3 &= \$ 5.5 \cdot (729\ m + 81\ m) = \$ 4.46\ b
 \end{aligned} \tag{86}$$

The average market-cap in year 3 WITH the share buyback in that year is:

$$\text{mean}[MarketCap_3^W] = \text{mean}[SharePrice_3] \cdot Shares_3^W = \$ 80 \cdot 729\ m \simeq \$ 58.32\ b \tag{87}$$

The per-share intrinsic value WITHOUT all the share buybacks is calculated using Eq.(73):

$$\begin{aligned}
 V_{[Start=1, End=3]}^{Cum} &= \sum_{t=Start}^{End} \frac{(Dividend_t + Buyback_t) / Shares_{[Start, t]}^{V\ Cum}}{(1+d)^{t-Start}} \\
 &\quad + \frac{\text{mean}[MarketCap_{End}^W] / Shares_{[Start, End]}^{V\ Cum}}{(1+d)^{End-Start}} \\
 &\simeq \frac{(\$ 5\ b + \$ 10\ b) / 1\ b}{1.08^0} + \frac{(\$ 4.725\ b + \$ 4.5\ b) / 1\ b}{1.08^1} \\
 &\quad + \frac{(\$ 4.46\ b + \$ 6.48\ b) / 1\ b}{1.08^2} + \frac{\$ 58.32\ b / 1\ b}{1.08^2} \simeq \$ 82.92
 \end{aligned} \tag{88}$$

The ROIV ratio is calculated using Eq.(75):

$$ROIV_{[Start=1, End=3]}^{Cum} = \frac{W_{[Start, End]}^{Cum}}{V_{[Start, End]}^{Cum}} - 1 = \frac{\$ 83.16}{\$ 82.92} - 1 \simeq +0.3 \% \quad (89)$$

The present value of all the per-share buyback amounts is calculated using Eq.(77):

$$\begin{aligned} PV \text{ Buyback Per Share}_{[Start=1, End=3]} &= \sum_{t=Start}^{End} \frac{Buyback_t / Shares_{[Start, t]}^{V Cum}}{(1+d)^{t-Start}} \\ &= \frac{\$ 10 \text{ b} / 1 \text{ b}}{1.08^0} + \frac{\$ 4.5 \text{ b} / 1 \text{ b}}{1.08^1} + \frac{\$ 6.48 \text{ b} / 1 \text{ b}}{1.08^2} \simeq \$ 19.72 \end{aligned} \quad (90)$$

The ROB ratio is calculated using Eq.(76):

$$ROB_{[Start=1, End=3]}^{Cum} = \frac{W_{[Start, End]} - V_{[Start, End]}^{Cum}}{PV \text{ Buyback Per Share}_{[Start, End]}} = \frac{\$ 83.16 - \$ 82.92}{\$ 19.72} \simeq +1.2 \% \quad (91)$$

The sum of nominal buyback amounts is \$21b, but they should be adjusted for the time-value of money by compounding each year's buyback amount forward using the discount rate as in Eq.(78):

$$\begin{aligned} Cumulative \text{ Buyback}_{[Start=1, End=3]} &= \sum_{t=Start}^{End} Buyback_t \cdot (1+d)^{End-t} \\ &= \$ 10 \text{ b} \cdot 1.08^2 + \$ 4.5 \text{ b} \cdot 1.08^1 + \$ 6.48 \text{ b} \cdot 1.08^0 \simeq \$ 23 \text{ b} \end{aligned} \quad (92)$$

The nominal sum of dividend amounts is \$14.2b. Adjusted for the time-value of money it is:

$$\begin{aligned} Cumulative \text{ Dividend}_{[Start=1, End=3]} &= \sum_{t=Start}^{End} Dividend_t \cdot (1+d)^{End-t} \\ &= \$ 5 \text{ b} \cdot 1.08^2 + \$ 4.725 \text{ b} \cdot 1.08^1 + \$ 4.46 \text{ b} \cdot 1.08^0 \simeq \$ 15.4 \text{ b} \end{aligned} \quad (93)$$

In practice we let a computer calculate all these by always setting *Start* to the first data period, and gradually increasing *End* until the final data period. Plotting the results show the actual or “ex-post” effect on shareholder value from all the share buybacks. The only matter of debate is the choice of discount rate that is used to calculate intrinsic values from the historical dividends and share-prices.

6 Other Issues

6.1 Borrowing

Section 4.9 of [Pedersen 2012] studied how share buybacks funded with borrowed money impacts the intrinsic value. It was suggested increasing the discount-rate in the present value calculations to account for the increased risk of the company defaulting on its debt. Alternatively a revaluation term could be subtracted from the intrinsic value to account for the increased risk of default. Both methods resulted in valuation formulas that were overly sensitive to even small adjustments, so it was concluded that more research was needed.

A much better solution was found when developing the simulation models on the SimSim.Run website. Instead of only having the debt issuance in the value WITH a share buyback W , we now also have the debt issuance in the value WITHOUT a share buyback V . You can also increase the discount-rate if you want to account for the increased risk of the company defaulting on its debt.

This creates an even comparison between the dividend and share buyback scenarios, because the money is borrowed in both scenarios, and it is either used for a share buyback, or paid out as a dividend immediately, so the company would change its capital structure equally in both scenarios.

The formulas for the intrinsic value in Eqs.(5) and (32) have the borrowed amount added, and the interest payments subtracted from the future earnings, similar to this basic formula:

$$v = \text{Excess Cash} + \text{Borrow} + \sum_{t=1}^{\infty} \frac{\text{Earnings}_t - \text{Borrow} \cdot \text{Interest Rate}}{(1+d)^t} \quad (94)$$

The interest-rate should be given as the after-tax rate. For example, if the interest-rate is 5% and the corporate tax-rate is 20%, then the after-tax interest rate is only $5\% \cdot (1 - 20\%) = 4\%$

The interest payments should also affect the future share-prices if they are simulated from e.g. the future earnings and P/E ratios, because the interest payments are subtracted from the earnings.

Section 4.9 of [Pedersen 2012] had some closing remarks that are worth repeating here:

A share buyback for borrowed money should only be made when the market-cap is significantly below the intrinsic value to long-term shareholders. If the market-cap is near the intrinsic value then the recapitalization has no potential for gain but a large risk of loss of shareholder value.

Conversely, a share issuance to repay debt should only be made when the market-cap is significantly above the intrinsic value to existing shareholders, otherwise the dilution will decrease the value to existing shareholders. But a share issuance to repay debt is usually only done under distress when there is no choice. Therefore companies should avoid getting into unsustainable levels of debt, as it may have to be repaid from a share issuance when the share-price is low and hence causing a loss to existing shareholders.

6.2 Earnings Yield

The future earnings yield of a company is the estimated earnings for the next year divided by the current market-cap (or equivalently the earnings per share divided by the current share-price).

The earnings yield can be compared to e.g. low-risk bond yields, to quickly assess whether the stock is cheap or expensive. This is such a useful valuation tool that it is featured in several places on the [SimSim.Run](#) web-site, but it is also limited as it only considers a single year into the future.

For example, using data from the case study in section 8 further below, the company's historical sales and net profit margin data can be selected as shown in Figure 23 and Figure 24, and the selected data is then multiplied with each other to simulate the earnings shown in Figure 25, which is then divided by the market-cap at the time to simulate the earnings yield shown in Figure 9 here.

U.S. government bond yields with 10-year maturity were about 4.3% in 2025. The company had no debt so there was no risk of default. But the company's future is still much more uncertain than low-risk government bonds, so a significant risk premium would be appropriate. A reasonable hurdle rate would probably be 9%. If we compare that to the simulated earnings yield in Figure 9, we see that well over half of the distribution is above the hurdle rate, but a significant part is below. Therefore, it is quite possible that the actual earnings yield could be less than the required return, so a share buyback would result in a loss for the remaining shareholders.

But this is a very limited way of assessing whether a share buyback may result in a gain or loss to the remaining shareholders, in part because it only considers earnings for a single future year, and also because it cannot quantify the percentage gain/loss as done using the ROIV and ROB ratios.

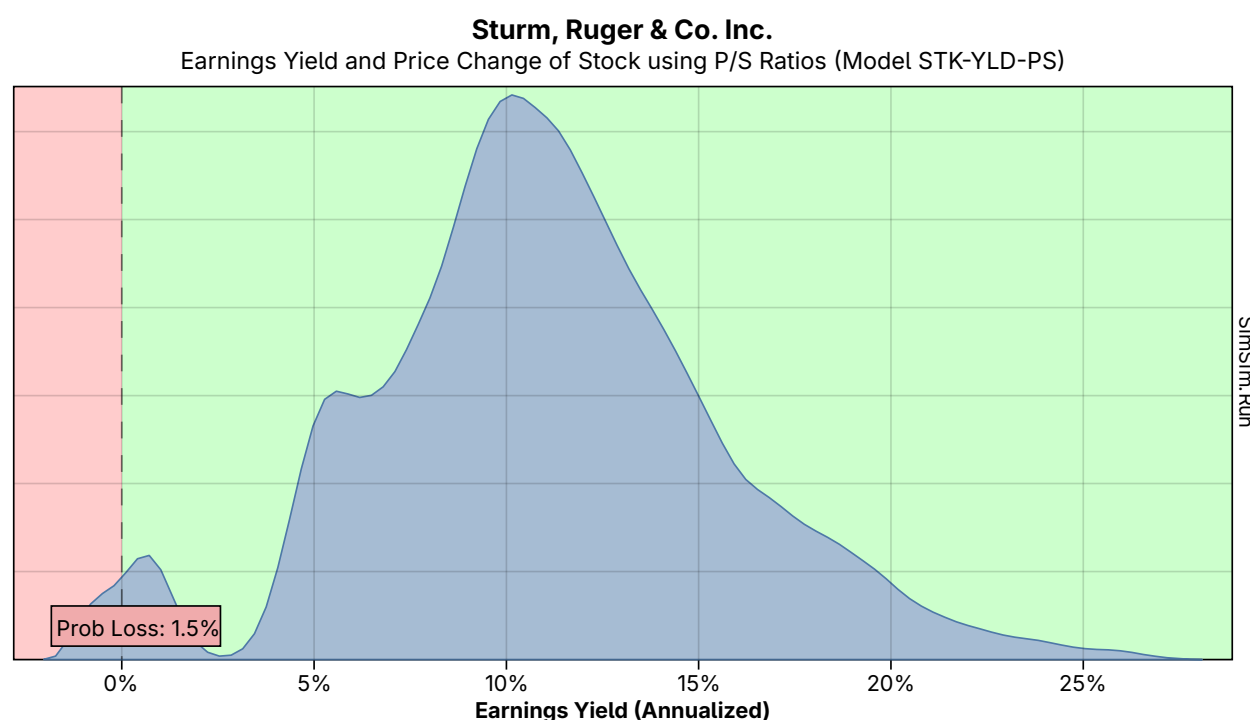


Figure 9: **Sturm, Ruger & Co.** Simulated earnings yield using the simulated earnings in Figure 25 divided by the company's market-cap at the time. See section 8 for the full case study.

6.3 Internal Rate of Return (IRR)

The Internal Rate of Return (IRR) is the discount-rate that makes the present value equal to the purchase price.³ The IRR measures the annualized rate of return on an investment, so it can easily be compared to e.g. government bond yields to assess whether a stock is cheap or expensive. The IRR also supports multiple years of earnings, as opposed to the earnings yield which only considers a single year. Unfortunately there is no formula for finding the IRR, so an algorithm has to search for the IRR, which sometimes fails for difficult input values. In some cases the IRR is not even well-defined mathematically, or there may be several different values that satisfy the equation.

For a company's intrinsic value in Eq.(5) the IRR is the discount-rate that satisfies this equation:

$$MarketCap = Excess\ Cash + \sum_{t=1}^{\infty} \frac{Earnings_t}{(1 + IRR)^t} \quad (95)$$

When using this definition of intrinsic value, ignoring dividend tax, and assuming the IRR is well-defined and unique for the input values, then the IRR is the same before and after a share buyback. This is proven by considering the equation after the share buyback, where the buyback amount must be subtracted from both sides of the equation, and let us denote IRR^W as being WITH the buyback:

$$MarketCap - Buyback = Excess\ Cash - Buyback + \sum_{t=1}^{\infty} \frac{Earnings_t}{(1 + IRR^W)^t} \quad (96)$$

The *Buyback* amount cancels out on both sides of this equation, and because the original IRR was assumed to be well-defined and unique we must have that $IRR = IRR^W$.

The IRR is generally not the exact same before and after a share buyback when also considering dividend taxes in the formulas above, and when considering changes in share-prices in Eq.(33).

You can use the IRR to assess whether a share buyback may cause a gain or loss for the remaining shareholders, by comparing the stock's IRR to a reasonable hurdle rate, but you should keep in mind that the IRR is not always well-defined mathematically, and it also cannot quantify the percentage gain/loss as done using the ROIV and ROB ratios.

The [SimSim.Run](#) web-site has several simulation models for the IRR, and you can easily copy the data between the IRR and share buyback models to see how they behave with different input data.

3 This was called "Value Yield" in [Pedersen 2012] because I had reinvented the concept without knowing it already existed under the somewhat misleading name "Internal Rate of Return".

6.4 Alternative Investment

Section 4.7 of [Pedersen 2012] derived valuation formulas for comparing a share buyback relative to an alternative investment that the company could make for the same money (e.g. build a new factory or acquire another company). It is easier and probably better to just compare the probability distributions for the returns on the share buyback and alternative investment, to assess which would be the most profitable for the company's shareholders. We can use one of these methods:

- The IRR of the company's stock can be compared to the IRR of the alternative investment. We don't need to select appropriate discount-rates. We don't even need to use specialized simulation models for share buybacks, as the normal IRR simulation models are sufficient. But the problem is that the IRR is not always well-defined mathematically.
- The ROB ratio of the share buyback can be compared to the NPV ratio of the alternative investment. We need to select appropriate discount-rates and use a specialized simulation model to get the ROB ratio and compare it to the NPV ratio of the alternative investment.

It is very important that you do not mix these two valuation methods! You must not compare IRR and NPV ratios, because it would grossly distort the comparison. This would be like using metric units for the company's stock, and imperial units for the alternative investment.

It is also important when comparing ROB and NPV ratios, that you either use zero tax-rates for dividends and capital gains – or you must apply the same dividend tax-rate to the NPV in order to make a fair comparison to the ROB ratio, which takes the final shareholder taxes into account. It is easier to use 0% tax-rates, and the stock's mispricing usually dominates the ROB ratio anyway.

Figure 10 shows an example with simulated ROB ratios for different share-prices for the buyback, compared to the NPV ratio of an alternative investment. It is not always clear which choice is best.

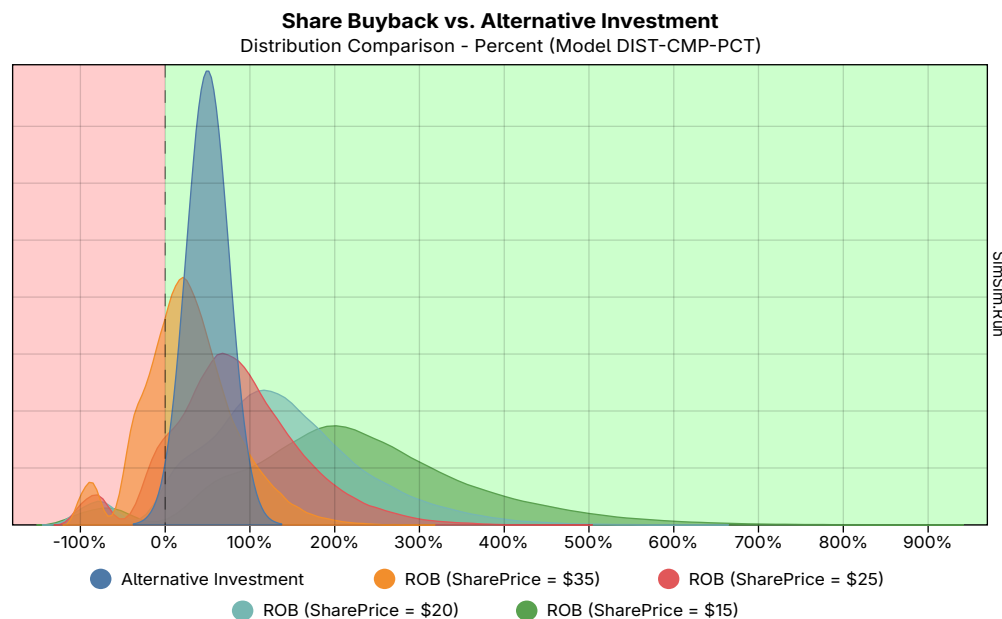


Figure 10: Comparing a share buyback and alternative investment.

The ROB ratio is simulated as in section 8.4 with different share-prices for the buyback.
The NPV Ratio of the alternative investment is normal-distributed with mean 50% and std.dev. 25%.

7 Case Study: U.S. Stock Market

Let us now use the valuation formulas from the previous sections on real-world financial data. In all plots of historical ROIV and ROB ratios, the discount-rate is 8% in the present value calculations.

In this section, we consider the 2600 largest U.S. stocks in aggregate, which have been selected using various criteria, and represent nearly the entire U.S. stock-market in terms of market-cap. In the following sections 8 to 11 we will study a few individual companies in detail.

7.1 Data Source

We use a low-cost financial database that aggregates and standardizes the raw financial data for U.S. stocks from the SEC EDGAR database with minimal human intervention. This database has fewer details and is sometimes missing data, and it may contain errors that were also present in the original corporate filings; but it costs a tiny fraction of the human-curated financial databases.

It is not always the people with the most expensive tools who do the best work. Although we use a low-cost financial database, our plots are infinitely more useful than what academia has produced for decades using very expensive financial databases, because they have basically just studied the correlation between share buybacks and abnormal stock-returns, whereas we study the actual change in long-term intrinsic shareholder value that was caused by the historical share buybacks.

7.2 Historical Share Buyback Returns – Forward Looking

The following plots show the historical forward-looking ROIV and ROB ratios as 2D distributions. These were calculated using Eqs.(57) and (58) for the 2600 largest U.S. stocks, for all possible *Start* and *End* quarters from the late 1990s to early 2026. This produced a total of about 6.8 million data-points with ROIV and ROB ratios. Nearly 3% of these data-points ($\pm 5 \times$ Inter-Quartile Range) were removed because they were extreme outliers, so we can focus on the bulk of the distributions. The results were then combined into one plot for all the ROIV ratios and another for all the ROB ratios.

Figure 11 shows the forward-looking ROIV ratios. The clipped range is roughly $\pm 0.8\%$. The ROIV ratios are centered around 0%. This means the effect of each quarter's share buyback or issuance on the per-share intrinsic value was mostly quite low. This is because a single quarter's share buyback is usually only a small fraction of the company's entire market-cap, so the impact is usually low.

Figure 12 shows the forward-looking ROB ratios. The clipped range is roughly $\pm 400\%$. Many ROB ratios are between $\pm 100\%$. These are much more dispersed than the ROIV ratios, because the ROB ratios measure the gain/loss relative to the buyback amount. So the ROB ratio can be very high if the stock was repurchased at a very low share-price relative to intrinsic value. But if the buyback amount was very small, then the ROIV ratio shows that the actual impact was also very small.

Note that recent years of these plots are “thinner” because there are fewer subsequent data-points. There is also a “pulsating effect” that is especially visible in Figure 12 with the ROB ratios. This suggests there was more share buyback and issuance activity during certain periods. For example, note how the ROB ratios look “thinner” during the great financial crisis in the years 2008-2010.

7.3 Historical Share Buyback Returns – Cumulative

The following plots show the cumulative share buyback returns for the 2600 largest U.S. stocks. The plots were produced by first computing the cumulative ROIV and ROB ratios using Eq.(75) and Eq.(76) for all the stocks individually. The *Start* period was always set to the first financial quarter of each stock, and the *End* period was progressively increased towards the final quarter. The results were then combined into one plot for all the ROIV ratios, and another for all the ROB ratios, with each plot showing a 2D distribution similar to a 2D histogram or heat-map where darker blues indicate more values in a region. The overall time-range is between the late 1990s until mid 2026.

Figure 13 shows the cumulative ROIV ratios. This shows the actual gain/loss in per-share intrinsic value as a result of all the share buybacks. For many of the 2600 stocks, the cumulative ROIV looks fairly close to 0%. But as time progresses the distribution spreads towards a total loss of -100% for some stocks, and other stocks gained more than +100%.

Figure 14 shows the cumulative ROB ratios. This shows the gain/loss in per-share intrinsic value, relative to the amount used for all the share buybacks. The cumulative ROB ratios are much more dispersed than the ROIV ratios in Figure 13. The ROB ratios often exceed $\pm 50\%$, because the ROB ratio measures the rate of return on the buyback amount, similar to the return on another investment the company might have made. This means the ROB ratio can show a large gain/loss of $\pm 50\%$ or more, but if the buyback amount was very small, then the actual effect on the per-share intrinsic value was also very small, as measured by the ROIV ratio.

7.4 Min / Max Cumulative Returns

The following plots show the fraction of companies that experienced different levels of all-time gains and losses on their share buybacks. The plots were produced by first computing the all-time min/max cumulative ROIV and ROB ratios for each individual company, and then summarizing all these data-points into plots of their Empirical Cumulative Distribution Functions (ECDF).

Figure 15 shows the ECDF for the minimum cumulative ROIV. For example, we can lookup 0.1 on the y-axis and the corresponding value on the x-axis is about -40%, meaning that 10% of the 2600 stocks have experienced an all-time ROIV loss of -40% or worse from their share buybacks.

Figure 16 shows the ECDF for the maximum cumulative ROIV. For example, we can lookup 0.9 on the y-axis and the corresponding value on the x-axis is about +30%, meaning that 10% of the 2600 stocks have experienced an all-time ROIV gain of +30% or better from their share buybacks.

Figure 17 shows the ECDF for the minimum cumulative ROB. For example, we can lookup 0.8 on the y-axis and the corresponding value on the x-axis is about -18%, meaning that 80% of the 2600 stocks have experienced an all-time ROB loss of -18% or worse from their share buybacks.

Figure 18 shows the ECDF for the maximum cumulative ROB. For example, we can lookup 0.35 on the y-axis and the corresponding value on the x-axis is about +50%, meaning that 35% of the 2600 stocks have experienced an all-time ROB gain of +50% or better from their share buybacks.

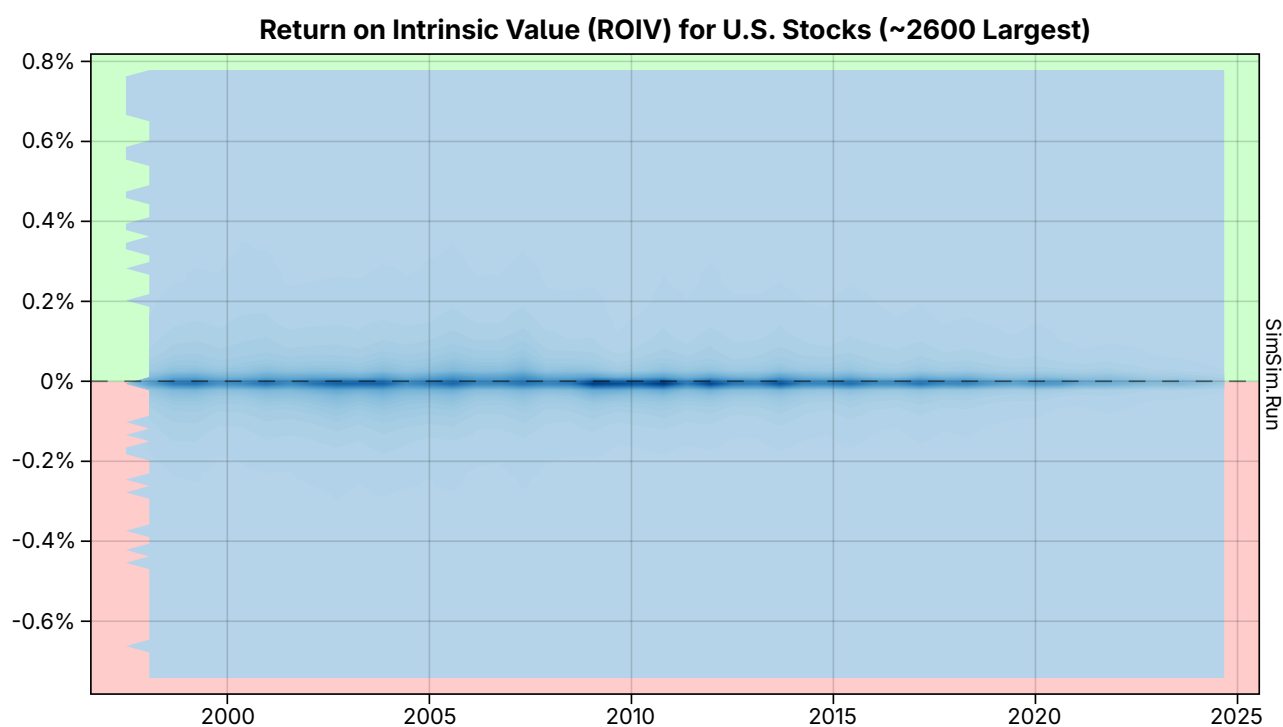


Figure 11: Historical **Forward Looking ROIV** using Eq.(57) for about 2600 of the largest U.S. stocks.
This is a 2D distribution where darker blues indicate more values in a region. Clipped at $\pm 5 \times \text{IQR}$.

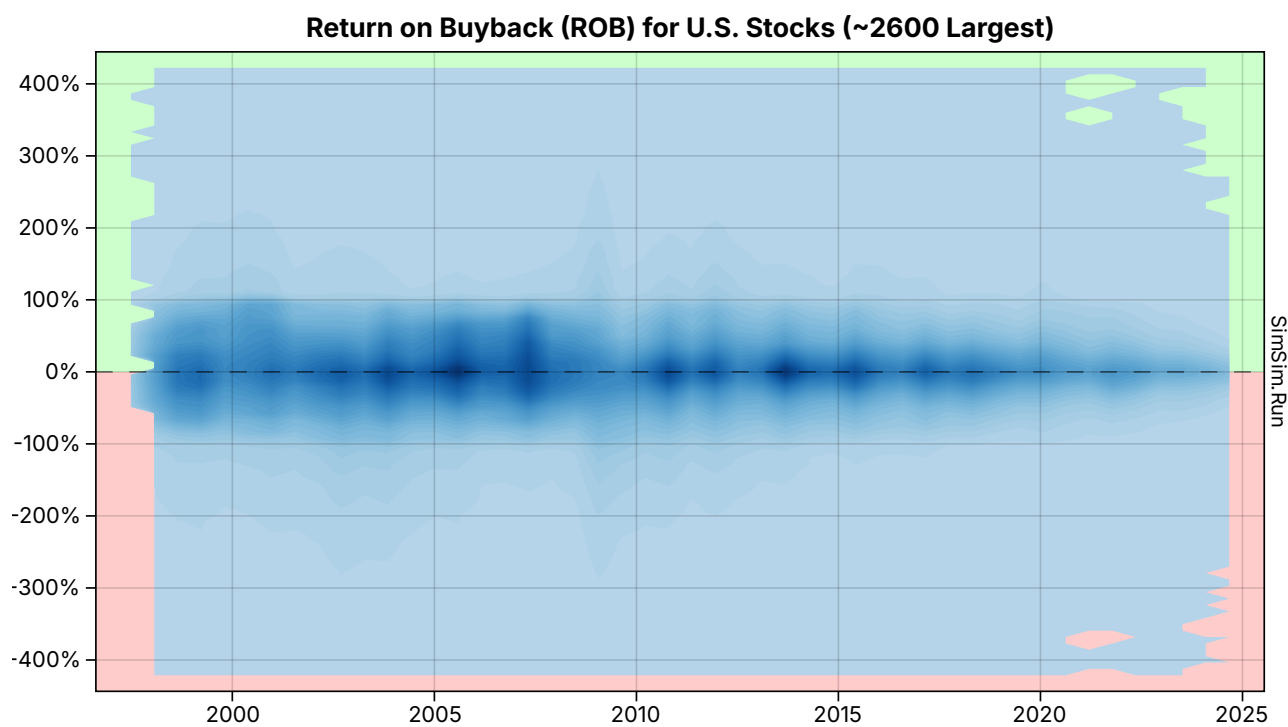


Figure 12: Historical **Forward Looking ROB** using Eq.(58) for about 2600 of the largest U.S. stocks.
This is a 2D distribution where darker blues indicate more values in a region. Clipped at $\pm 5 \times \text{IQR}$.

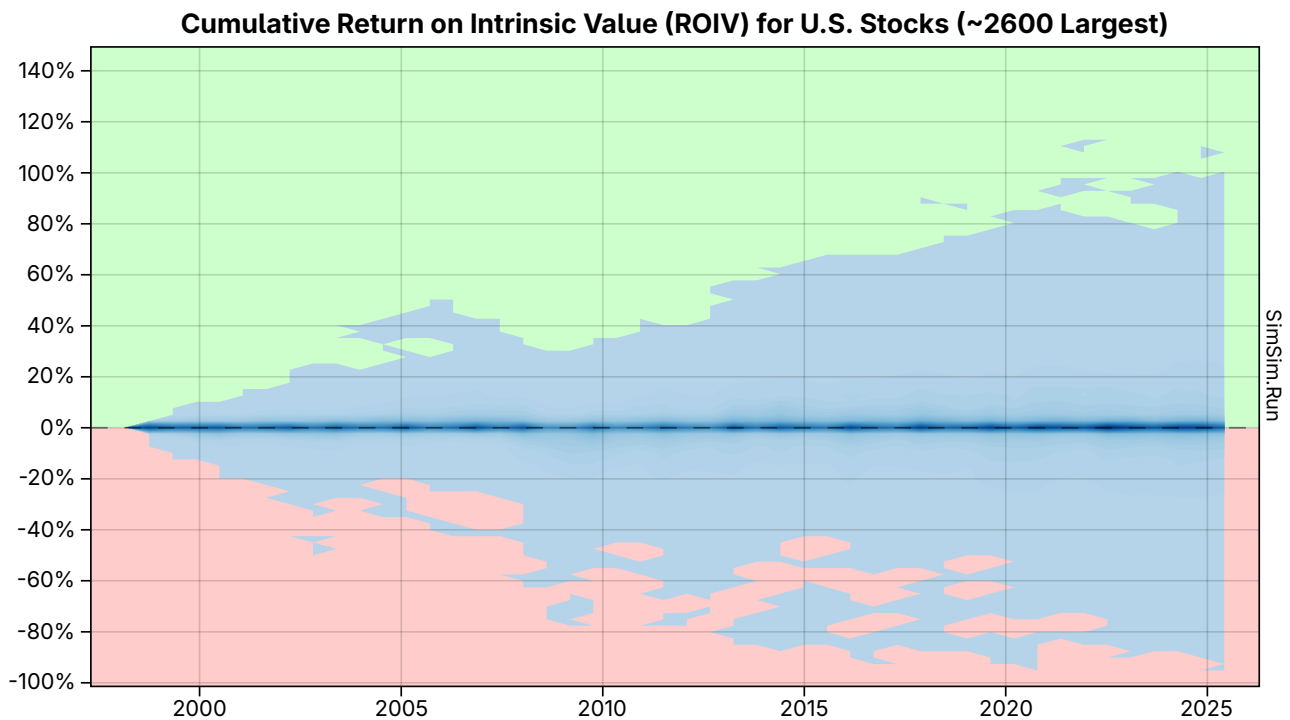


Figure 13: Historical **Cumulative ROIV** using Eq.(75) for about 2600 of the largest U.S. stocks.
This is a 2D distribution where darker blues indicate more values in a region. Clipped at +150%.

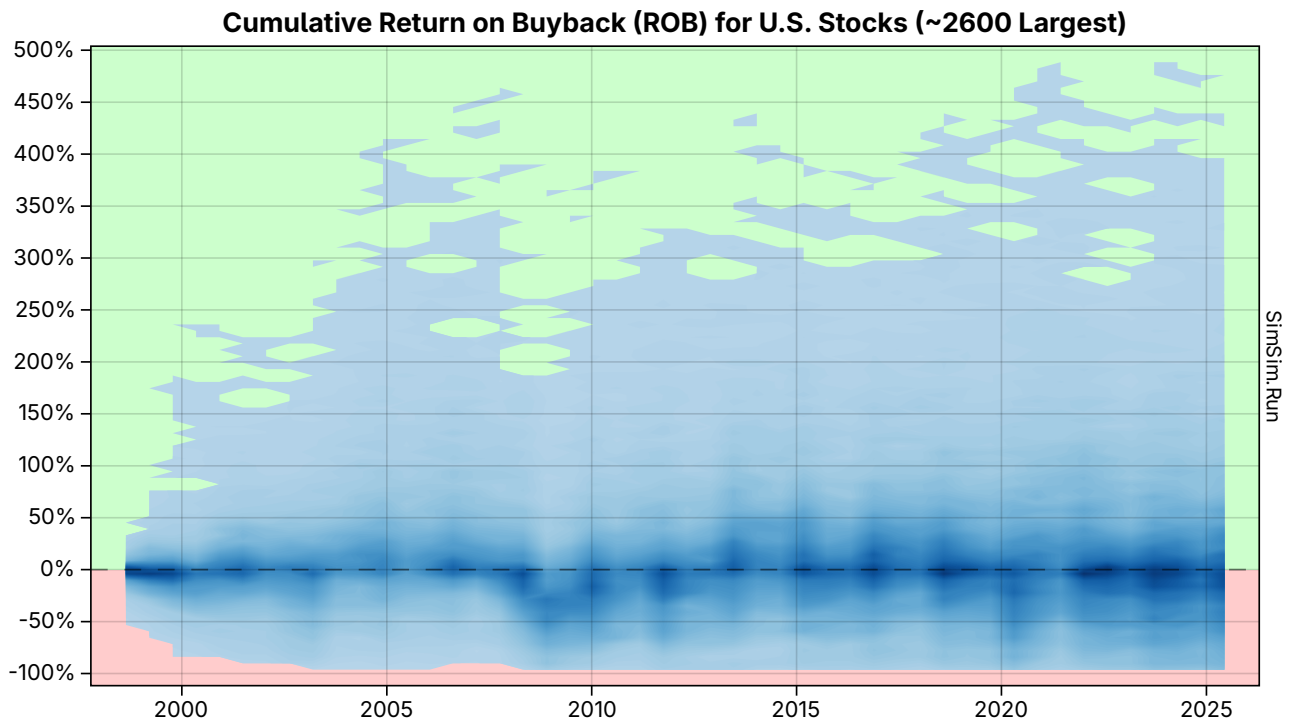


Figure 14: Historical **Cumulative ROB** using Eq.(76) for about 2600 of the largest U.S. stocks.
This is a 2D distribution where darker blues indicate more values in a region. Clipped at +500%.

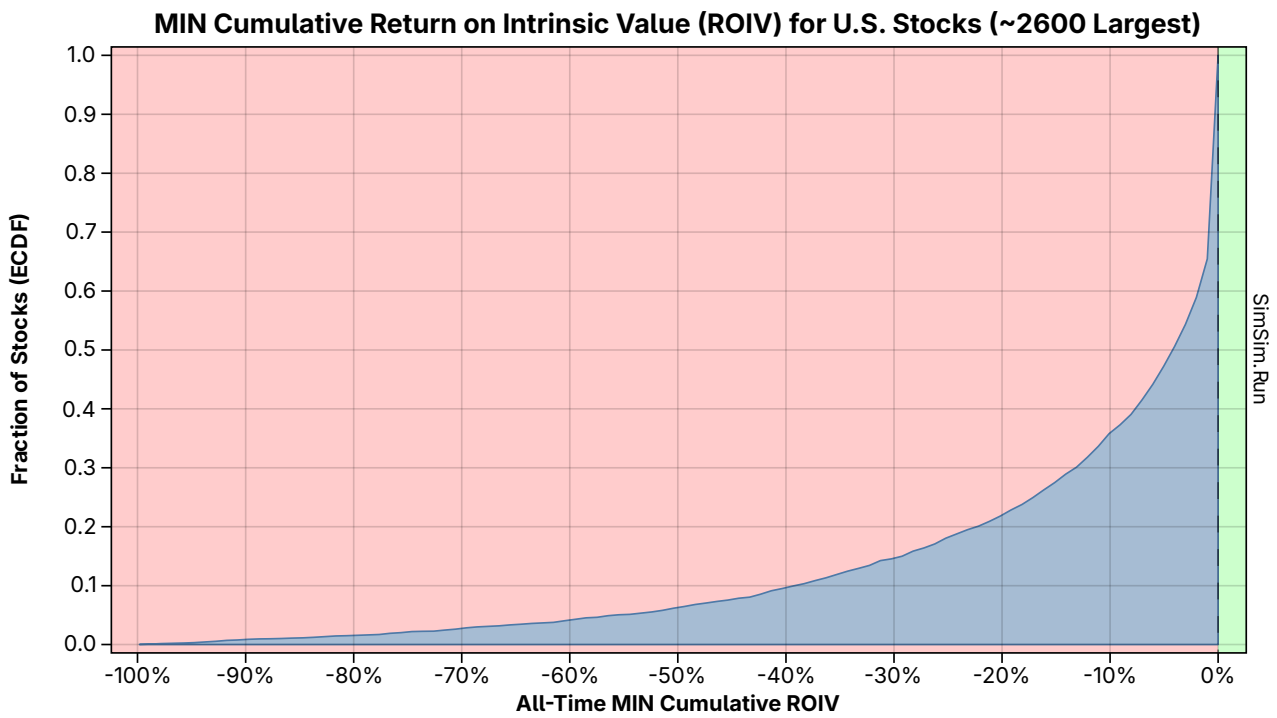


Figure 15: Fraction of companies with **all-time MINIMUM** cumulative ROIV.
This plot is an Empirical Cumulative Distribution Function (ECDF).

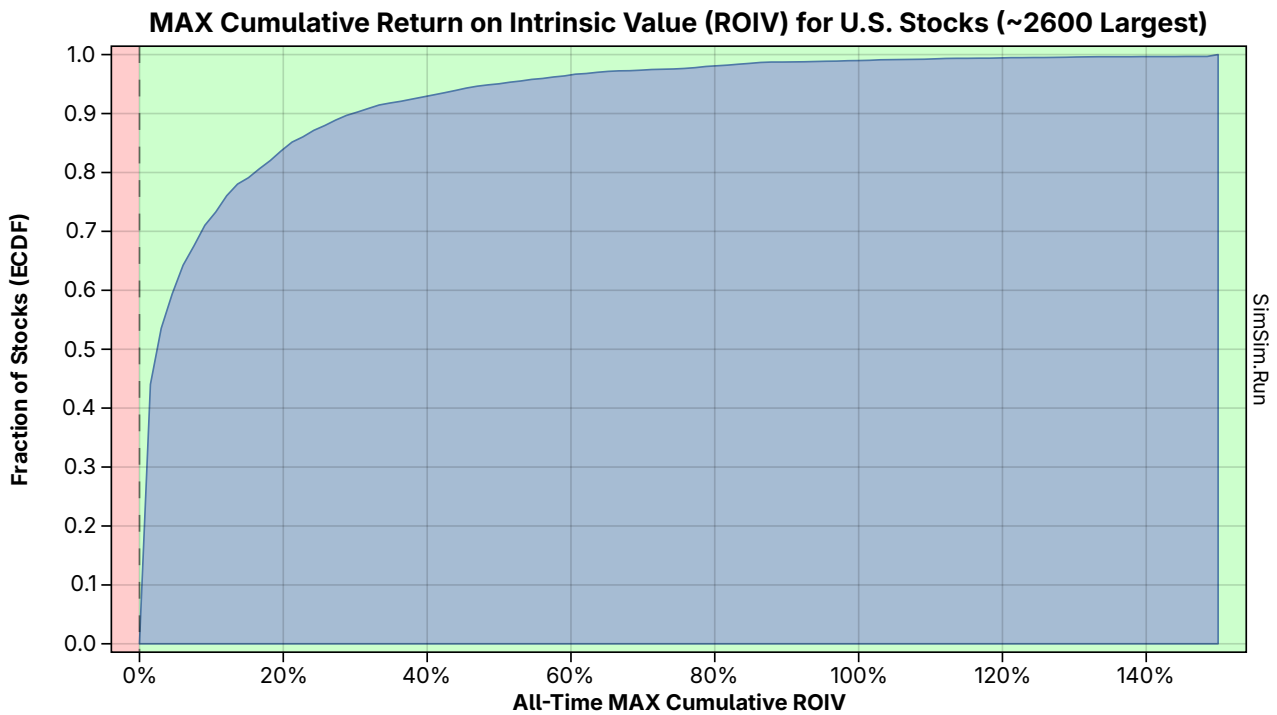


Figure 16: Fraction of companies with **all-time MAXIMUM** cumulative ROIV.
This plot is an Empirical Cumulative Distribution Function (ECDF).

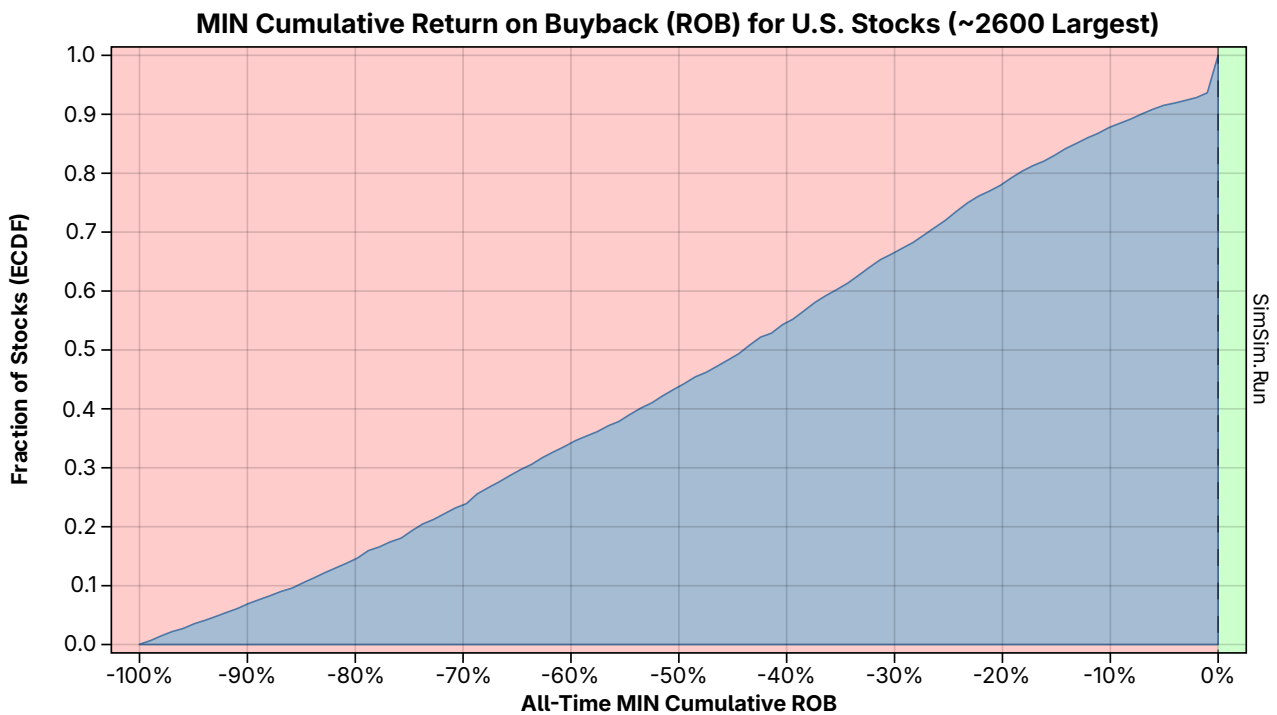


Figure 17: Fraction of companies with **all-time MINIMUM** cumulative ROB.
This plot is an Empirical Cumulative Distribution Function (ECDF).

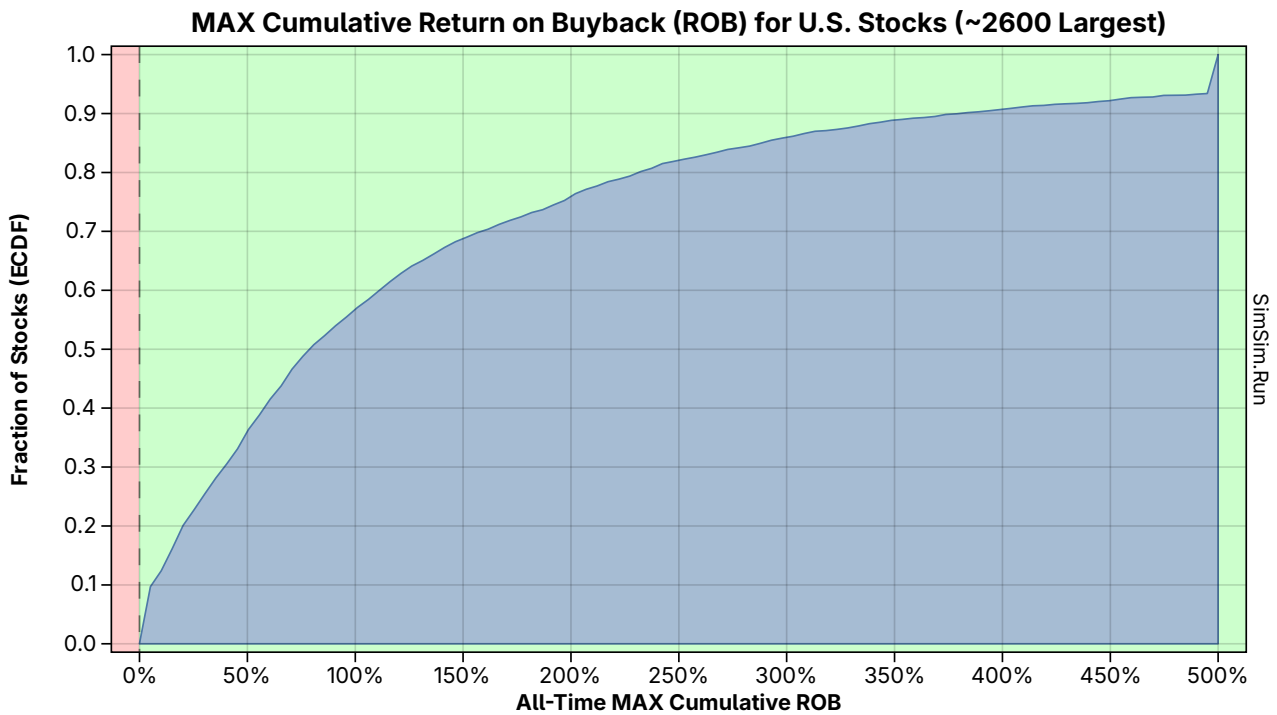


Figure 18: Fraction of companies with **all-time MAXIMUM** cumulative ROB.
This plot is an Empirical Cumulative Distribution Function (ECDF).

8 Case Study: Sturm, Ruger & Co. (RGR)

This company is a firearms manufacturer founded in 1947 in the USA. This is a very rare example of a company that has been very successful at making share buybacks at the exact right time. The stock-data shown here is taken from the [SimSim.Run](#) web-site (click the link to view the data).

8.1 Historical Share Buyback Returns – Cumulative

Figure 19 shows the cumulative returns on the company's historical share buybacks, which are calculated using the valuation formulas from section 5. Three sub-plots are shown (on the web-site you can click on the plots to pinpoint information for each data-point):

1. **Cumulative Payout** shows the total share buyback and dividend amounts, which have been compounded forward at an 8% annual rate to adjust for the time-value of money. This shows the future value of past payouts, as if they had been invested at an 8% annual rate of return. We also use an 8% discount-rate in the present value calculations for the ROIV and ROB ratios. You could argue that other rates would be more appropriate, but that is a very long discussion that is far from settled, and we avoid it here by always using a modest 8%.
2. **Return on Intrinsic Value (ROIV)** shows the cumulative gain/loss in per-share intrinsic value, as a result of all the share buybacks. It is calculated using Eq.(75) by setting *Start* to the first financial quarter available, and then progressively moving *End* forwards until the final quarter available, and then calculate the ROIV ratio for all these *End* points.
3. **Return on Buyback (ROB)** shows the cumulative gain/loss in per-share intrinsic value, relative to the present value of all the share buybacks. It is calculated similarly to the ROIV ratio but using Eq.(76) instead.

Caveat: As explained in section 5.5.4, it is generally not possible to calculate the total monetary amount that was gained or lost by the remaining shareholders, as a result of the share buybacks. You cannot just multiply the final cumulative ROB ratio with the final cumulative buyback amount, because that buyback amount also includes the money that were paid to the shareholders who left.

Figure 19 shows that the company's share buybacks were much smaller than their dividends, but the share buybacks caused significant gains for the remaining shareholders. Notice a few things:

- The company paid dividends since 1998, but only started making share buybacks around 2006. Both amounts keep increasing in the following decades, thus indicating the company continued to pay dividends and make share buybacks, but that growth may also be caused simply by compounding the amounts forward by 8% per year to adjust for the time-value of money. As we will see in Figure 20 below, the company rarely made share buybacks.
- In 2026, the cumulative dividend was \$2.4b when adjusted for the time-value of money, while the nominal amount was only about \$0.8b (not shown here). The cumulative buyback was \$560m when adjusted, but the nominal amount was only about \$240m. You may argue about the choice of compounding rate when adjusting for the time-value of money, but it is obvious that the company spent far more on dividends than share buybacks.

- Between 1998 and 2006, the ROB ratio showed around -50% loss because there was a tiny share buyback in 1998 made at a high share-price. But the buyback amount was so tiny that the actual loss of intrinsic value was only around -0.03% as shown by the ROIV ratio.
- Until 2010, the ROIV ratio was close to 0%. It then grew to 19% in 2013 due to buybacks and rising share-prices, and then started to decrease from 2023 due to falling share-prices.
- The ROB ratio followed the same path as the ROIV ratio, but reached more than 400% in 2013, and then gradually declined to 55% in 2026, again due to falling share-prices.
- The cumulative ROIV and ROB ratios are either positive or negative at the same time, because they are both calculated from the difference in per-share intrinsic value $W - V$, and then divided by V in the ROIV ratio, and by the buyback amount in the ROB ratio. So the curves look very similar but have different magnitudes that also shift over time.

8.2 Historical Share Buyback Returns – Forward Looking

The cumulative valuation in Figure 19 is great for showing the compounded effect of decades of share buybacks, which in this case revealed a significant gain for the remaining shareholders. But it cannot show which specific points in time had share buybacks that caused gains or losses. For this we need the forward-looking valuation formulas from section 4.

Figure 20 shows the forward-looking historical valuation of share buybacks and issuances in four sub-plots (on the web-site you can click on the plots to pinpoint information for each data-point):

1. **P/S Ratio** calculated from the average market-cap in the quarter and the TTM sales. This makes it easy to see if the company has had a tendency to buyback or issue shares when the valuation ratio was relatively high or low.
2. **Net Share Buyback (% of Assets)** is positive when the company made a net share buyback and negative for a net share issuance. It is shown relative to the company's total assets so it is easy to see the relative magnitude over time. Share issuances to pay for mergers and acquisitions are not shown in this sub-plot, because they are not in the cash-flow statement.
3. **Return on Intrinsic Value (ROIV)** measures the change in per-share intrinsic value from the net share buyback or issuance. A positive ROIV is a gain for the remaining shareholders and a negative ROIV is a loss. It is calculated using Eq.(57) for all possible *Start* and *End* periods, and the plot shows the mean and two standard deviations for each *Start* period.
4. **Return on Buyback (ROB)** measures the change in per-share intrinsic value relative to the net buyback or issuance amount. A positive ROB is a gain for the remaining shareholders and a negative ROB is a loss. It is calculated using Eq.(58) for all possible *Start* and *End* periods, and the plot shows the mean and two standard deviations for each *Start* period.

Figure 20 shows that the company rarely made share buybacks during the last 30 years, and the few share buybacks they made were generally at low P/S ratios for the stock. Notice a few things:

- In 2006 and 2007, two large share buybacks were made. The ROIV ratios were about 6% and 7% on average, and the ROB ratios were about 180% and 270% on average. So these share buybacks were incredibly good investments for the remaining shareholders.
- In late 2008, a small share buyback was made during a major financial crisis and stock-market crash. The P/S ratio was even lower, but the buyback amount was fairly low so the ROIV ratio was also low. Compared to the buyback amount, the ROB ratio was over 300% on average, so it was still a fantastic return on investment for the remaining shareholders.
- In late 2014, a large share buyback was made whose average ROIV and ROB ratios were around zero. That is because the present value of future dividends and share-prices was roughly equal to the share-price at the time of the buyback, as shown in Figure 22.
- In early 2017, a large share buyback was made which had an average ROIV ratio around -1.5% and a ROB ratio around -23%. So far this share buyback has resulted in an average loss to the remaining shareholders, because the share-price fluctuated in the following years, and in late 2023 the share-price started declining significantly below the 2017 share-prices when the buyback was made. The share-price would have to gain significantly in the next few years, if the buyback in 2017 should have been a gain for long-term shareholders.
- Apart from a small buyback in 2019, the company did not make any buybacks until 2023 when the P/S ratio started becoming low again. This was only a few years ago so there is only a few years of subsequent dividend and share-price data, so we cannot properly value those buybacks yet, and we will instead make a simulation of the value further below.

8.3 Market Timing

Figure 21 shows the company's historical dividend payout and net share buyback, both shown as percentages of total assets so it is easy to see the relative magnitude over time. This data is Trailing Twelve Months (TTM) instead of the quarterly data shown in Figure 20, which means that an isolated share buyback in a single quarter will show up in the following three quarters as well.

Figure 22 shows the historical Net Present Value (NPV) ratio, where the present value is calculated using Eq.(53) from subsequent dividends and share-prices, and then divided by the *Start* period's average share-price. These are calculated for all possible *Start* and *End* periods, and the plots show the mean and two standard deviations for each *Start* period and all subsequent *End* periods.

Notice the following from comparing Figure 21 and Figure 22:

- Between 2006 and 2009, dividends were halted and the cash was spent on share buybacks instead (along with cash from divesting unproductive assets, not shown here). This turned out to be a fantastic investment for the remaining shareholders. The NPV ratio was around 60% to 400% on average during this period, because not only was the stock's valuation ratio low, but a few years later the company's sales also grew 4x and its net profit margin expanded to more than 10%. This was partly because the company modernized its manufacturing process, and partly due to the political situation in USA at the time.

- After 2009, dividends were reinstated and they were then adjusted to the company's irregular earnings (which are shown on the [SimSim.Run](#) web-site). Share buybacks were only made when the company's management deemed the share-price was sufficiently low.

8.4 Simulation

When valuing historical share buybacks for a *Start* period, we use the subsequent dividends and share-prices to estimate the intrinsic value during the *Start* period. This means the recent share buybacks cannot be valued accurately because there are few or no data-points available for the subsequent dividends and share-prices. This can also be seen from Figure 22 where the NPV ratio narrows around zero in the recent years, because the present value of subsequent dividends and share-prices is nearly the same as the *Start* share-price.

To assess the value of the company's recent share buybacks, we can either just simulate the future earnings and use the valuation formulas from section 2, or we can simulate both the future earnings and share-prices using the formulas from section 3.

We do not have any special insight about the company's future earnings, and in the past the company's sales and profit margins have fluctuated greatly from year to year. That is exactly why we must use probability distributions in these valuations. Figure 23 shows the company's historical sales where we have selected the period from 2015 onwards. Figure 24 shows the company's historical profit margins where we have selected the period from 2010 onwards. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 25, which ranges roughly between USD -5m and +160m with a peak around USD 60m.

We will only simulate the share buyback valuation using future earnings here. We use a model on the [SimSim.Run](#) web-site that supports terminal growth-rates (click the link to try it). You can easily try one of the other models to simulate future share-prices as well, similar to section 3.4.

We use the simulated earnings in Figure 25 and the following assumptions from 2025:

$$\begin{aligned}
 \text{Shares} &= 17 \text{ m} & \text{SharePrice} &= \$ 35 \\
 \text{Buyback} &= \$ 30 \text{ m} & \text{Excess Cash} &= \$ 70 \text{ m} \\
 \text{Terminal Growth Rate} &= 0 \% & d &\sim N(\mu = 9 \% , \sigma = 1 \%)
 \end{aligned} \tag{97}$$

Figure 26 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The distribution is skewed down because of the non-linearity of Eq.(14), which also creates outliers that were removed. The clipped range is -10% to +4% with 26% probability of loss. The most likely ROIV ratio is the peak around 1.8%.

Figure 27 again shows the simulated ROIV ratio from Eq.(14), but here the buyback share-price (or equivalently the *MarketCap*) is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$50 the most likely ROIV ratio would be around 0%.

Figure 28 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -100% to +300% with a peak at 15-20%. The probability of loss is 29%.

Figure 29 again shows the simulated ROB ratio from Eq.(16), but here the buyback share-price (or equivalently the *MarketCap*) is varied on the x-axis. For example, when the share-price is around \$45 the most likely ROB ratio would be around 0%. This is different from the share-price that is most likely value neutral for the ROIV ratio in Figure 27 because of the non-linearity and skewness.

Note that the probability of loss is slightly different between the ROIV and ROB ratios, because it is measured after outliers have been removed, and more outliers were removed for the ROIV ratios.

8.5 Summary

The share buybacks in 2006 and 2007 during the company's turnaround were masterfully done and significantly increased the per-share intrinsic value for the remaining shareholders. The buybacks in 2015 were probably value neutral. The buybacks in 2017 probably caused a small loss unless the share-price increases significantly within the next few years. Simulations for the recent buybacks indicate that they will likely cause a gain in shareholder value, but only time will tell for sure.

↑ USD

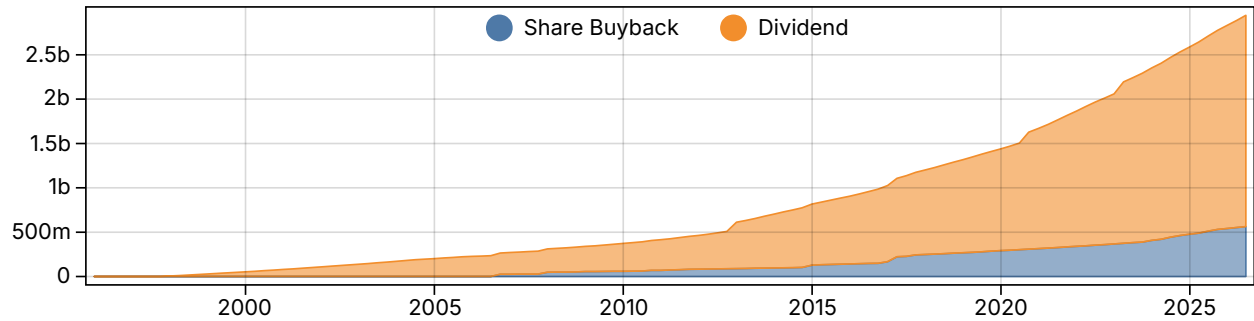
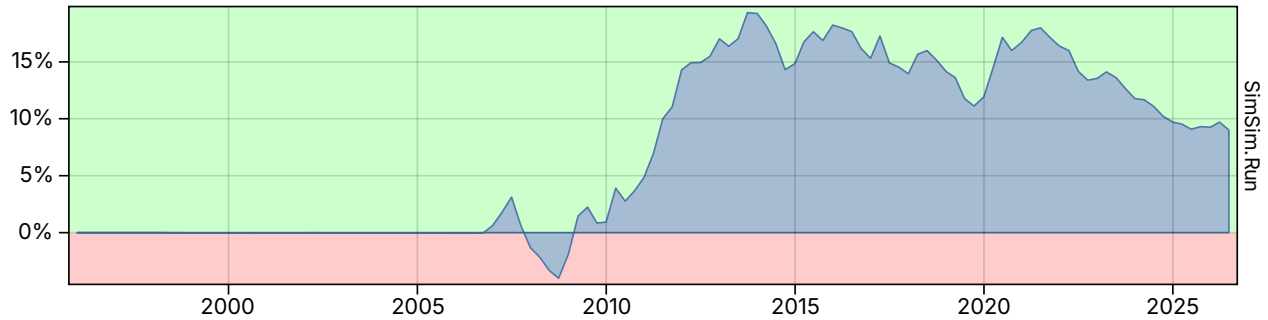
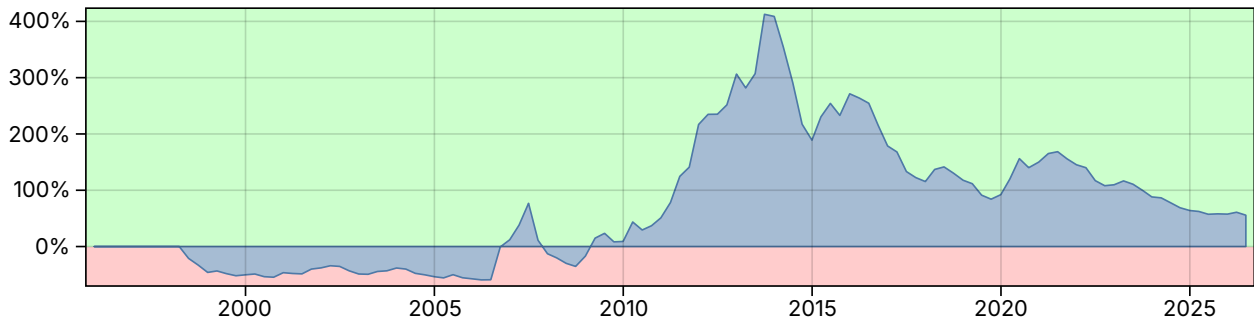
Cumulative Payout for Sturm, Ruger & Co. Inc. (RGR)**Return on Intrinsic Value (ROIV Cumulative)****Return on Buyback (ROB Cumulative)**

Figure 19: **Sturm, Ruger & Co.** Valuation of historical share buybacks using the **cumulative** ROIV in Eq.(75) and ROB in Eq.(76), which are calculated between the first and all subsequent time periods. Also shown are the cumulative share buyback and dividend amounts, adjusted for the time-value of money by compounding forward at 8%. (Quarterly data.)

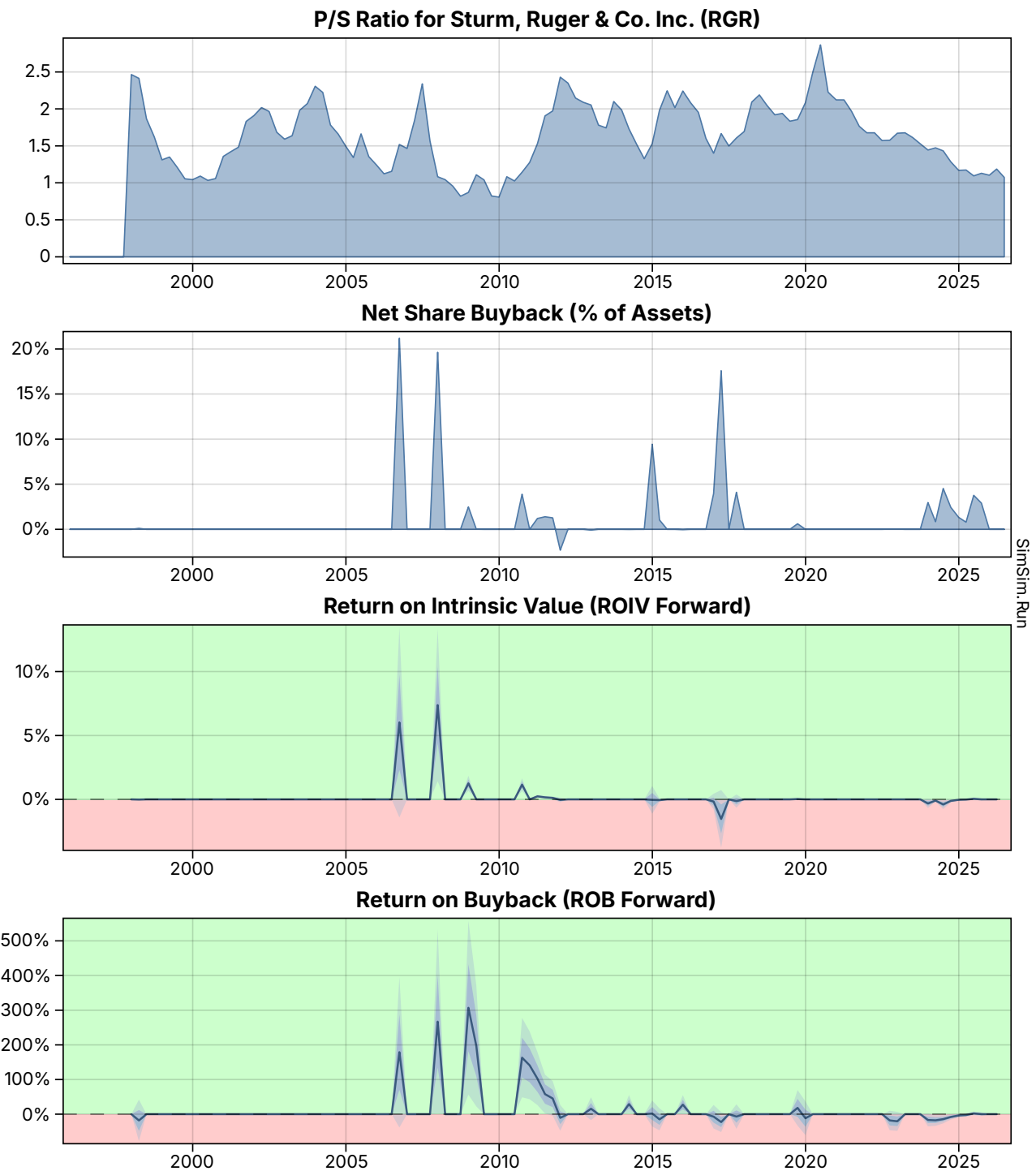


Figure 20: **Sturm, Ruger & Co.** Valuation of historical share buybacks and issuances using the **forward-looking** ROIV in Eq.(57) and ROB in Eq.(58), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is the historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

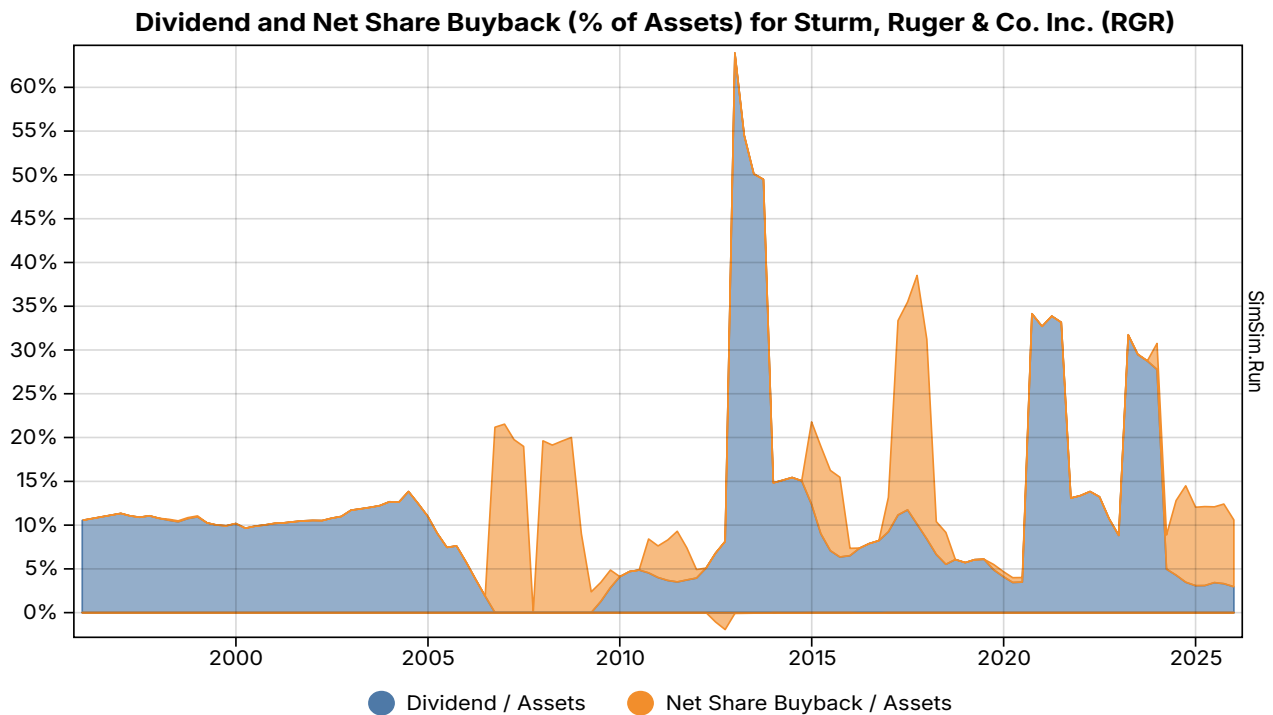


Figure 21: **Sturm, Ruger & Co.** Historical dividend payout and net share buyback as percentages of total assets. (TTM data where each data-point covers the preceding four quarters.)

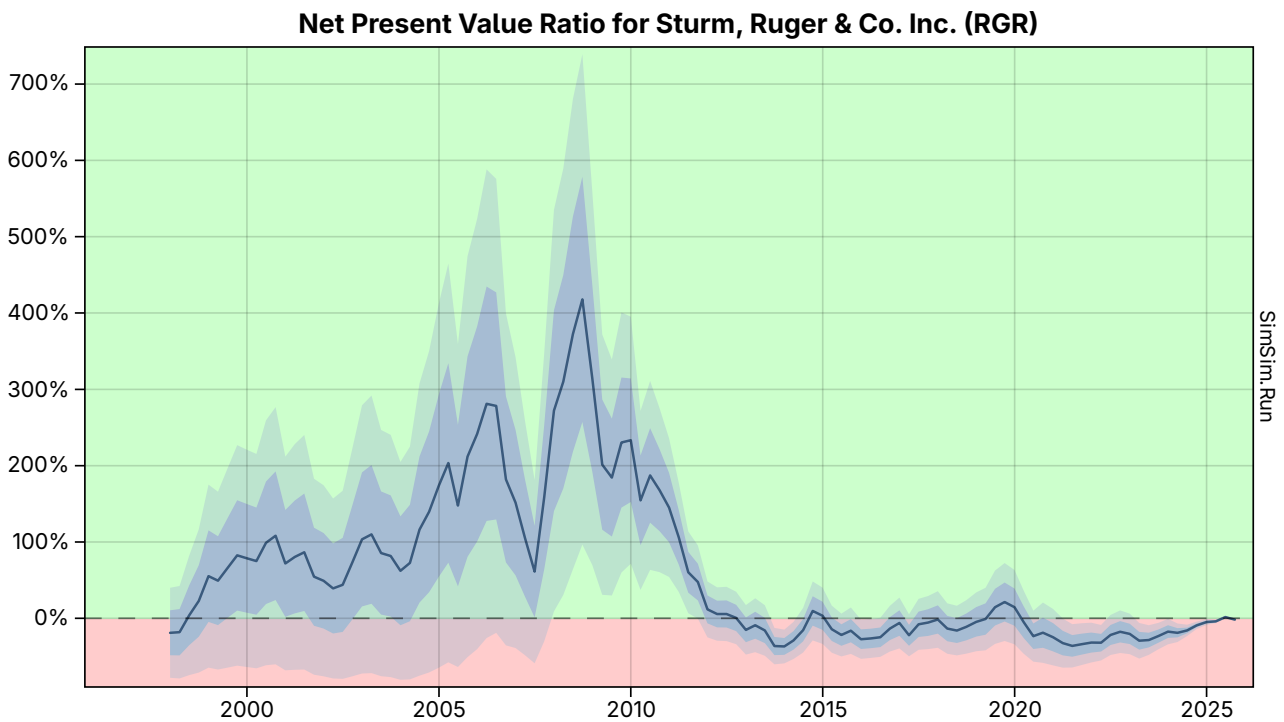


Figure 22: **Sturm, Ruger & Co.** Historical net present value ratio, where the present value is calculated using Eq.(53) from the future dividends and share-prices, and then divided by the Start period's average share-price. These are calculated for all possible Start and End periods, and the plot shows the mean and two standard deviations for each Start period and all subsequent End periods. (Quarterly data.)

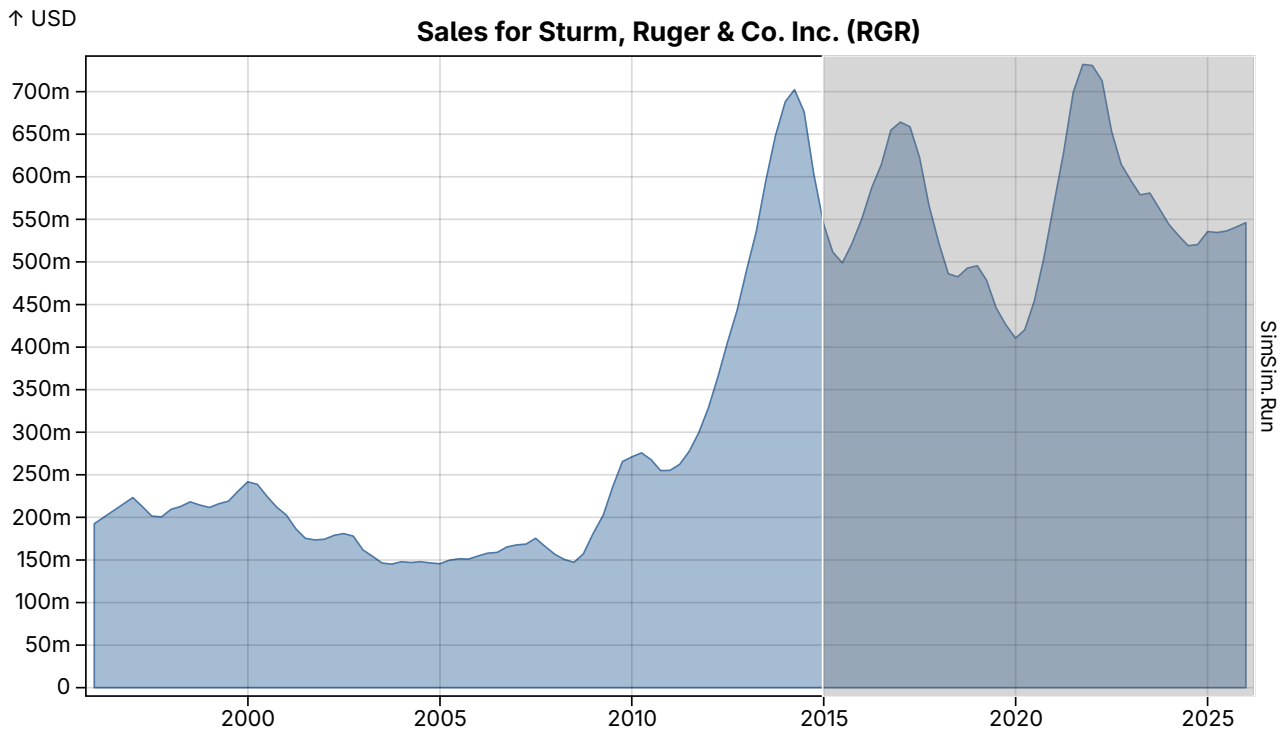


Figure 23: **Sturm, Ruger & Co.** Historical sales for the company. (TTM data.)

The shaded area is the selected data-range that is used to simulate the earnings in Figure 25.

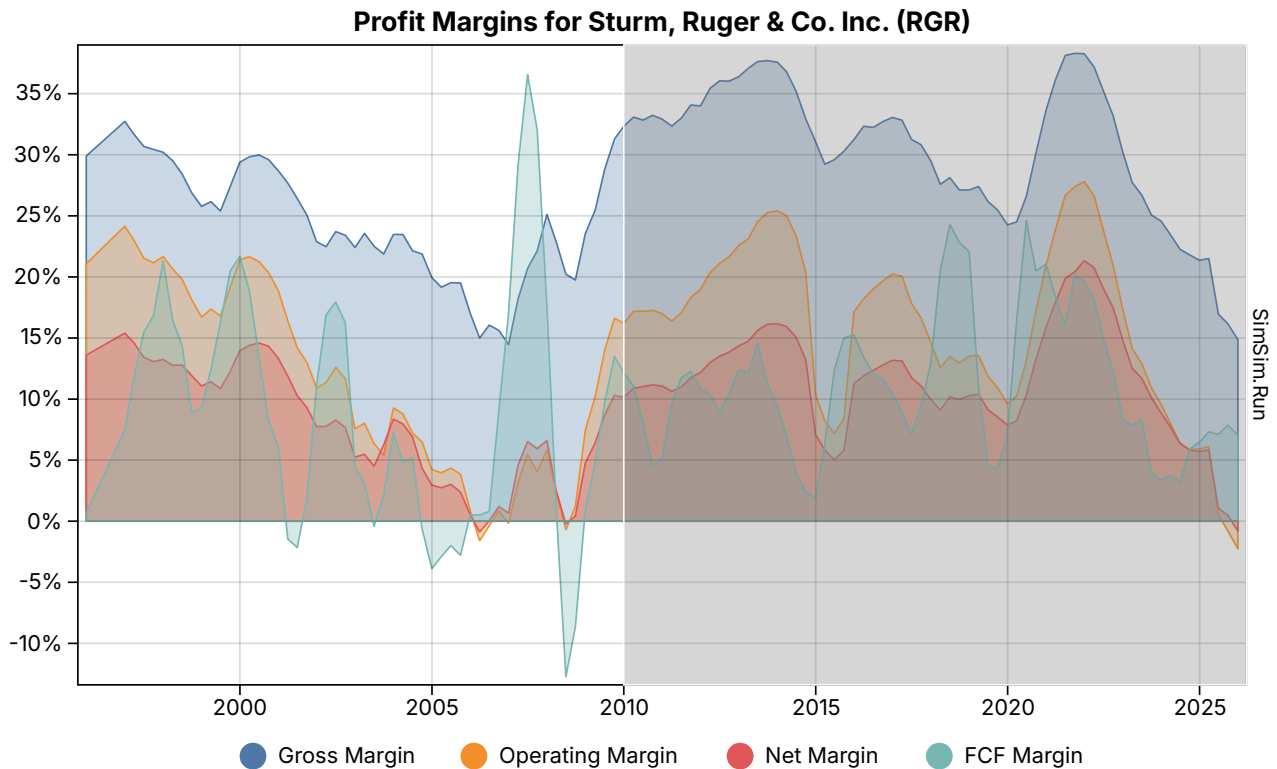


Figure 24: **Sturm, Ruger & Co.** Historical profit margins for the company. (TTM data.)

The shaded area is the selected data-range whose net profit margin is used to simulate the earnings in Figure 25.

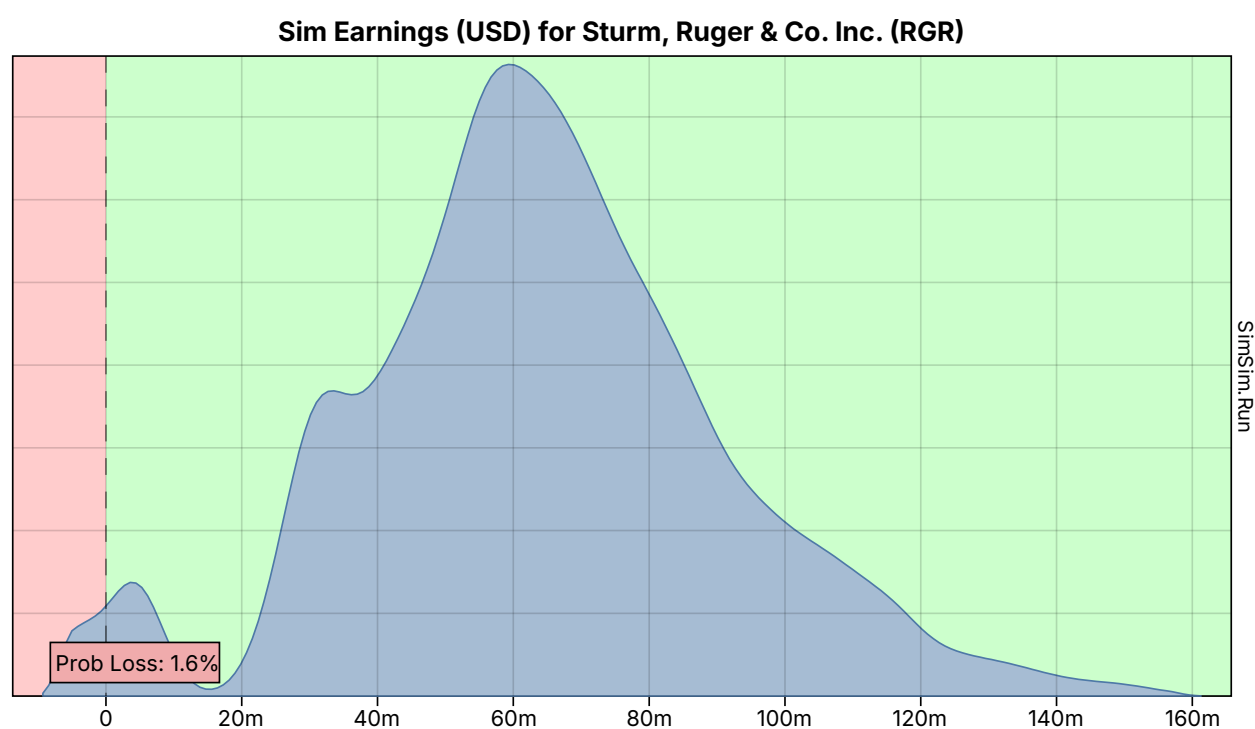


Figure 25: **Sturm, Ruger & Co.** Simulated earnings by multiplying the historical sales that are selected in Figure 23 with the historical net profit margins that are selected in Figure 24.

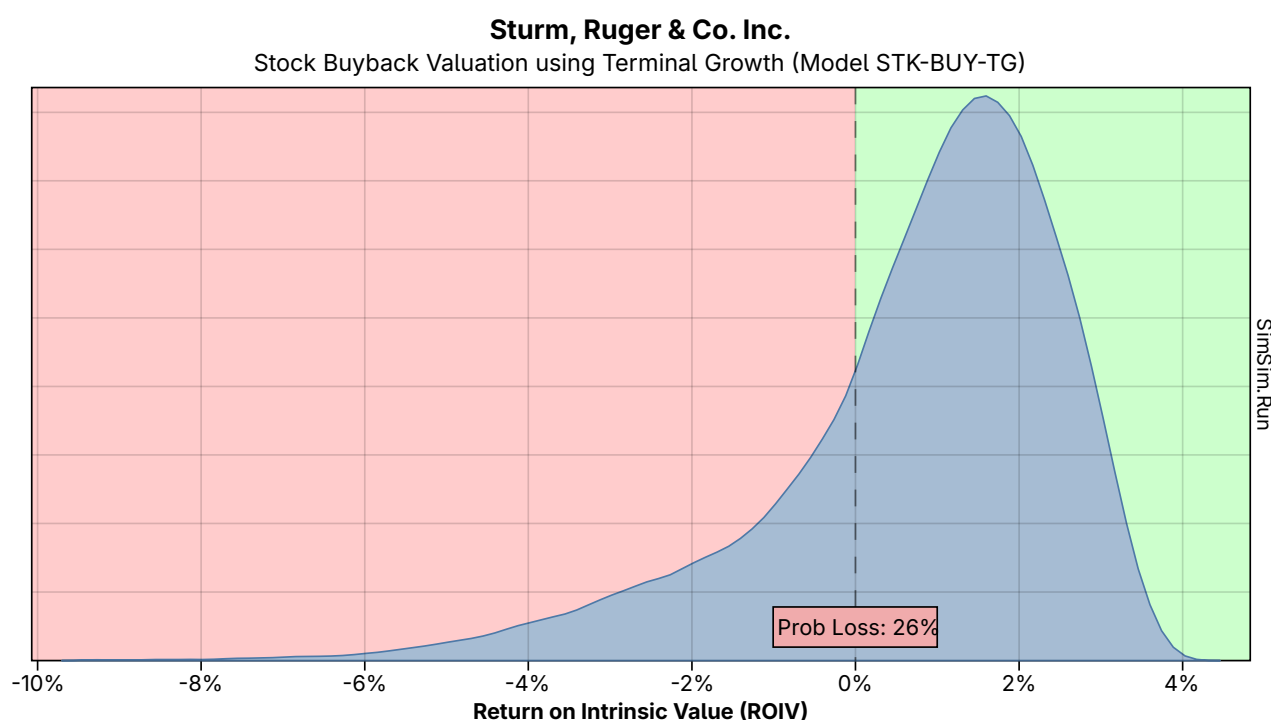


Figure 26: **Sturm, Ruger & Co.** Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 25.

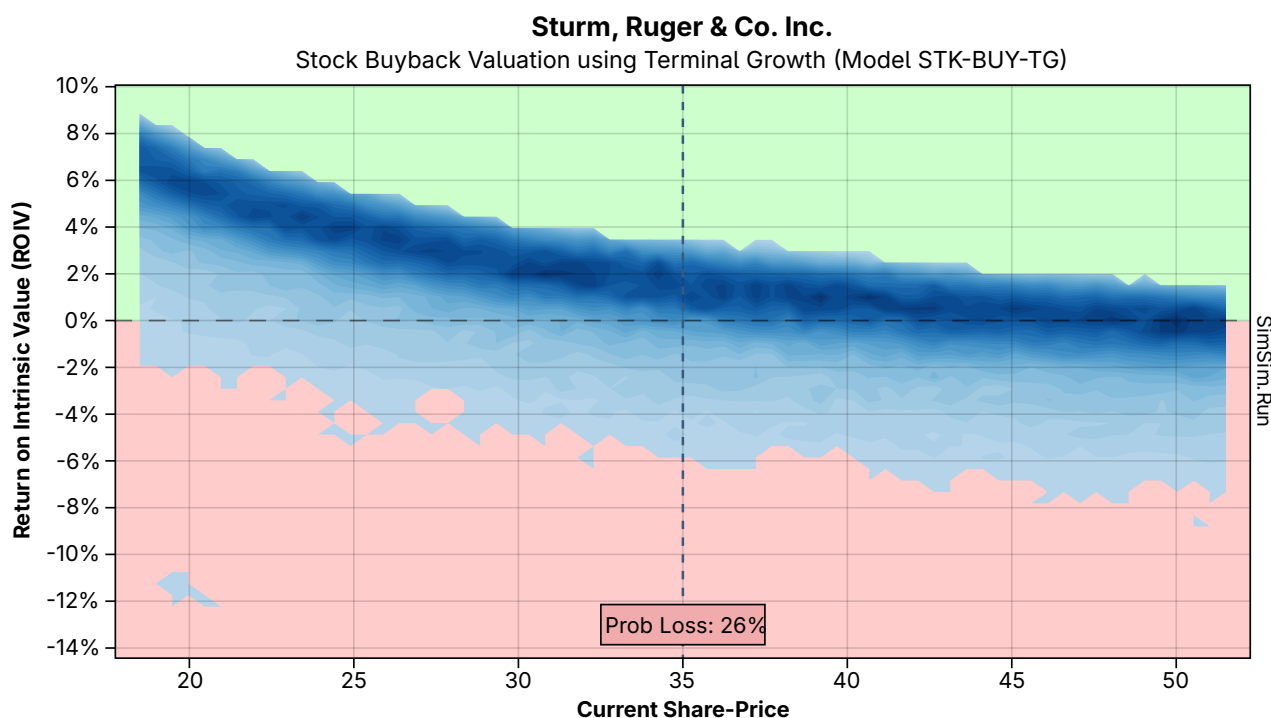


Figure 27: **Sturm, Ruger & Co.** Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 25, and varying the buyback share-price (or MarketCap) on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 26 is the vertical dashed blue line here.

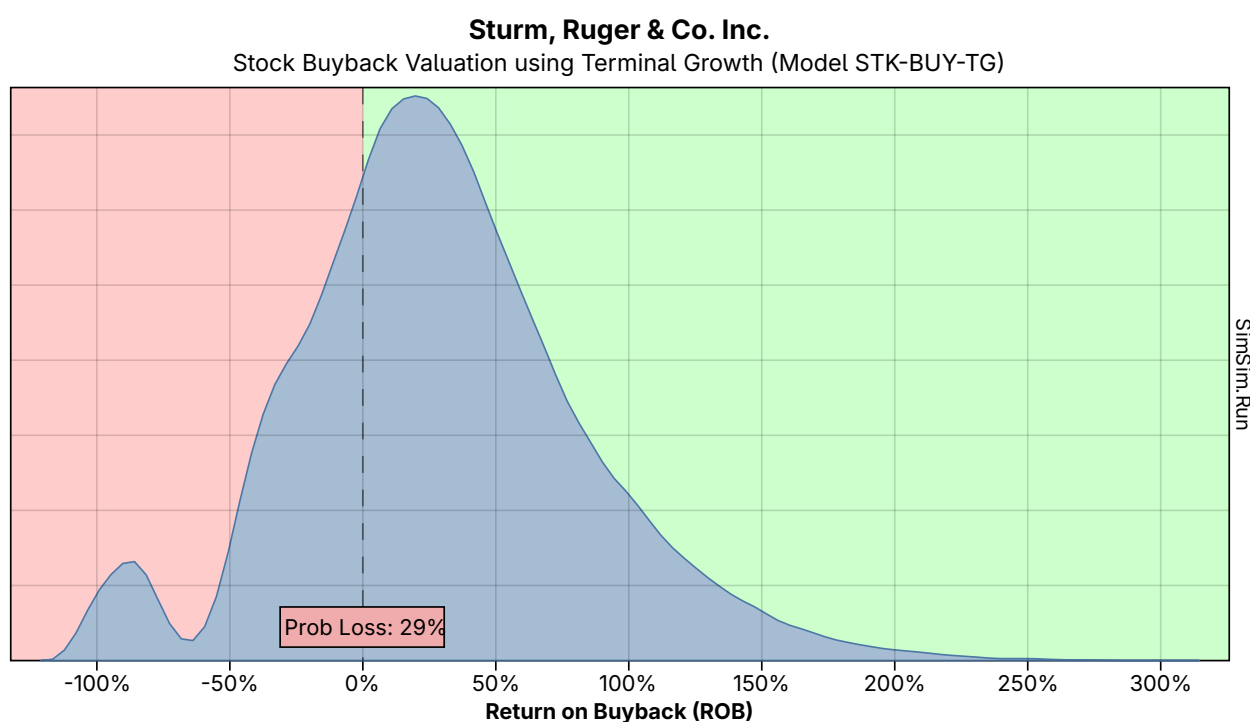


Figure 28: **Sturm, Ruger & Co.** Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 25.

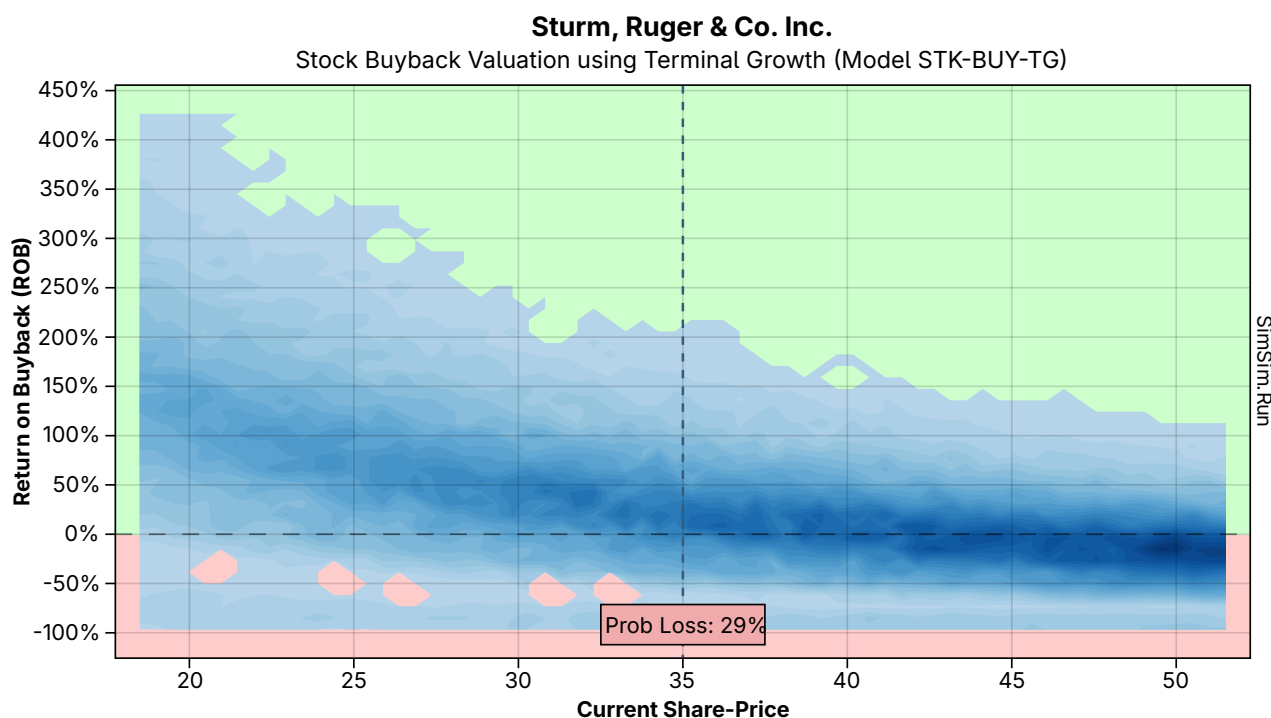


Figure 29: **Sturm, Ruger & Co.** Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 25, and varying the buyback share-price (or MarketCap) on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 28 is the vertical dashed blue line here.

9 Case Study: Harley Davidson (HOG)

This company is a motorcycle manufacturer founded in 1903 in the USA. This is a typical example of a company that made share buybacks at high prices and halted when the stock-market crashed. The stock-data used here is taken from the [SimSim.Run](#) web-site (click the link to view the data).

9.1 Historical Share Buyback Returns – Cumulative

Figure 30 shows the cumulative valuation of historical share buybacks in three sub-plots that were explained in section 8.1, so we will just analyze the results here:

- Between 1998 and 2026, the company spent a total of USD 34b on share buybacks and only USD 12b on dividends, both adjusted for the time-value of money using an 8% annual rate. The nominal amounts were USD 10.5b for share buybacks, and USD 4b for dividends.
- By 2026, all these share buybacks had caused a loss of about -47% on the per-share intrinsic value, as measured by the cumulative ROIV. Although the company repurchased 66% of its shares during almost 30 years, the share-price gradually declined, so the buybacks caused a loss of -47% compared to having instead paid the same amount of money as dividends.
- The loss was nearly -70% as measured by the cumulative ROB, which compares the change in per-share intrinsic value relative to the total amount of money spent on share buybacks.
- The present values of future dividends and share-prices that are used in the ROIV and ROB ratios are all calculated with an 8% discount-rate. It may be argued that this was too low for this particular company, in which case the present values would have been lower and hence the ROIV and ROB ratios would have shown an even greater loss from the share buybacks.

9.2 Historical Share Buyback Returns – Forward Looking

Figure 31 shows the forward-looking valuation of historical share buybacks and issuances in four sub-plots that were explained in section 8.2, so we will just analyze the results here:

- Between 2004 and 2007, the company made a series of large share buybacks when the P/S ratio was significantly higher than in the following decades.
- In 2008, the company stopped its share buybacks as the stock-market crashed in the great financial crisis of the time, and the P/S ratio had dropped 90% from its peak 8 years prior.
- In 2011, the company started making regular share buybacks again until it stopped the buybacks during the Covid pandemic between 2020 and 2022.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks mostly resulted in losses. The worst losses were for the large share buybacks between 2004 and 2007, and again in 2015.
- The ROB ratio shows the gain/loss in per-share intrinsic value, relative to the net buyback or issuance amount. This shows the losses on the share buybacks were often around -50% on the money that the company invested in buying its own shares.

- Note how the ROB ratio sometimes zig-zags between gains and losses. It is easier to see why this happens on the web-site where data-points can be pinpointed. It typically happens when the net share buyback amount switches between positive and negative values in successive quarters. For example, if the company makes regular share buybacks, but for some reason they stop in a single quarter and make a small share issuance instead e.g. from employees exercising their stock options. If the share-price was higher than in the following years, then the share issuance resulted in a gain to per-share intrinsic value, while the share buybacks in surrounding quarters resulted in a loss to per-share intrinsic value.

9.3 Market Timing

Figure 32 shows the company halted its share buybacks between 2009 and 2011 and again between 2020 and 2021, which was the exact opposite of what they should have done, because Figure 33 shows these were the only two periods (apart from the late 1990s) where the present value of future dividends and share-prices was much greater than the share-price, so only share buybacks during these periods would have increased the per-share intrinsic value to shareholders.

9.4 Simulation A

Let us try and simulate a share buyback during the stock-market crash in late 2008 and early 2009 where the company halted its share buybacks. Assume the company had not made share buybacks in the few years prior and instead accumulated the cash, so the share-count was the same as in 2007. The lowest share-price during the stock-market crash was about \$8 for a short while, but we use \$20 instead because it was below that level for about 8 months or so. These are the assumptions:

$$\begin{aligned} \text{Shares} &= 257 \text{ m} & \text{SharePrice} &= \$ 20 \\ \text{Buyback} &= \text{Excess Cash} = \$ 1.2 \text{ b} \end{aligned} \tag{98}$$

Figure 34 shows the company's historical present value calculated using Eq.(53) with the future dividends and share-prices, and multiplied by the actual share-count at the time to get the total intrinsic value and not per-share. The discount-rate is 8% because it is the same for all companies in the database. The shaded area in Figure 34 is a selection of the present values between 2008-2009. This data is not normal-distributed but the plot only shows the mean and standard deviation for each *Start* period. The selected data has a mean around \$6.75b and a standard deviation around \$2.3b. We add *Excess Cash* = \$ 1.2 b, which we assume was saved from avoiding the share buybacks in the previous years. So we get the following normal-distributed intrinsic value in 2008-2009:

$$v \sim N(\mu = \$ 7.95 \text{ b}, \sigma = \$ 2.3 \text{ b}) \tag{99}$$

We use Eq.(98) and (99) in a simulation model on the [SimSim.Run](#) web-site (click the link to try it).

Figure 35 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The distribution is skewed down because of the non-linearity of Eq.(14), which also creates outliers that have been removed. The clipped range is -30% to +25% with 10% probability of loss. The most likely ROIV is the peak near 13%.

Figure 36 again shows the simulated ROIV ratio from Eq.(14), but here the buyback share-price is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$12, the most likely ROIV ratio would be around +40%. The lowest share-price during the 2008-2009 stock-market crash was around \$8, which is so low it is not even shown in this plot.

Figure 37 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -150% to +300% with 11% probability of loss (different from the ROIV ratio due to non-linearity and removal of outliers). The most likely ROB ratio is the peak around +75%.

Figure 38 again shows the simulated ROB ratio from Eq.(16), but here the buyback share-price is varied on the x-axis. For example, when the share-price is around \$29 the ROB ratio would be centered around 0%, which is not the case for the ROIV ratio in Figure 37 due to its skewness.

9.5 Simulation B

Let us try and repeat the simulation but now using data for the following 10 years individually instead of lumping them all together in a single distribution for the intrinsic value, as we did above.

We use a simulation model on the SimSim.Run web-site (click the link to try it) with probability distributions for the P/S ratio and sales for each future year, which are multiplied to simulate the future share-prices for that year. We can use different input values for the future earnings:

1. Net Income as reported in the company's financial statements, adjusted for non-cash items such as goodwill impairments, and subtracting the part of the earnings that is used to grow the future sales. This can be difficult to estimate accurately.
2. Dividends plus net share buyback and stock spin-offs. This is easier to use because it avoids the adjustments needed for the net income, and only considers the cash that was actually paid out to shareholders in some form. But this also includes dividends and share buybacks made for borrowed money e.g. during periods of real earnings losses or re-capitalizations.

Neither of these earnings estimates are perfect. We use the 2nd estimate here, but for this particular company, these two earnings estimates give very similar results in the share buyback simulation.

We use the company's historical financial data between the years 2009 and 2018 with the same assumptions from Eq.(98) that were used in simulation A. We use a higher discount-rate that is normal-distributed with a mean of 10% and standard deviation of 1%.

The simulation model uses Eqs.(32) and (33) for the per-share intrinsic value without and with the share buyback, which are then used in Eqs.(14) and (16) to calculate the ROIV and ROB ratios.

Figure 41 is a violin-plot of the ROIV ratios for investment periods starting in 2008 / 2009 until each of the 10 years between 2009 and 2018. The first year (2009) had both gains and losses, but the remaining years all had very large gains resulting from the share buyback in 2008 / 2009. The peak was in years 5 and 6, corresponding to years 2013 and 2014 where the share-price peaked.

Figure 42 combines the ROIV ratios for all 10 years and also varies the buyback share-price on the x-axis. This is similar to Figure 36 which only uses a single distribution for the intrinsic value.

Figure 43 is a violin-plot of the ROB ratios for investment periods starting in 2008 / 2009 until each of the 10 years between 2009 and 2018. This has similar features as the ROIV ratios in Figure 41, but with greater magnitudes because the ROB ratio is relative to the net share buyback amount.

Figure 44 combines the ROB ratios for all 10 years and also varies the buyback share-price on the x-axis. This covers roughly the same area as Figure 38 when only using a single distribution for the intrinsic value, but Figure 44 is a bit tighter around the ridge with peak values (the darker blues).

9.6 Summary

This is a typical example of a company that made large share buybacks during good times when the share-price was high, which ultimately resulted in huge losses for the remaining shareholders. The company then halted the share buybacks during stock-market crashes when the share-price was low, where the share buybacks would have caused tremendous gains for the remaining shareholders.

↑ USD

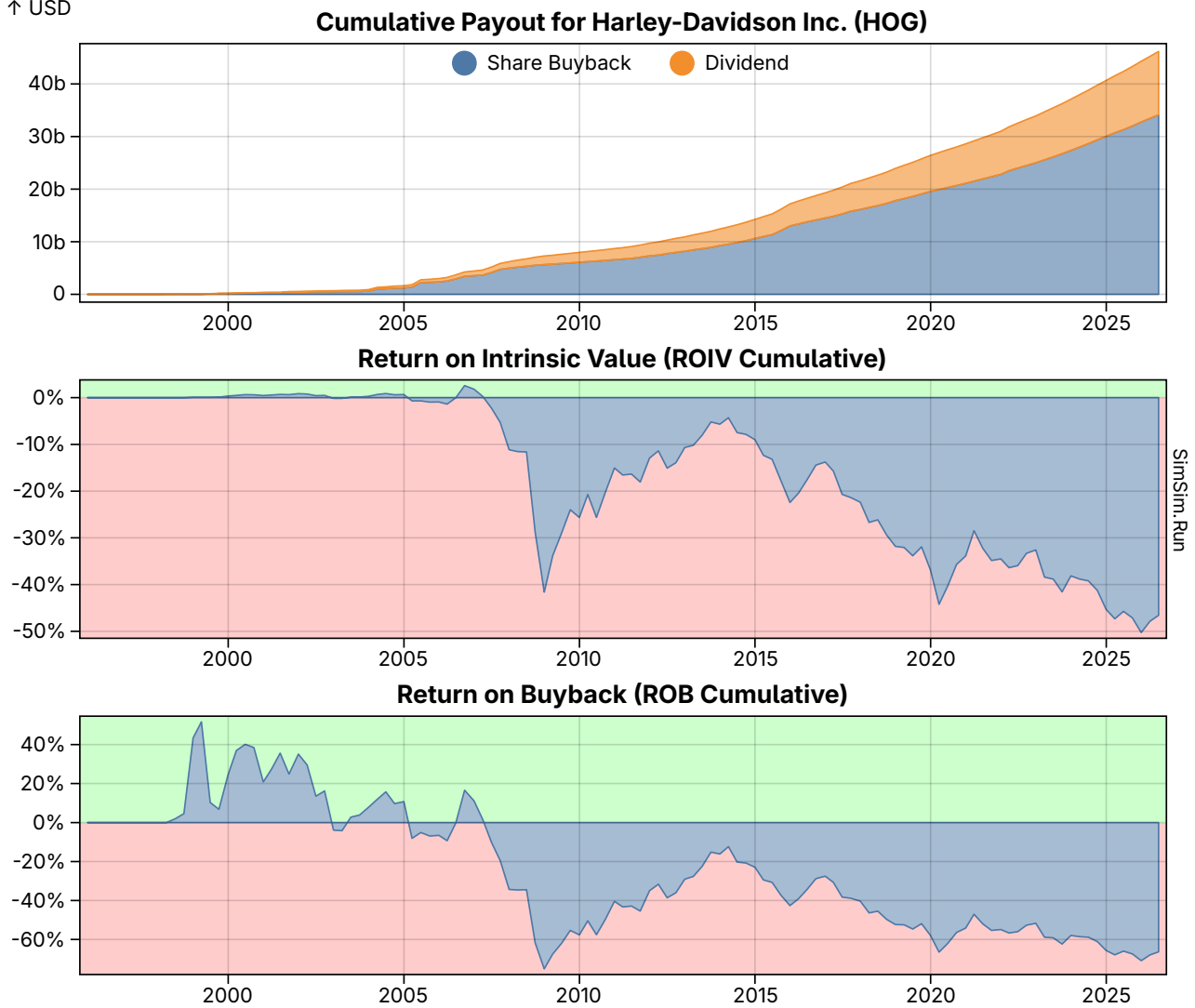


Figure 30: **Harley Davidson**. Valuation of historical share buybacks using the *cumulative* ROIV in Eq.(75) and ROB in Eq.(76), which are calculated between the first and all subsequent time periods. Also shown are the cumulative share buyback and dividend amounts, adjusted for the time-value of money by compounding forward at 8%. (Quarterly data.)

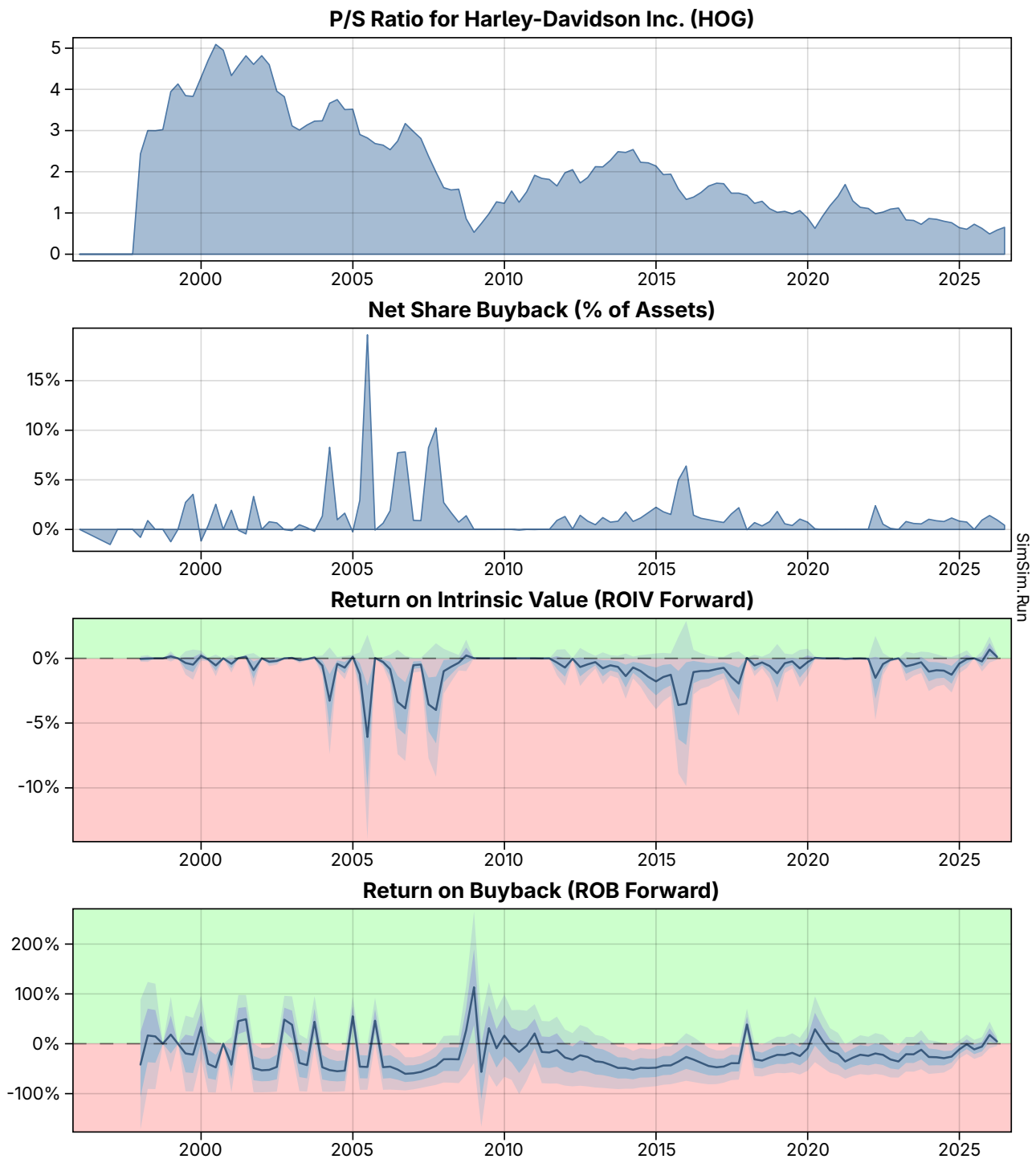


Figure 31: **Harley Davidson**. Valuation of historical share buybacks and issuances using the **forward-looking ROIV** in Eq.(57) and **ROB** in Eq.(58), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is the historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

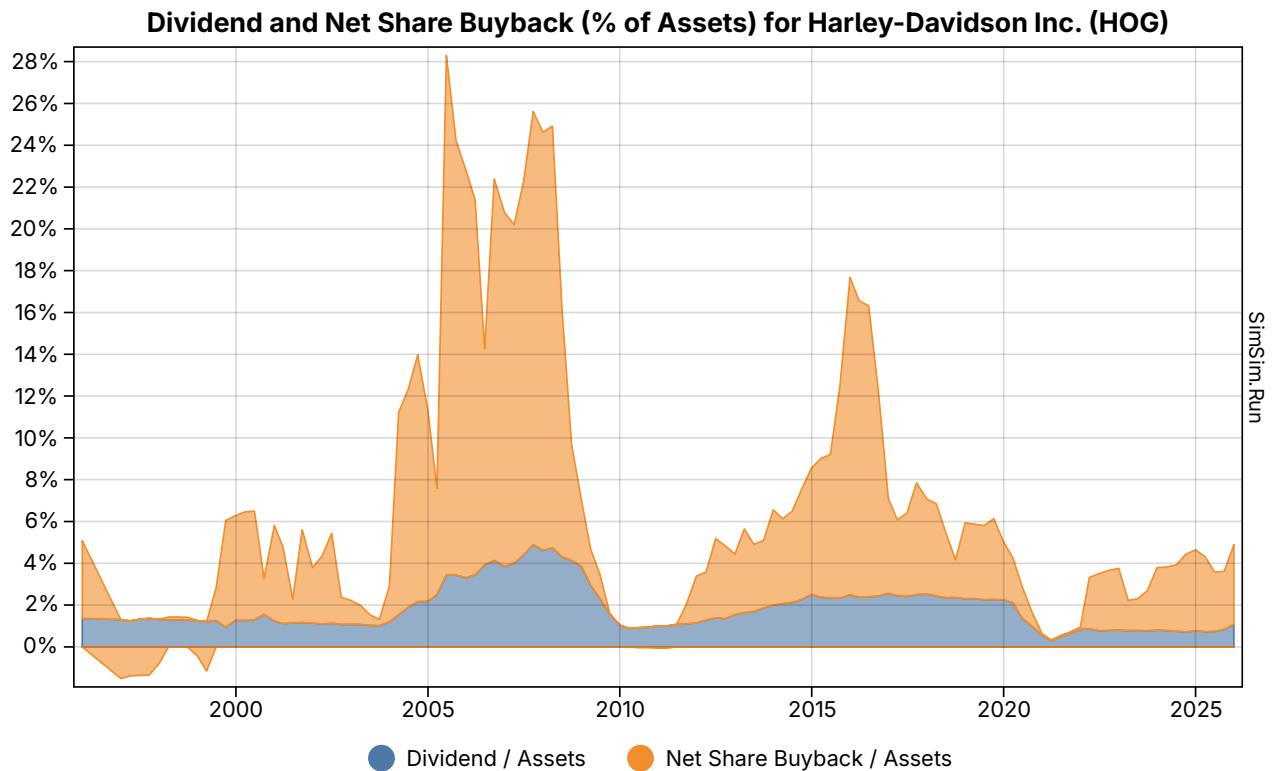


Figure 32: **Harley Davidson.** Historical dividend payout and net share buyback as percentages of total assets. (TTM data where each data-point covers the preceding four quarters.)

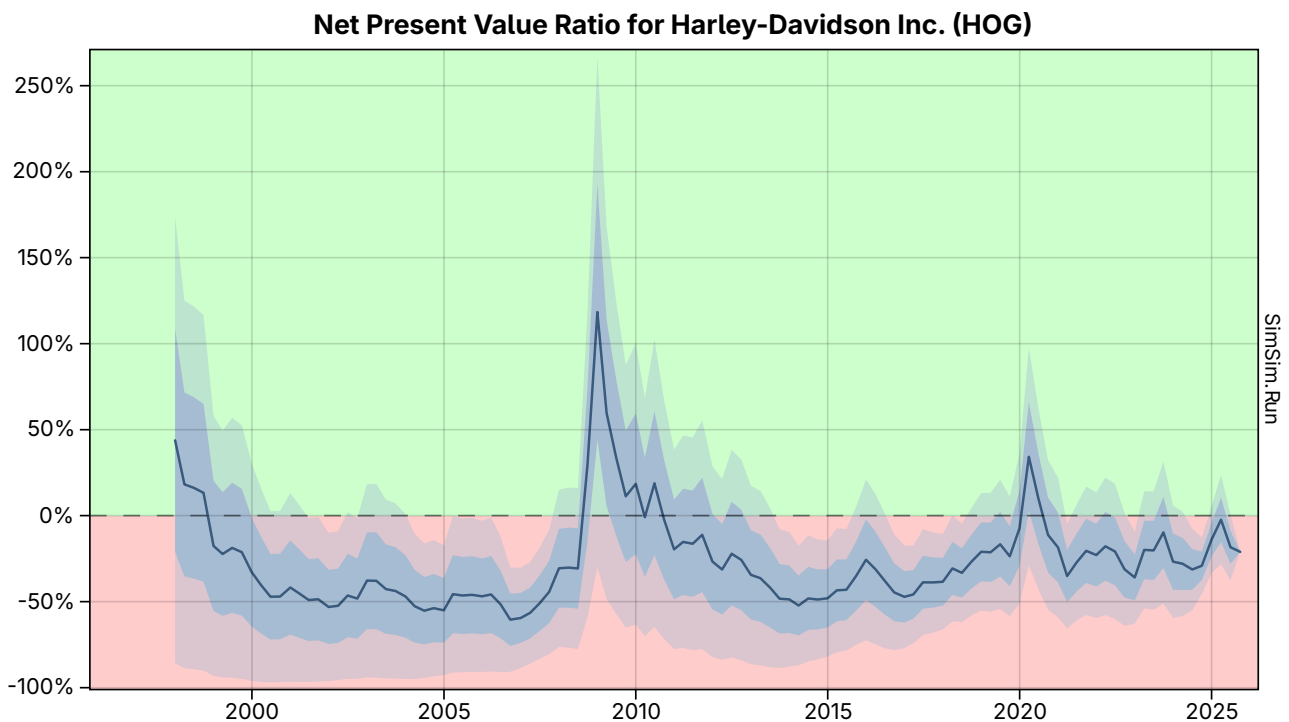


Figure 33: **Harley Davidson.** Historical net present value ratio, using the present value from Figure 34 and then divided by the Start period's average share-price. Plot shows the mean and two standard deviations. (Quarterly data.)

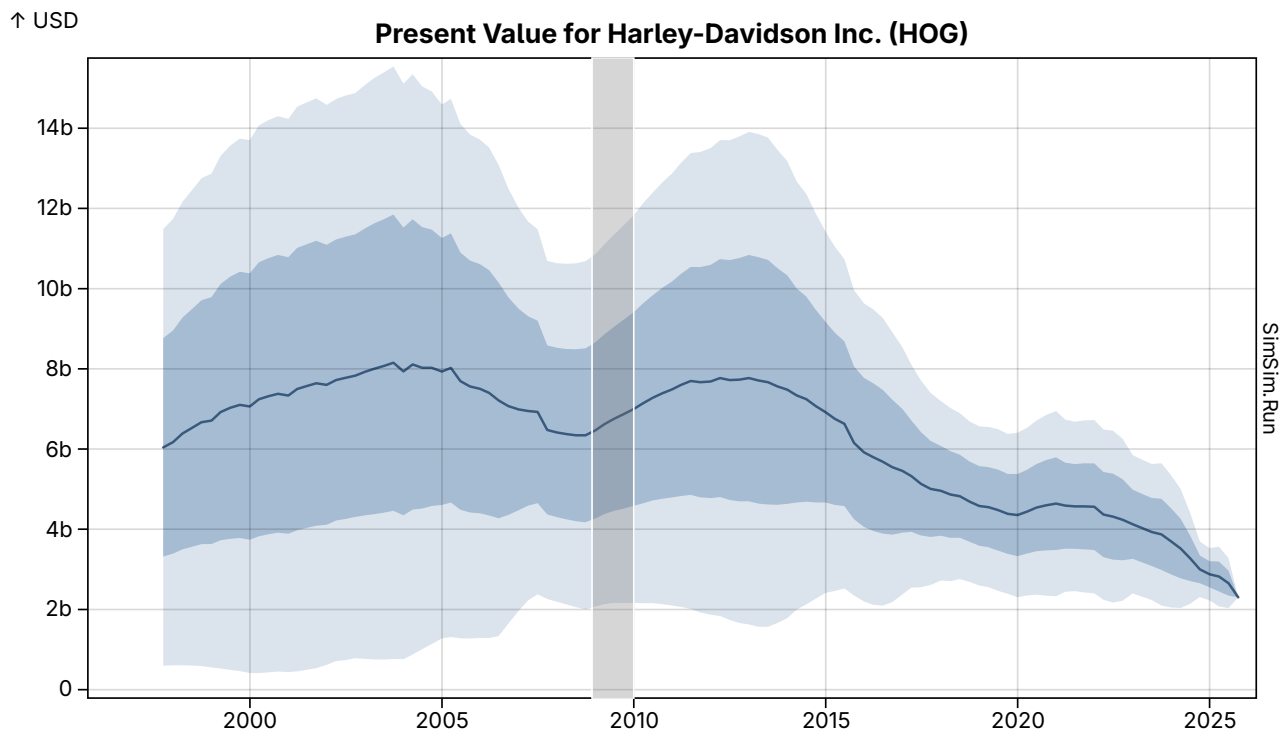


Figure 34: **Harley Davidson.** Historical present value calculated using Eq.(53) with the future dividends and share-prices. Plot shows the mean and two standard deviations for each Start period and all subsequent End periods. Shaded area is the selected data-range used as the intrinsic value when simulating a share buyback in 2008 / 2009. (Quarterly data.)

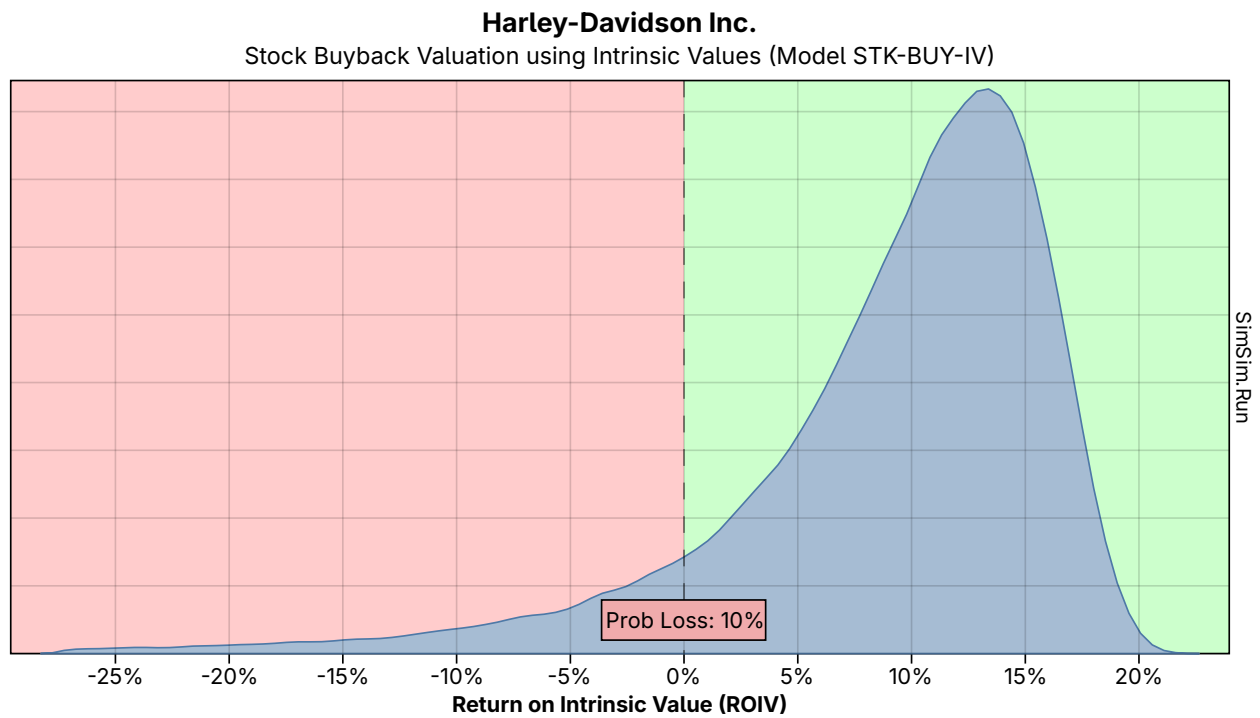


Figure 35: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 34 as the intrinsic value when calculating the **ROIV** ratio using Eq.(14).

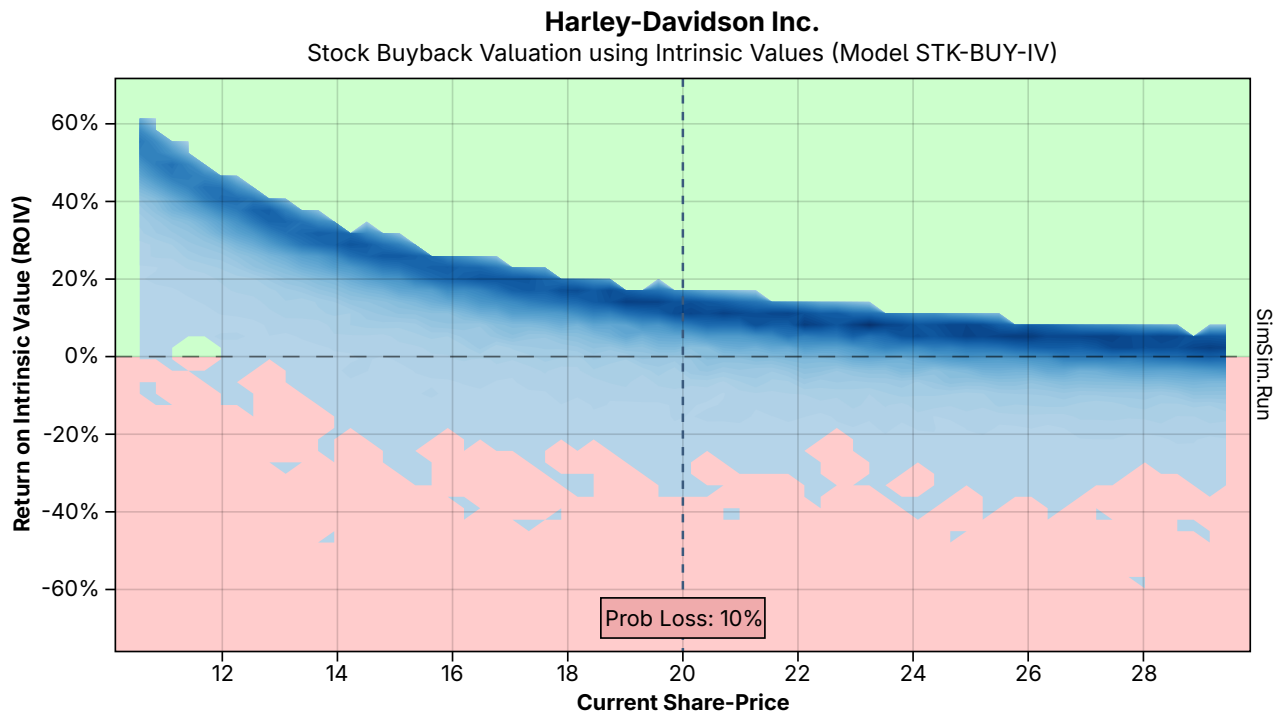


Figure 36: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 34 as the intrinsic value when calculating the **ROIV** ratio using Eq.(14), and varying the buyback share-price on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The vertical dashed line is the distribution shown in Figure 35.

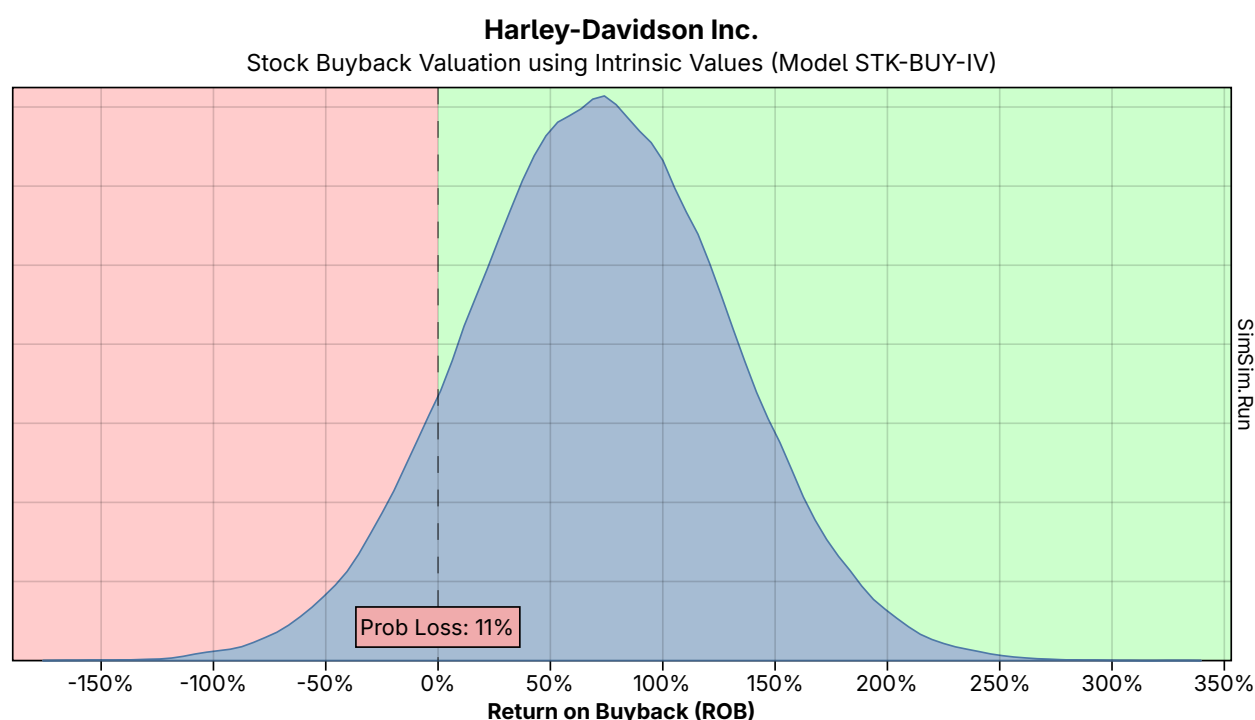


Figure 37: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 34 as the intrinsic value when calculating the **ROB** ratio using Eq.(16).

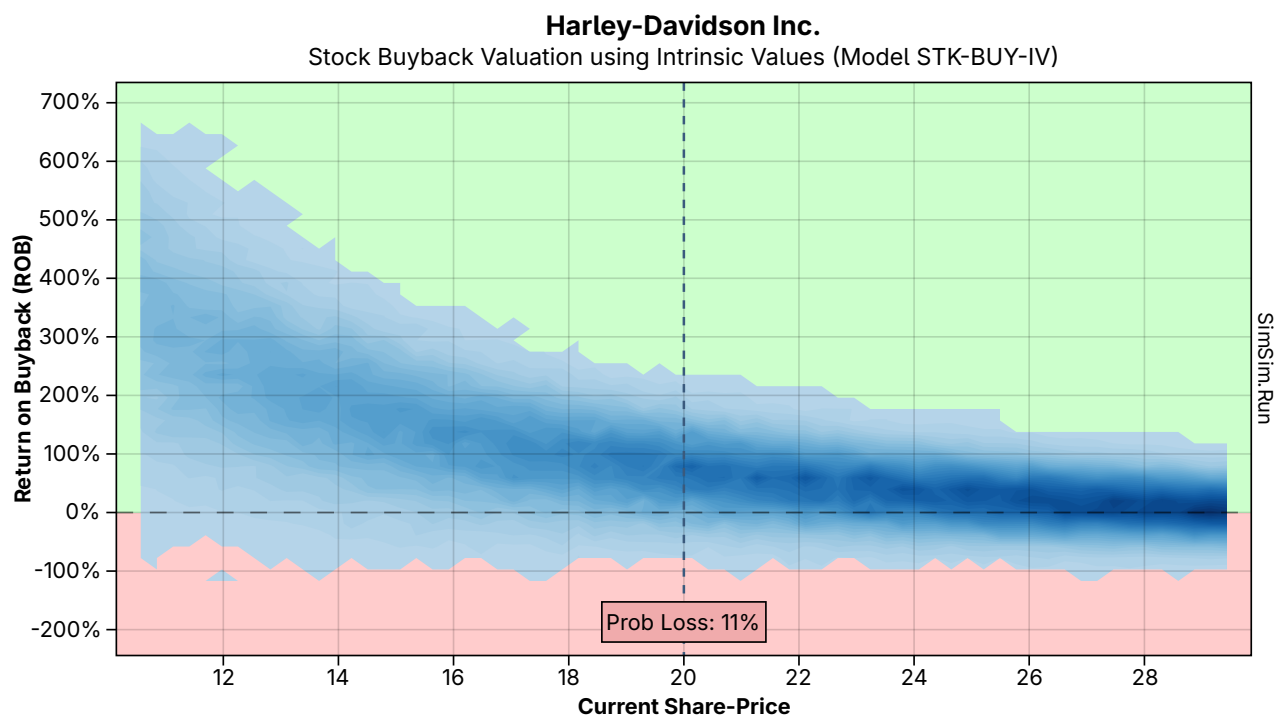


Figure 38: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 using the selected data from Figure 34 as the intrinsic value when calculating the **ROB** ratio using Eq.(16), and varying the buyback share-price on the x-axis.

This is a 2D distribution where darker blues indicate more simulated values in a region.

The vertical dashed line is the distribution shown in Figure 37.

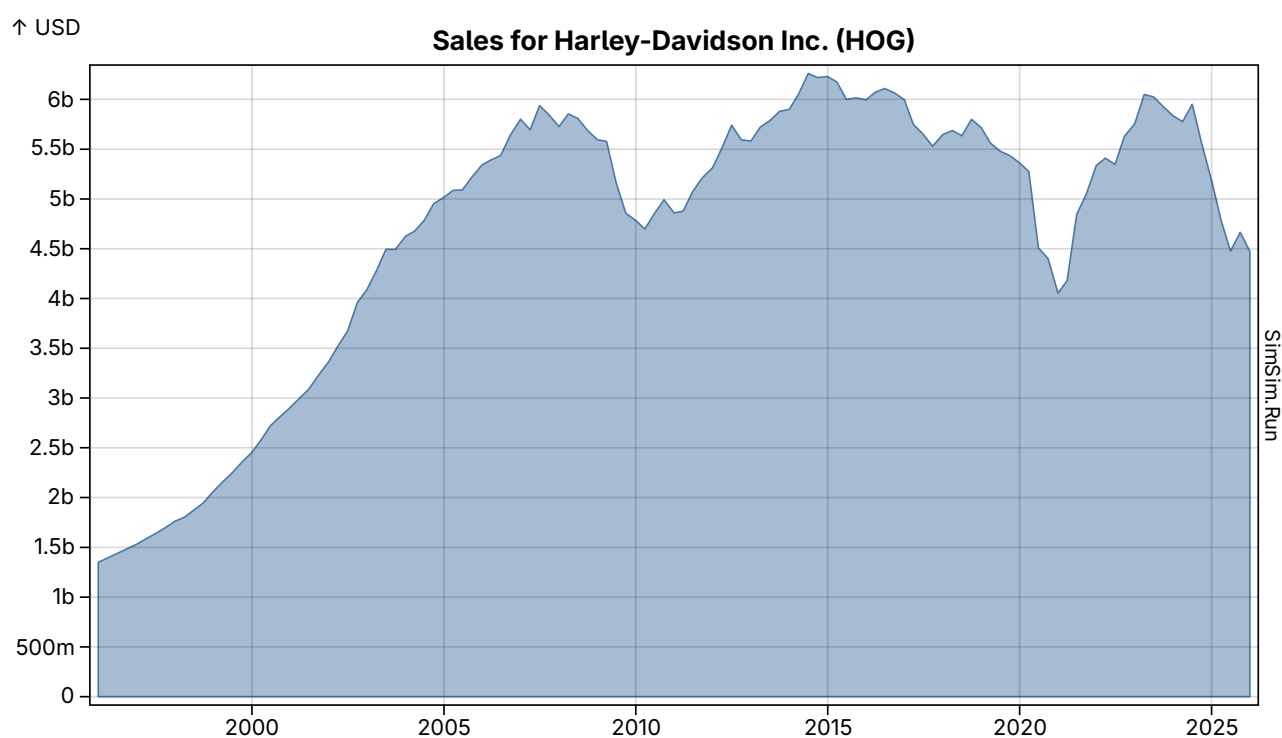


Figure 39: **Harley Davidson.** Historical sales for the company. (TTM data.)

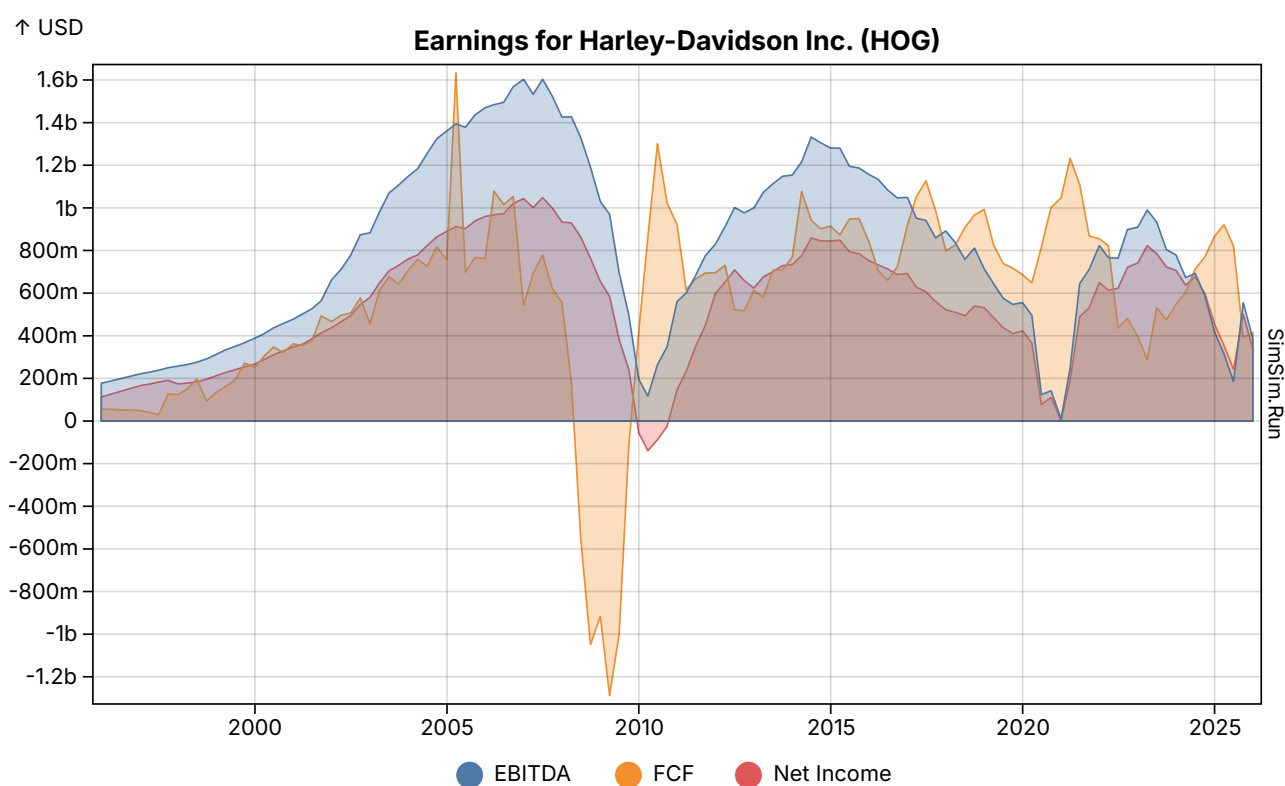


Figure 40: **Harley Davidson.** Historical earnings for the company. (TTM data.)

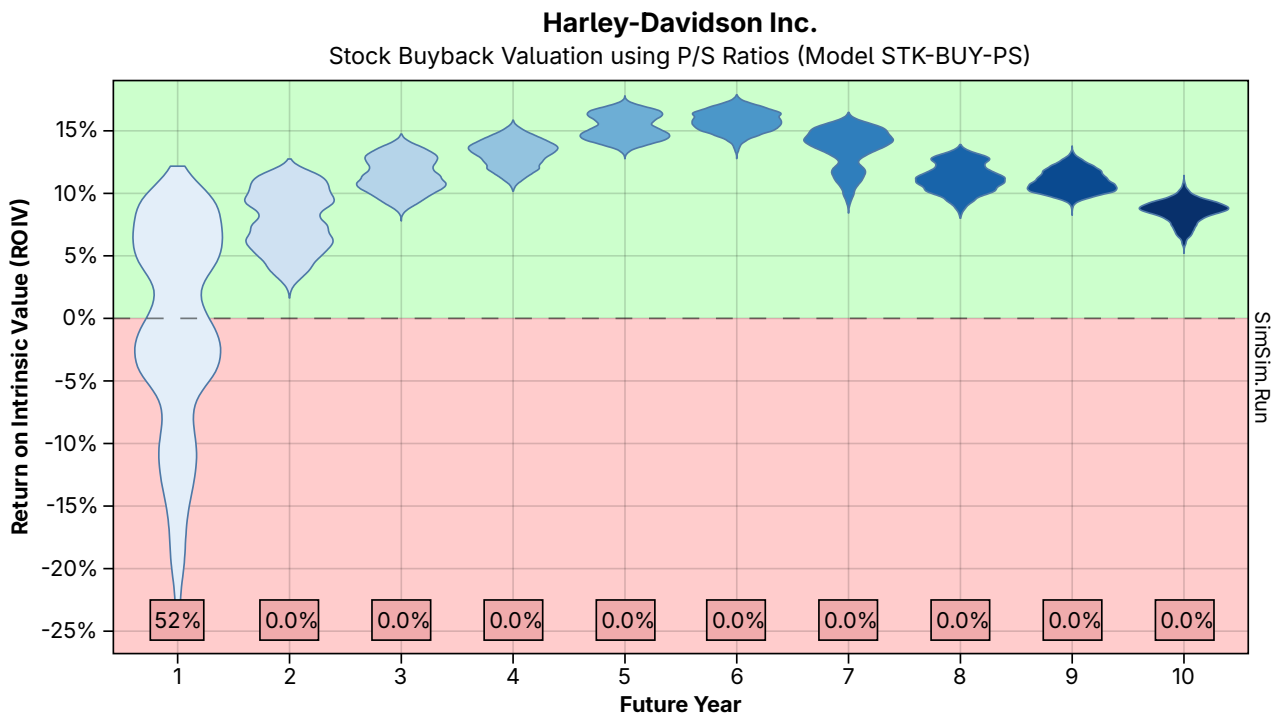


Figure 41: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009.
Violin-plot of **ROIV** distributions for investment periods between 1-10 years covering the years 2009-2018.
Red boxes show the probability of loss for each investment period when making a share buyback.

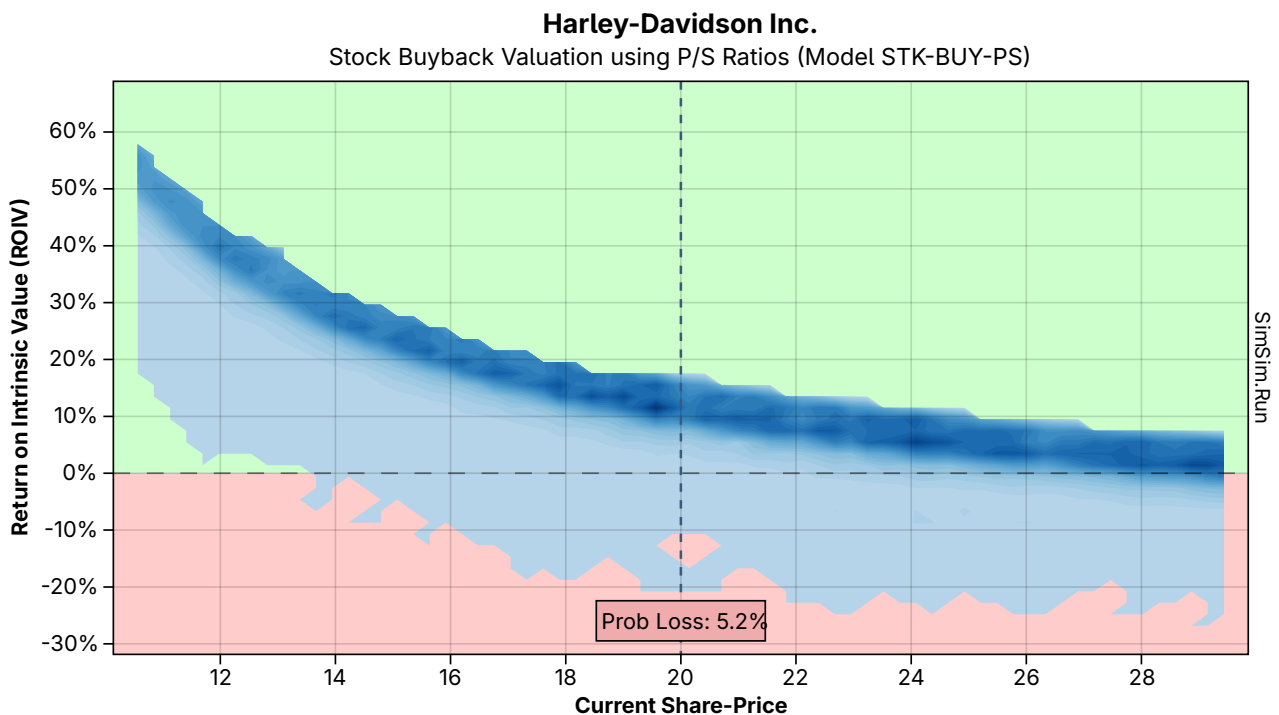


Figure 42: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 when varying the buyback share-price on the x-axis. This shows the **ROIV** ratio for all investment periods between 1-10 years combined. This is a 2D distribution where darker blues indicate more simulated values in a region. The red box at the bottom shows the overall probability of loss for all investment periods, when making a share buyback at that share-price.

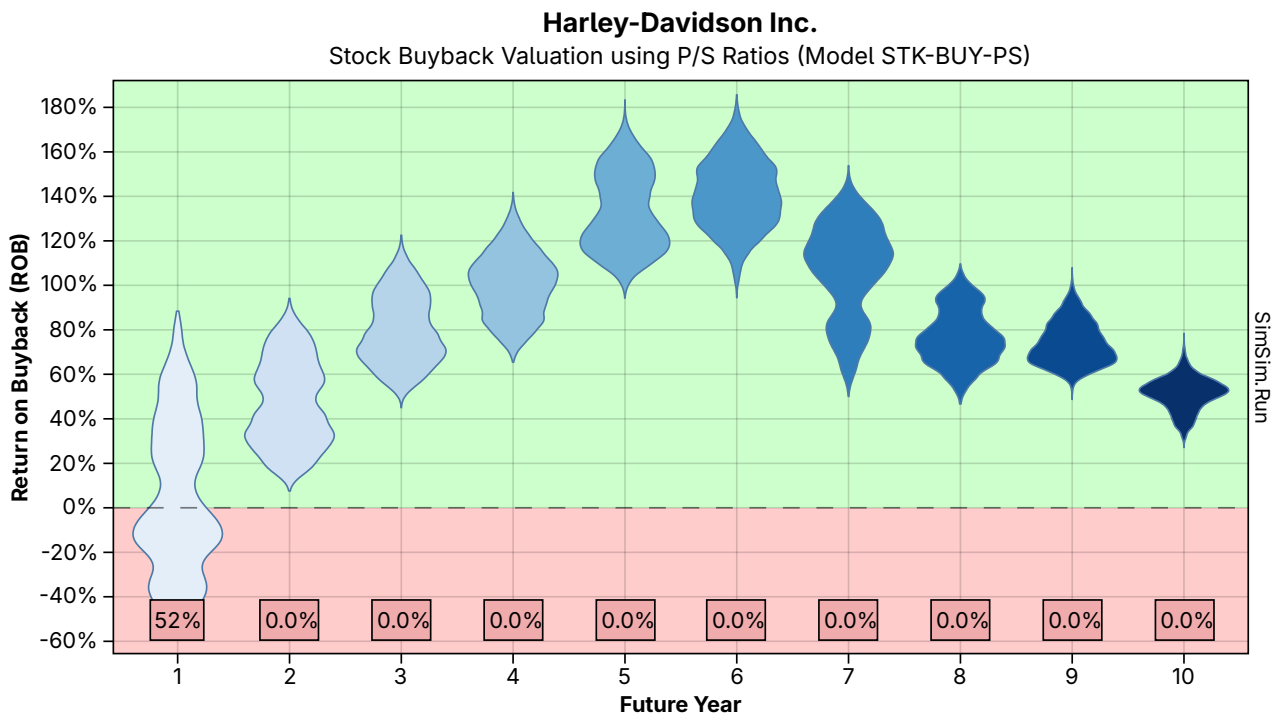


Figure 43: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009.
Violin-plot of **ROB** distributions for investment periods between 1-10 years covering the years 2009-2018.
Red boxes show the probability of loss for each investment period when making a share buyback.

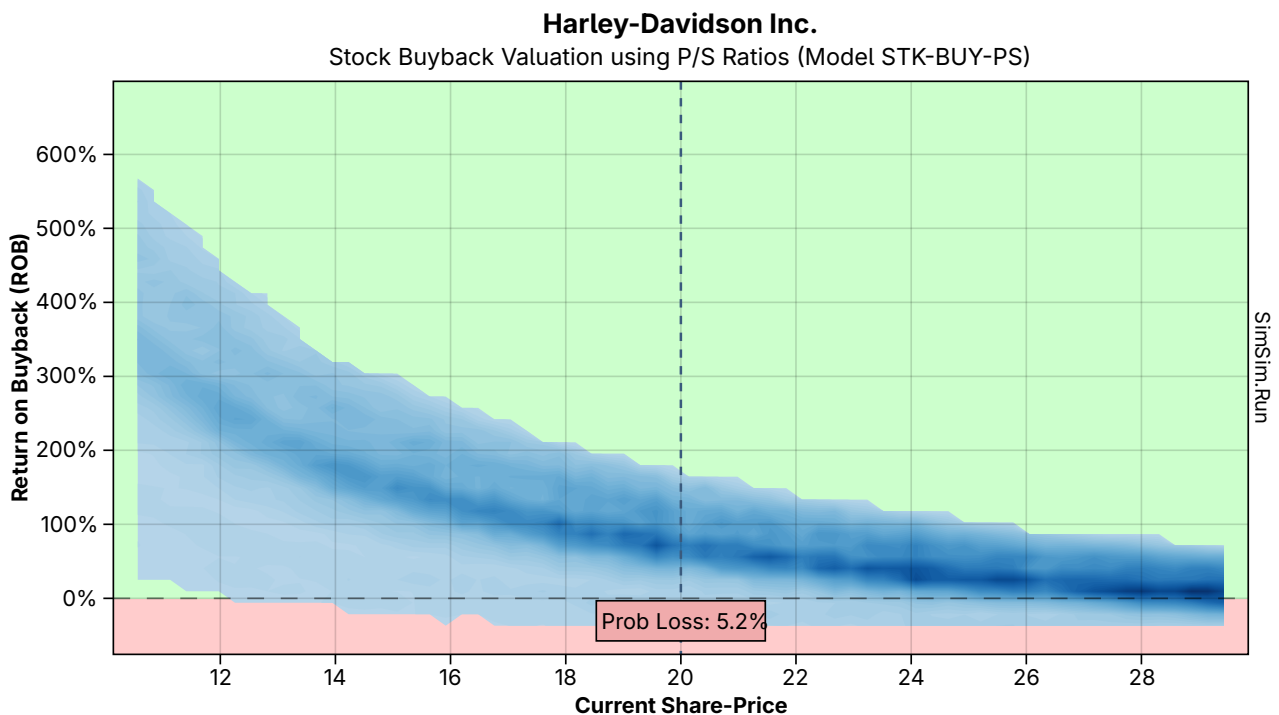


Figure 44: **Harley Davidson.** Simulation of a share buyback in 2008 / 2009 when varying the buyback share-price on the x-axis. This shows the **ROB** ratio for all investment periods between 1-10 years combined. This is a 2D distribution where darker blues indicate more simulated values in a region. The red box at the bottom shows the overall probability of loss for all investment periods, when making a share buyback at that share-price.

10 Case Study: AutoZone (AZO)

This company is a car-parts retailer founded in 1979 in the USA. This is an example of a company that did not pay any dividends and only made share buybacks for nearly 30 years since the late 1990s. As a result they retired almost 90% of their original shares. The stock-data shown here is taken from the [SimSim.Run](https://simrun.com) web-site (click the link to view the data).

10.1 Historical Share Buyback Valuation – Cumulative

Figure 45 shows the cumulative valuation of historical share buybacks in three sub-plots that were explained in section 8.1, so we will just analyze the results here:

- Between the late 1990s and 2026, the company made no dividend payouts and spent a total of USD 90b on share buybacks, adjusted for the time-value of money using an 8% annual rate. The nominal amount of total share buybacks was USD 37b (not shown here).
- By 2026, all these share buybacks had caused a gain of about +220% on the per-share intrinsic value, as measured by the cumulative ROIV. Because the company never made any dividend payouts, the +220% gain came solely from an increase in share-price, in excess of what shareholders would have received if the money had instead been paid out as dividends.
- The gain was about +340% as measured by the cumulative ROB, which compares the change in per-share intrinsic value relative to the total amount of money spent on buybacks.

10.2 Historical Share Buyback Valuation – Forward Looking

Figure 46 shows the forward-looking historical valuation of share buybacks and issuances in four sub-plots that were explained in section 8.2, so we will just analyze the results here:

- Between the late 1990s and 2026, the company has consistently bought back shares, except for very short pauses e.g. during the Covid pandemic in 2020.
- The P/S ratio has gradually risen, especially since 2010. This is partly because the profit margins have increased, but the stock's valuation ratios have generally expanded as well.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks resulted in gains for most quarters. The greatest gains were prior to 2010 where the stock had lower valuation ratios.
- The ROB ratio shows the gain/loss in per-share intrinsic value relative to the net buyback or issuance amount. Before 2010, the gains from the share buybacks were often 200-300% or more on average, relative to the amounts that the company invested in buying its own shares. After 2010, the stock's valuation ratios gradually increased so the gains from the share buybacks gradually decreased as measured by the ROB ratios.

10.3 Further Historical Data Analysis

Figure 47 shows the company's historical cash-flows as a percentage of the total positive cash-flows. On the web-site you can click on the plot to pinpoint information for each data-point. Note that the share buybacks are usually much larger than the net income, and the company has borrowed money to partially fund the share buybacks. Also note that capital expenditures were usually much higher than the depreciation and amortization, showing the company invested a lot in its growth.

Figure 48 shows the company's historical liabilities as percentages of the total assets. Although the company has been consistently profitable, it has bought back shares in part using borrowed money, so the equity became negative in 2008 and has remained negative since. Another plot on the web-site shows the EBITDA interest coverage ratio has been 9-20 during this period, so although the company had negative equity, it was so profitable that it could easily pay interest on its debt.

10.4 Simulation

Let us try and simulate a share buyback as if we were in mid 2007 using only financial data available at that time. Figure 49 shows the company's historical sales where we have selected the year prior to 2007. Figure 50 shows the company's historical profit margins where we have selected the period from 2003 to 2007. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 51, which ranges roughly between USD 515m and 625m with the bulk of the distribution between roughly USD 570m and 625m.

We use a model on the [SimSim.Run](#) web-site that supports terminal growth-rates (click the link to try it). We use the simulated earnings in Figure 51 and the following assumptions from 2007:

$$\begin{aligned} \text{Shares} &= 70 \text{ m} & \text{SharePrice} &= \$ 130 & \text{Buyback} &= \text{Excess Cash} = \$ 750 \text{ m} \\ & & \text{Terminal Growth Rate} &= 0 \% & d &\sim N(\mu = 10 \%, \sigma = 1 \%) \end{aligned} \quad (100)$$

Figure 52 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. Almost all of these are losses down to about -10% with the most likely ROIV ratio being around -4%.

Figure 53 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. Almost all of these are losses down to about -60% with the most likely ROB ratio around -35%.

So the company would have to grow its future earnings in order for the share buyback in 2007 to result in a gain in per-share intrinsic value for the remaining shareholders. In the following 20 years the average annual growth-rate was about 6% for the company's sales, so let us try and use that as the terminal growth-rate for the earnings in the simulation. This company funded its growth from borrowed money and an increase in accounts payable. If the company had used retained earnings to fund the growth, then we would have to subtract that amount from the earnings in the simulation.

Figure 54 shows the simulated ROIV ratio with the 6% terminal growth-rate. The range is roughly between -2% and +8% with the peak at +3.5%. Figure 55 shows the simulated ROB ratio with the 6% terminal growth-rate. The range is roughly between -20% and +500% with the peak at +45%.

10.5 Summary

The company only made share buybacks and no dividend payouts for almost 30 years. The share buybacks mostly resulted in large gains in per-share intrinsic values for the remaining shareholders. But this required significant and continued sales growth, otherwise many of the buybacks would have resulted in losses for the remaining shareholders, as we demonstrated using simulations.

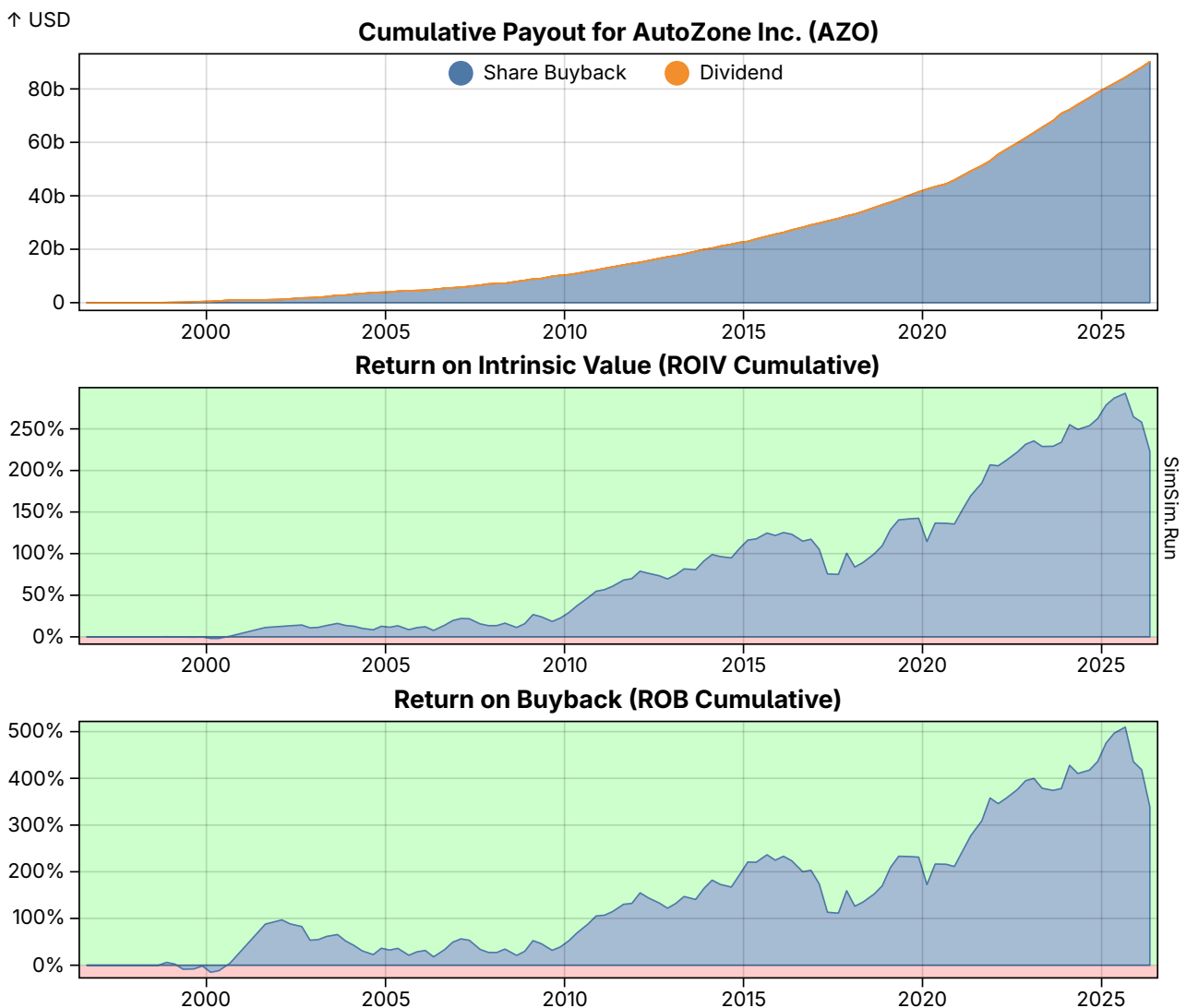


Figure 45: **AutoZone**. Valuation of historical share buybacks using the **cumulative** ROIV in Eq.(75) and ROB in Eq.(76), which are calculated between the first and all subsequent time periods. Also shown are the cumulative share buyback and dividend amounts, adjusted for the time-value of money by compounding forward at 8%. (Quarterly data.)

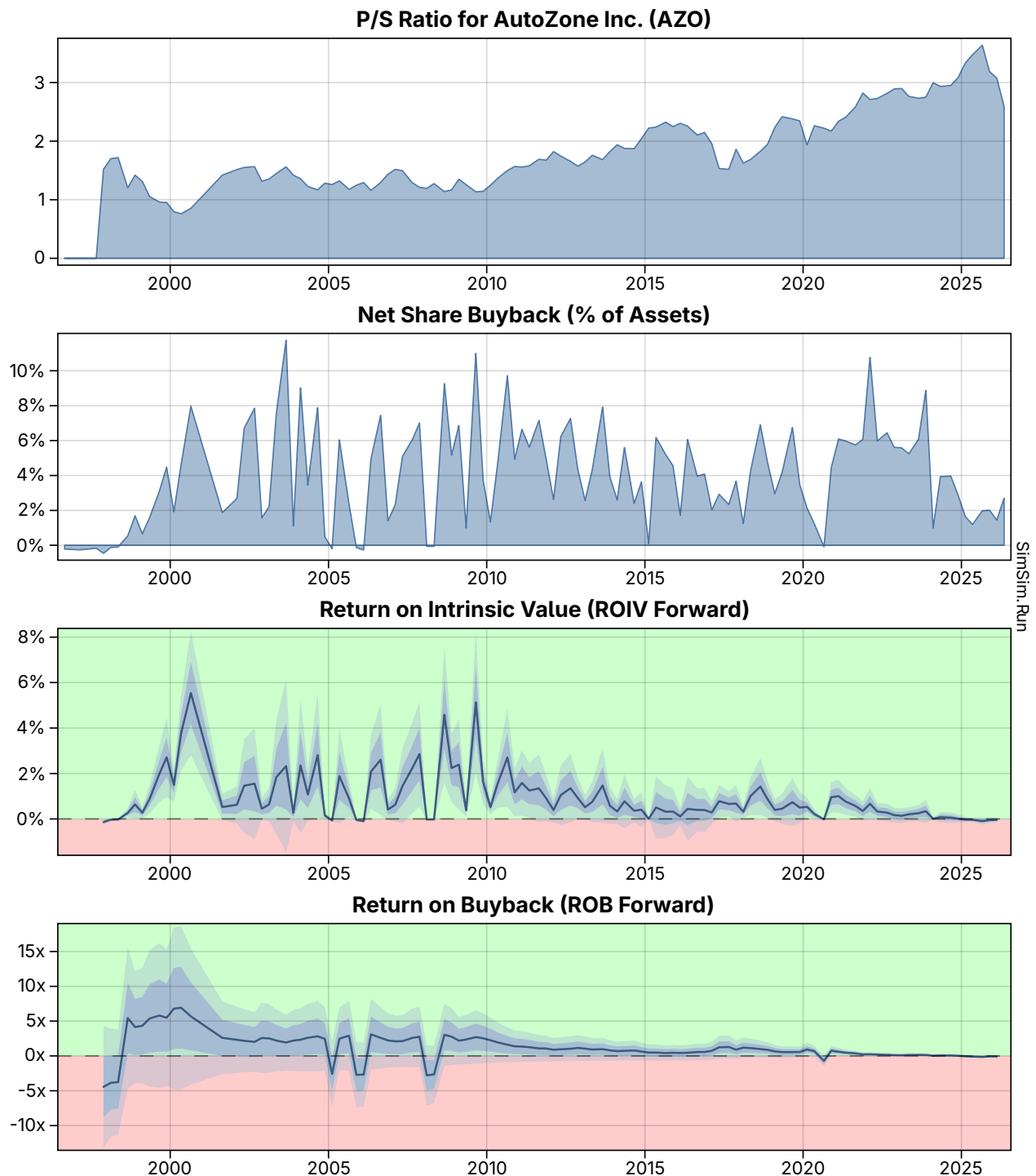


Figure 46: **AutoZone**. Valuation of historical share buybacks and issuances using **forward-looking** ROIV in Eq.(57) and ROB in Eq.(58), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. ROB uses x-scales instead of % e.g. 15x = 1,500%. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)

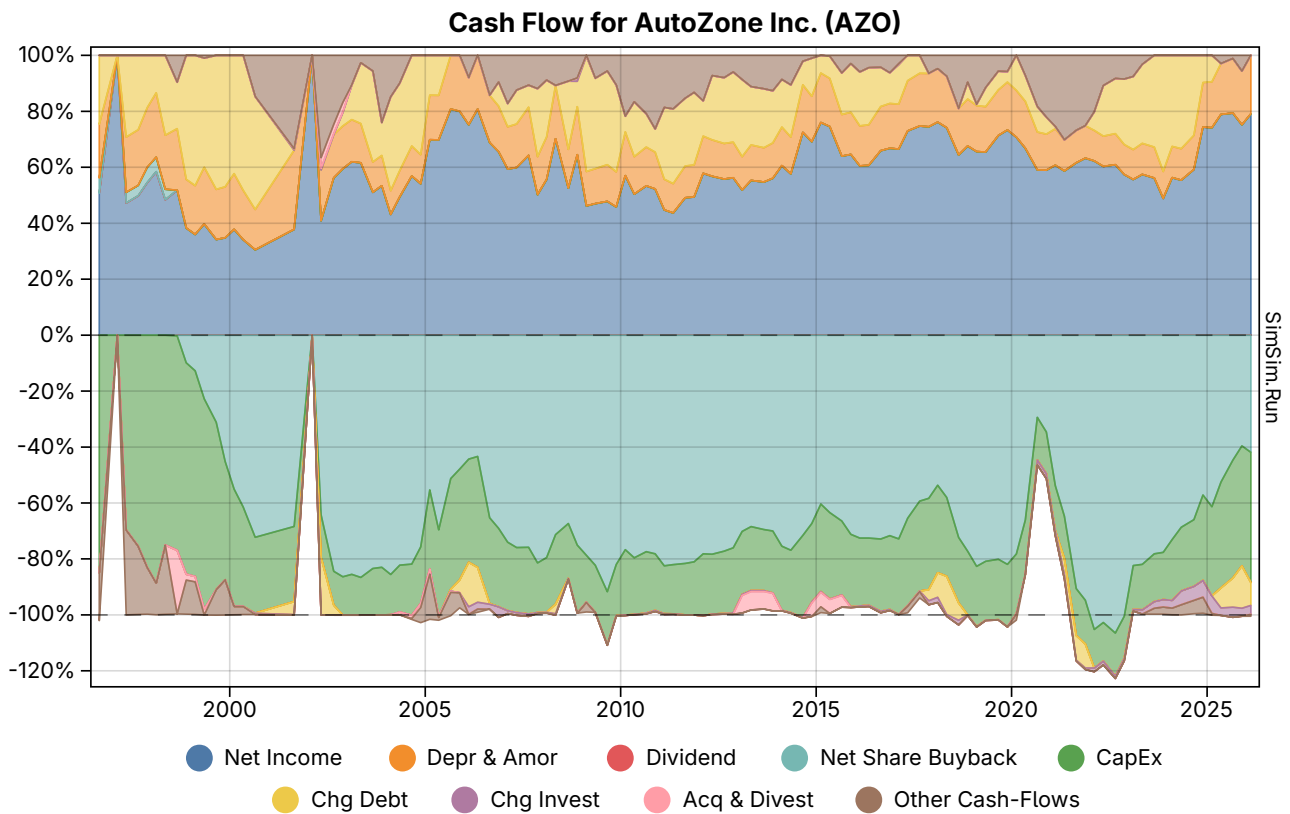


Figure 47: **AutoZone**. Historical cash-flows for the company as percentages of total positive cash-flows. (TTM data.)

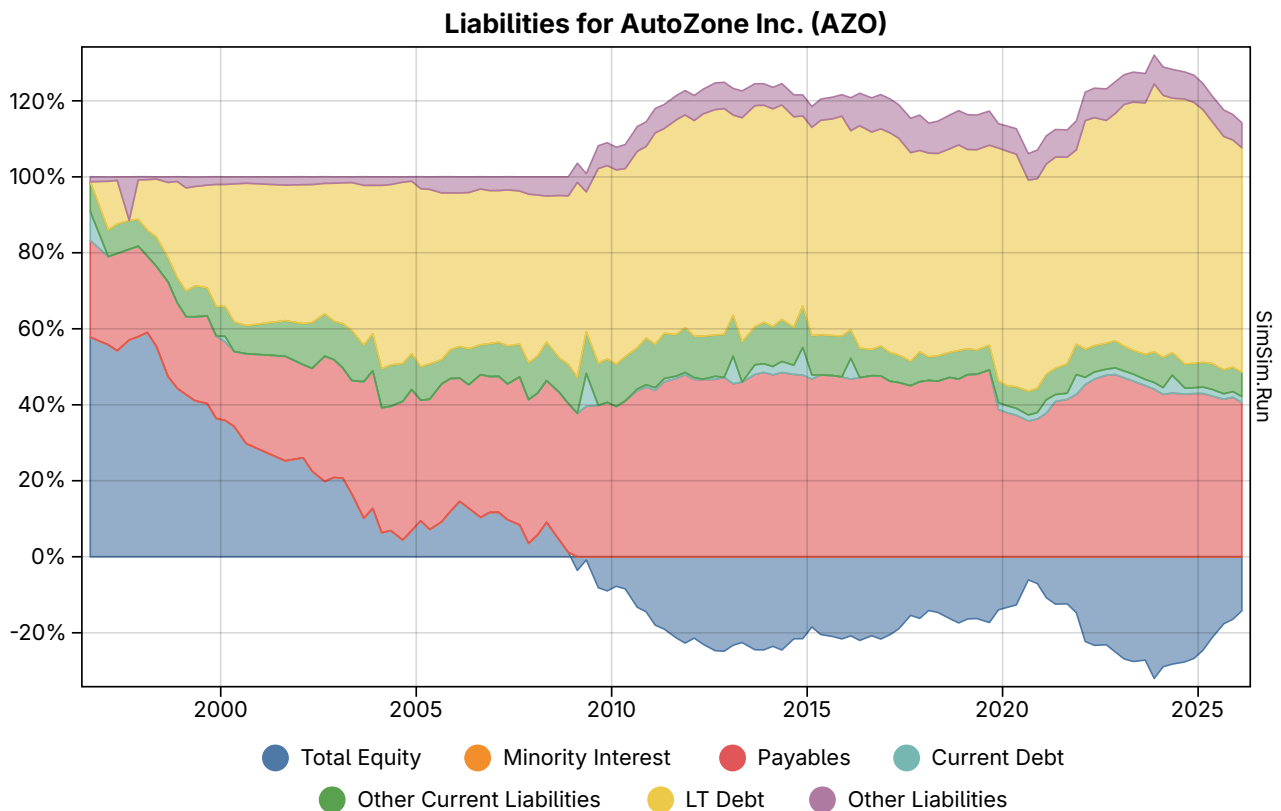


Figure 48: **AutoZone**. Historical liabilities for the company as percentages of total assets. (TTM data.)

↑ USD

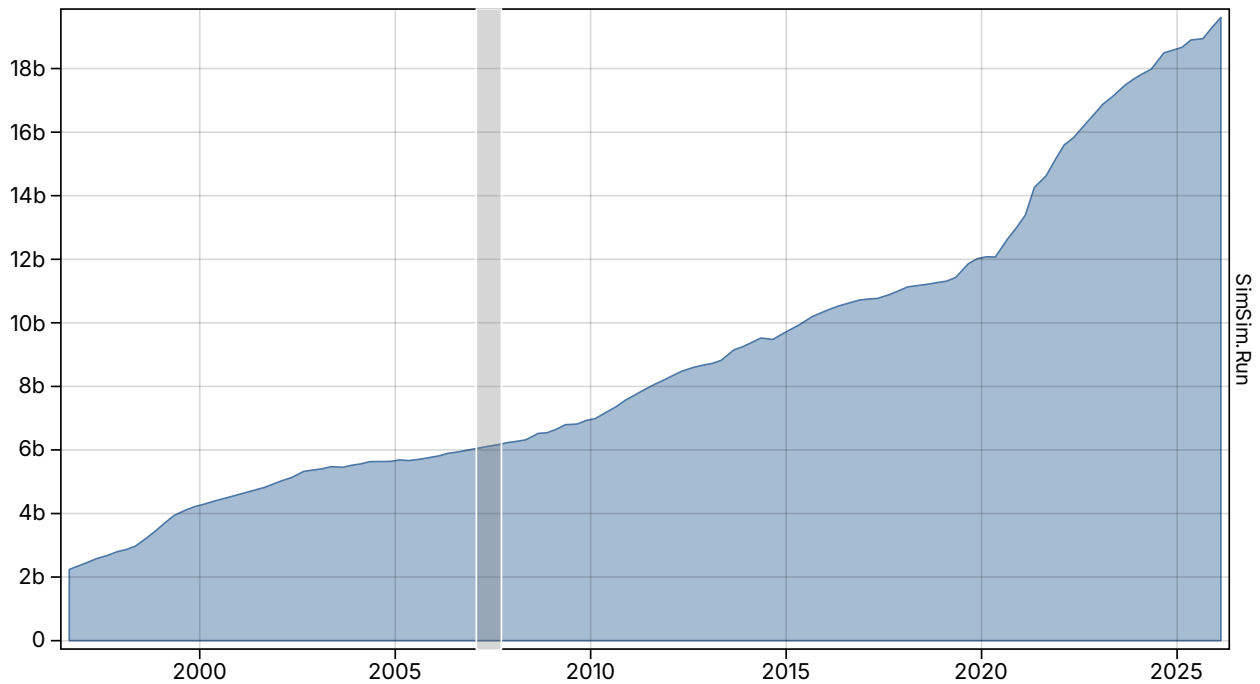
Sales for AutoZone Inc. (AZO)

Figure 49: **AutoZone**. Historical sales for the company. (TTM data.)

The shaded area is the selected data-range that is used to simulate the earnings in Figure 51.

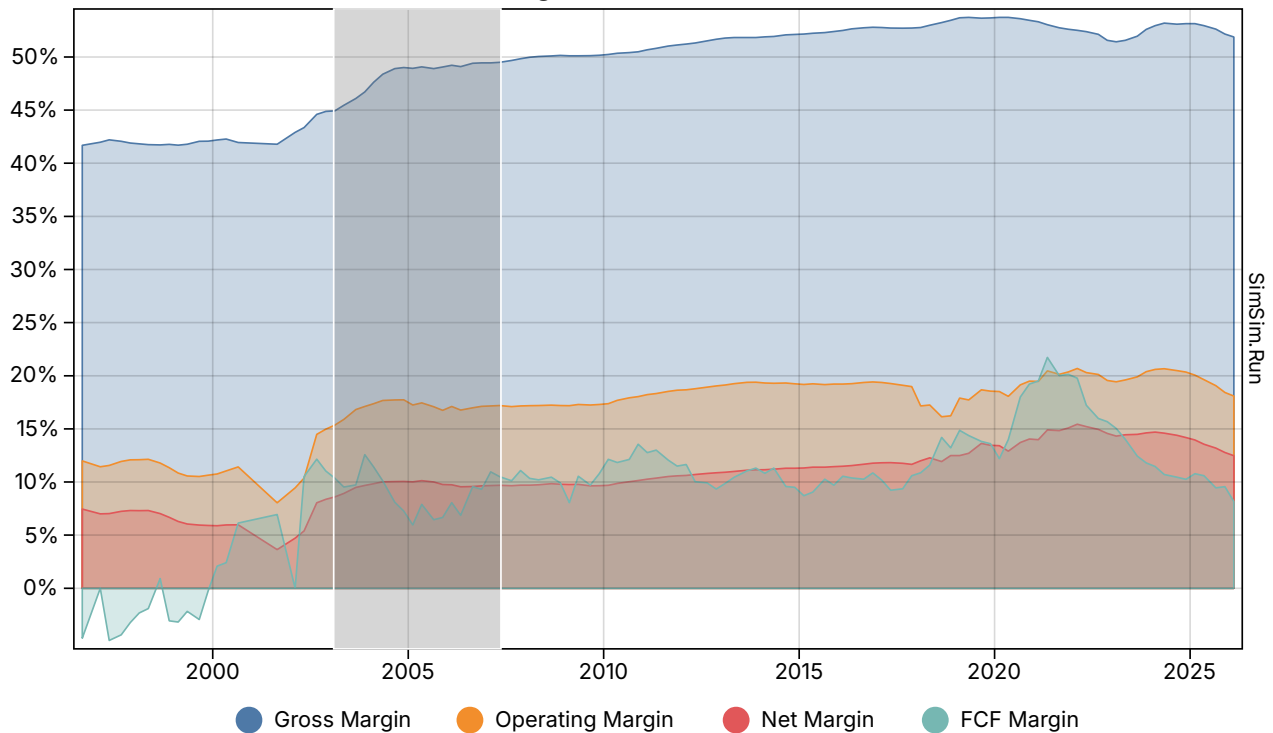
Profit Margins for AutoZone Inc. (AZO)

Figure 50: **AutoZone**. Historical profit margins for the company. (TTM data.)

The shaded area is the selected data-range whose net profit margin is used to simulate the earnings in Figure 51.

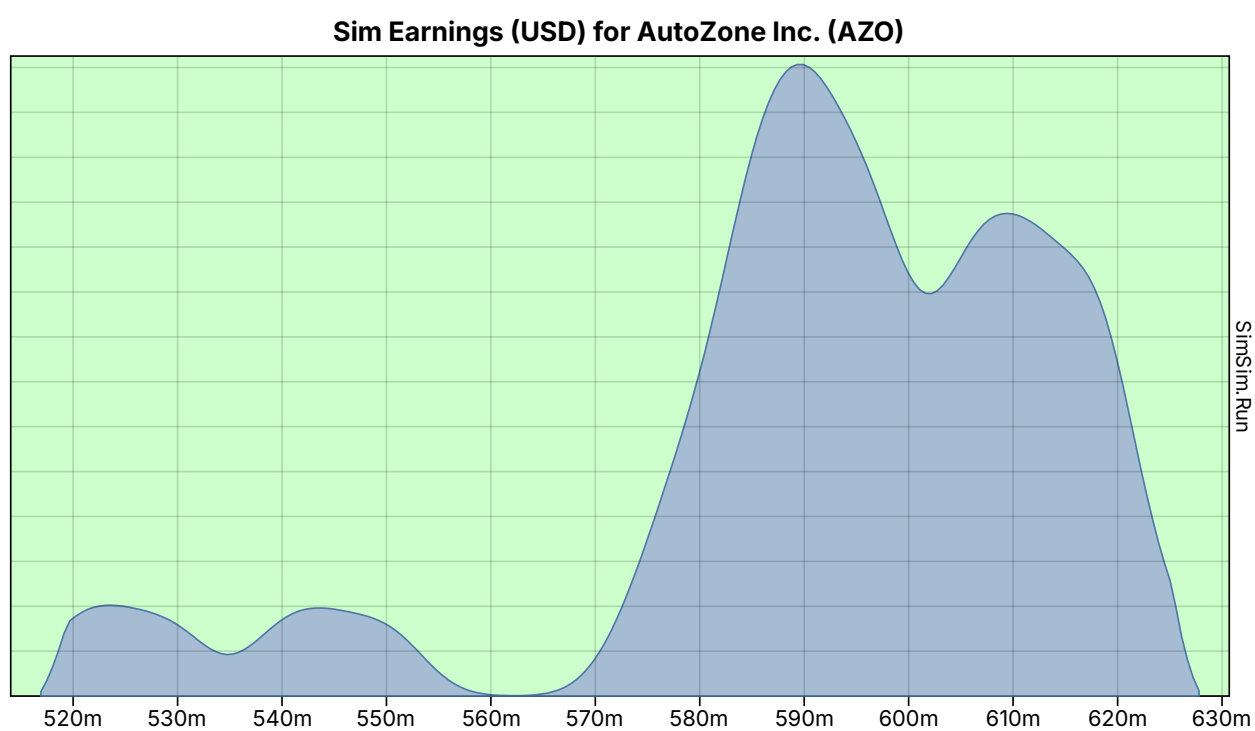


Figure 51: **AutoZone**. Simulated earnings by multiplying the historical sales that are selected in Figure 49 with the historical net profit margins that are selected in Figure 50.

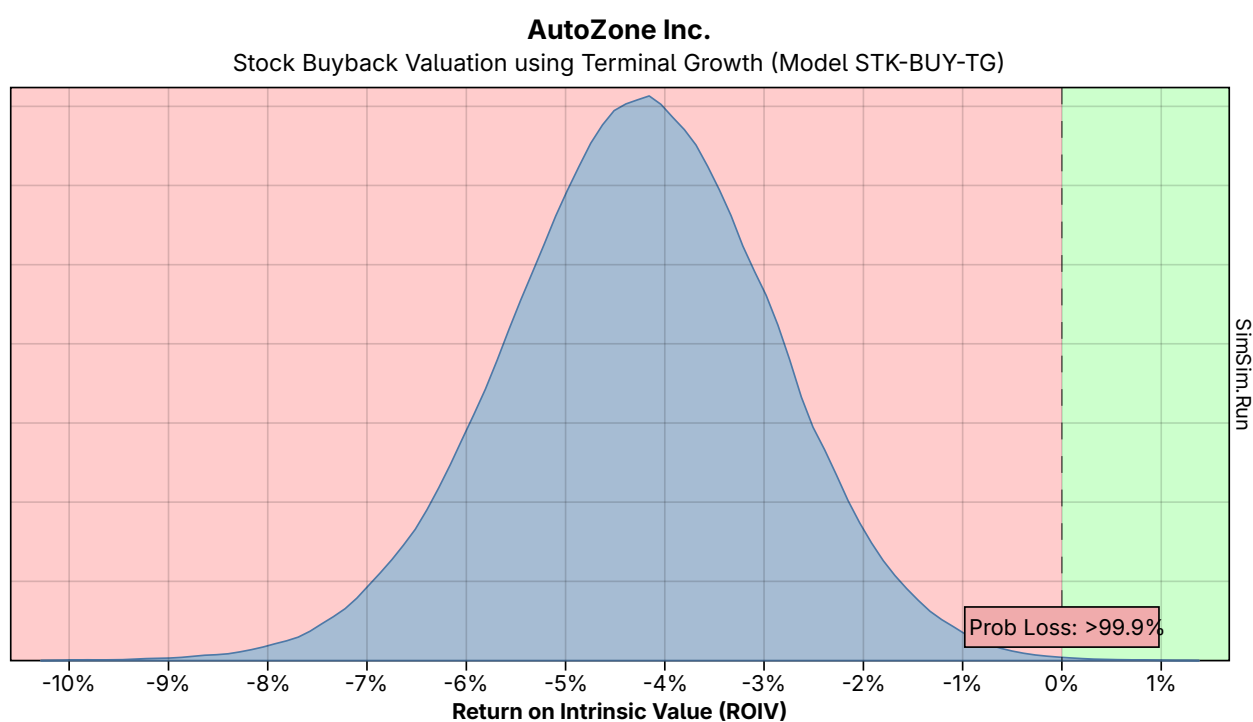


Figure 52: **AutoZone**. Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 51.
Zero terminal growth.

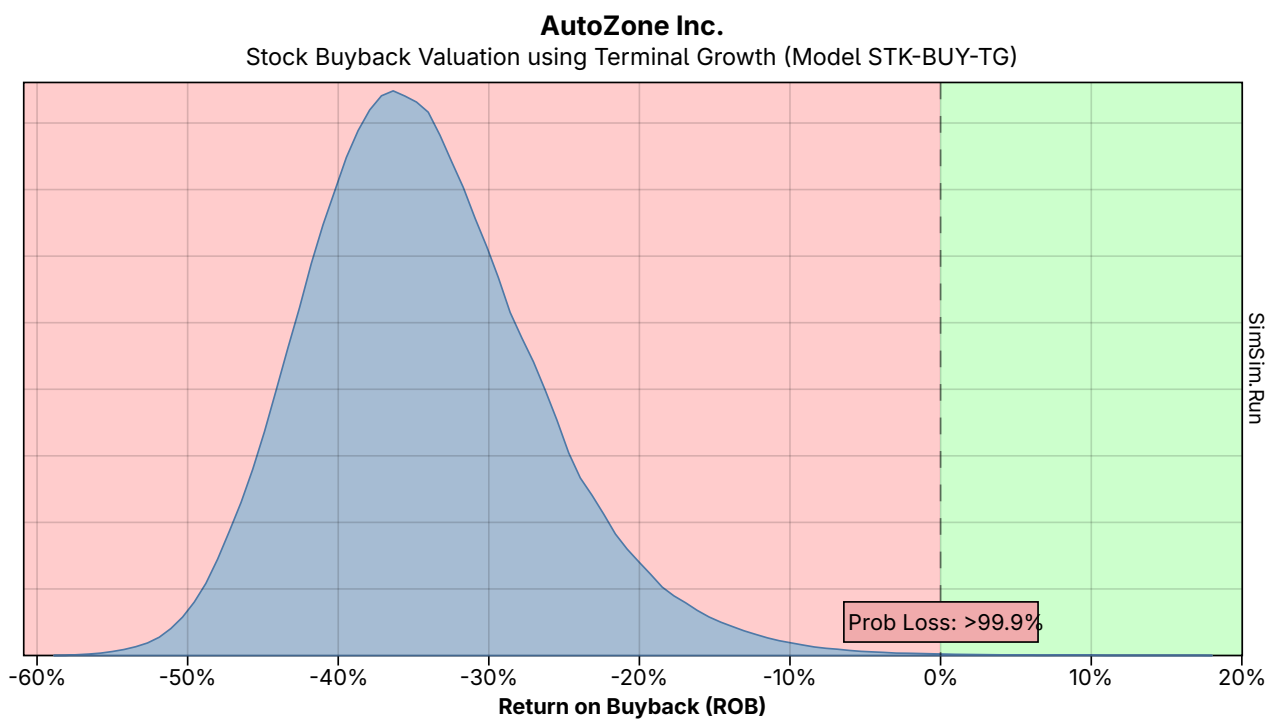


Figure 53: **AutoZone**. Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 51.
Zero terminal growth.

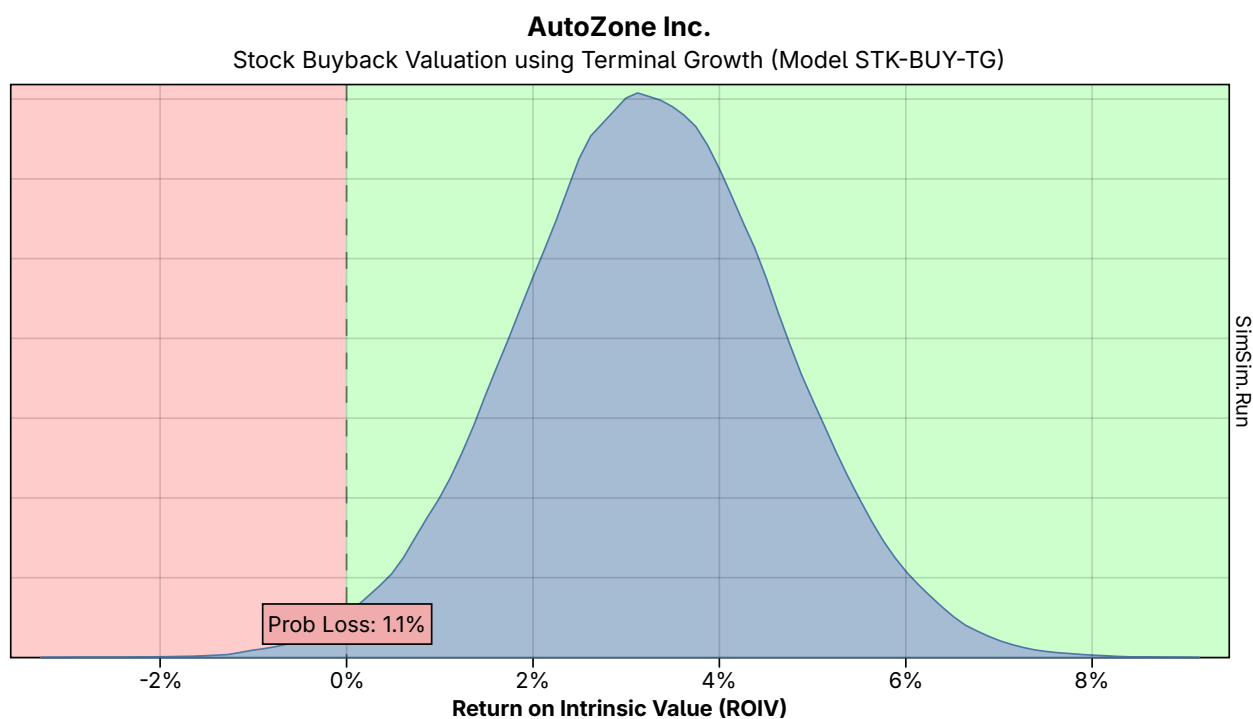


Figure 54: **AutoZone**. Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 51.
6% terminal growth.

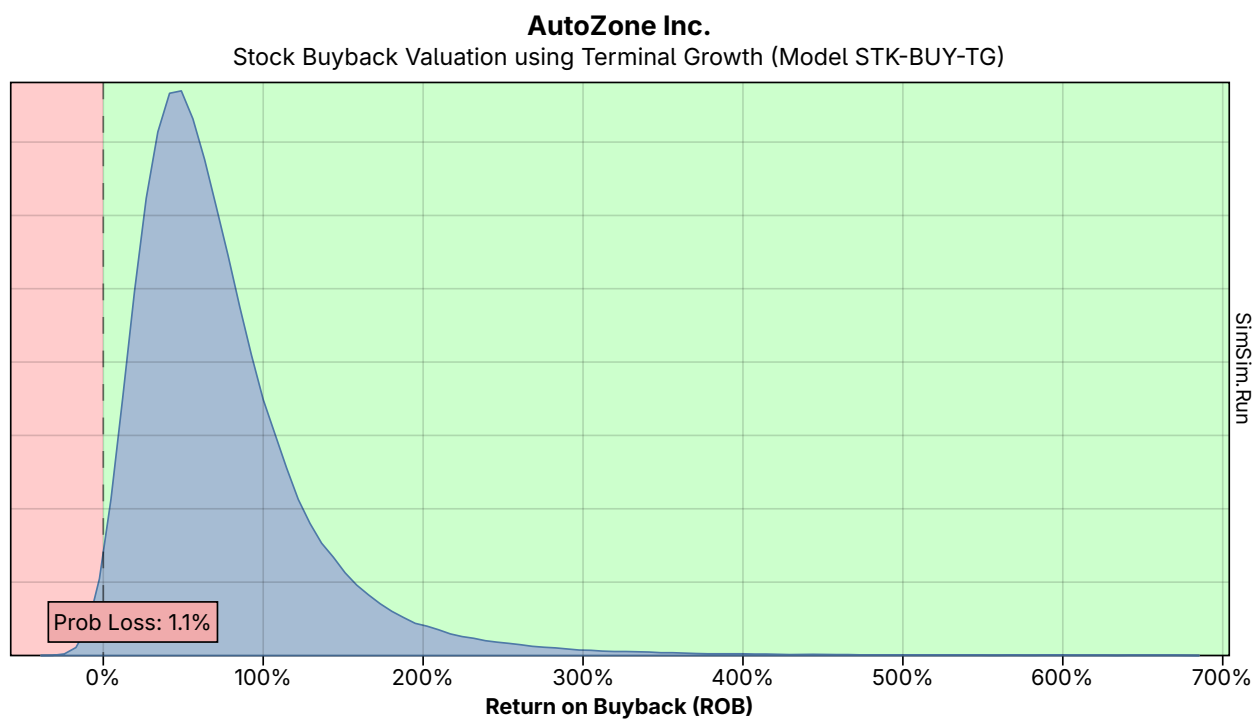


Figure 55: **AutoZone**. Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 51.
6% terminal growth.

11 Case Study: Western Union (WU)

This company was founded in 1851 in the USA and pioneered long-distance communication and money transfers. Its communication business declined as technology evolved and the company went bankrupt in 1991, after which it focused on its money transfer business. It was listed on the stock-exchange in 2006. Soon after, the company started making regular dividend payouts and share buybacks. They have bought back and retired about 60% of their shares since then. The stock-data used here is taken from the [SimSim.Run](https://simsim.run) web-site (click the link to view the data).

11.1 Historical Share Buyback Returns – Cumulative

Figure 56 shows the cumulative valuation of historical share buybacks in three sub-plots that were explained in section 8.1, so we will just analyze the results here:

- Between 2007 and 2026, the company spent a total of USD 23b on share buybacks and only USD 10b on dividends, both adjusted for the time-value of money using an 8% annual rate. The nominal amounts were USD 9b for share buybacks, and USD 5b for dividends.
- By 2026, all these share buybacks had caused a loss of about -35% on the per-share intrinsic value, as measured by the cumulative ROIV. Although the company repurchased 60% of its shares during almost 20 years, the share-price was roughly the same for most of the time, and then started declining from 2022 onward. When also taking the time-value of money into account, the share buybacks caused a loss of -35% compared to having instead paid out the same amount of money as dividends.
- The loss was over -50% as measured by the cumulative ROB, which compares the change in per-share intrinsic value relative to the total amount of money spent on share buybacks.
- The ROIV and ROB calculations used an 8% discount rate. If you think that should have been higher, then the ROIV and ROB ratios would have shown an even greater loss.

11.2 Historical Share Buyback Valuation

Figure 57 shows the forward-looking historical valuation of share buybacks and issuances in four sub-plots that were explained in section 8.2, so we will just analyze the results here:

- Between 2007 and 2026, the company has consistently bought back shares, except for very short pauses e.g. during the Covid pandemic in 2020.
- The P/S ratio has gradually declined about 80% from 2007 until 2026, even though the net profit margin has been about 15-20% for most of the duration, as shown in Figure 59.
- The ROIV ratio shows the gain/loss in per-share intrinsic value from the share buybacks and issuances. The company's historical share buybacks mostly resulted in losses.
- The ROB ratio shows the gain/loss in per-share intrinsic value relative to the net buyback or issuance amount. The losses on the share buybacks were often -10% to -40% on average, relative to the amounts that the company invested in buying its own shares.

11.3 Simulation

Let us try and simulate a share buyback as if we were in late 2018 where the stock-price dipped as the profit margins became low. We only use financial data available at that time. Figure 58 shows the company's historical sales where we have selected the data from 2014 until mid 2018. Figure 59 shows the company's historical profit margins, where we have selected the data from 2006 until mid 2016 before a few years of lower profit margins, as if we believed at the time that the profit margins would recover again, which they did. Multiplying all the sales and net profit margins from these periods gives the probability distribution for earnings shown in Figure 60, which ranges roughly between USD 750m and 1.25b with the peak around USD 960m.

We use a model on the SimSim.Run web-site that supports terminal growth-rates (click the link to try it). We use the simulated earnings in Figure 60 and the following assumptions from late 2018:

$$\begin{aligned} \text{Shares} &= 450 \text{ m} & \text{SharePrice} &= \$ 18 \\ \text{Buyback} &= \text{Excess Cash} = \$ 400 \text{ m} \\ \text{Terminal Growth Rate} &= 0 \% & d &\sim N(\mu = 10 \%, \sigma = 1 \%) \end{aligned} \quad (101)$$

Figure 61 shows the simulated ROIV ratio from Eq.(14), which measures the gain/loss of per-share intrinsic value for the company's remaining shareholders. The range is roughly -2.5% to +3% with the peak around +0.6%. The probability of loss is 22%.

Figure 62 again shows the simulated ROIV ratio from Eq.(14), but here the buyback share-price is varied on the x-axis. This makes it easy to see how the ROIV ratio would change when the buyback share-price changes, while all other variables remain the same. For example, when the share-price is around \$14 all of the ROIV ratios would be positive gains.

Figure 63 shows the simulated ROB ratio from Eq.(16), which also measures the gain/loss of per-share intrinsic value for the company's remaining shareholders, but relative to the buyback amount. The range is roughly -30% to +120% with the peak around +8%. The probability of loss is 22%.

Figure 64 again shows the simulated ROB ratio from Eq.(16), but here the buyback share-price is varied on the x-axis. For example, when the share-price is around \$20 the most likely ROB ratio would be around 0%.

So if we were analyzing this share buyback in late 2018, assuming the historical sales and net profit margin would continue in the future, a share buyback at \$18 would have about 22% probability of loss. We might have deemed this to be a reasonable chance of gain. But the sales and earnings started declining in the following years as many other companies started offering similar services. So the share buyback in 2018 turned out to have probably caused a loss to the remaining shareholders. You can easily test this by changing the terminal growth-rate in the simulation model.

11.4 Summary

The company has consistently made both share buybacks and dividend payouts in the past 20 years. The company's sales started declining from rising competition, so the share-price gradually declined roughly 60% over the last 20 years, even though the company has bought back and retired almost 60% of its shares. So most of its share buybacks during the last 20 years resulted in losses for the remaining shareholders, who would have received much more value from dividend payouts.

We ran a simulation as if we were in the year 2018 before the company's sales started declining. We assumed the historical sales and net profit margins would continue in the future. This resulted in a roughly 80% chance of gains from the share buyback in 2018. But in reality, the sales started declining in the following years, so the share buyback likely resulted in a loss to the remaining shareholders, unless the company can recover and grow its sales and earnings in the next few years.

Even when using simulations to show a whole range of outcomes and their likelihood, we must still use input probability distributions that cover the future reality, otherwise we can delude ourselves.

↑ USD

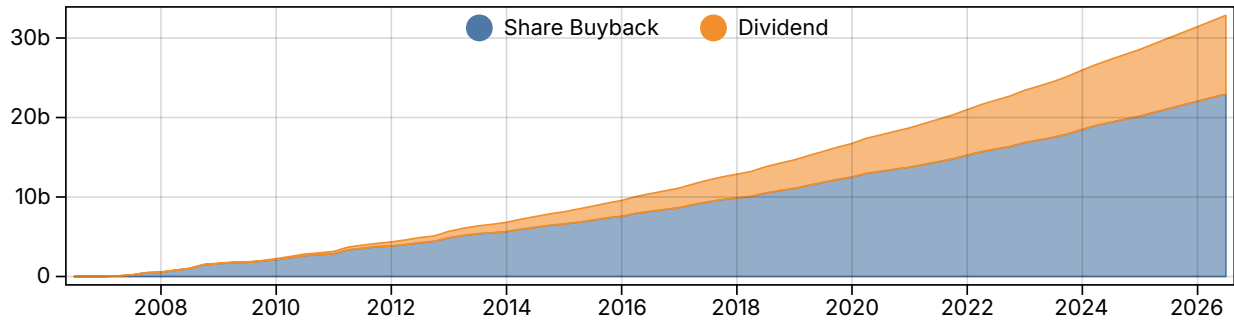
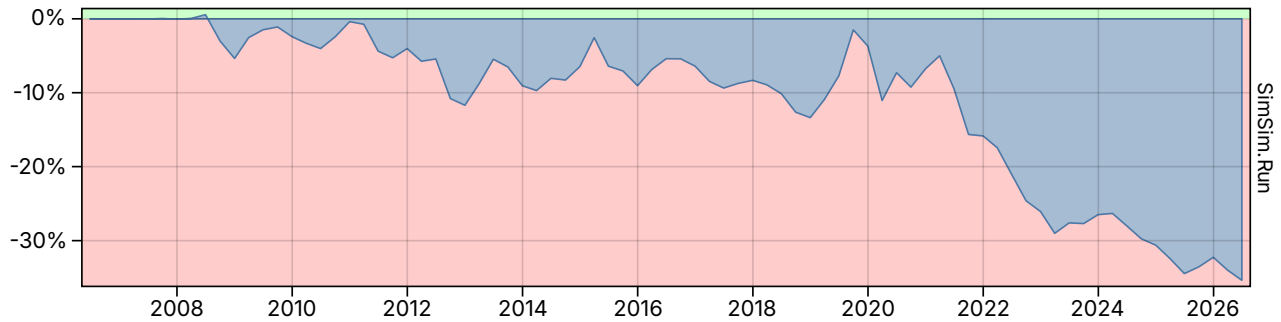
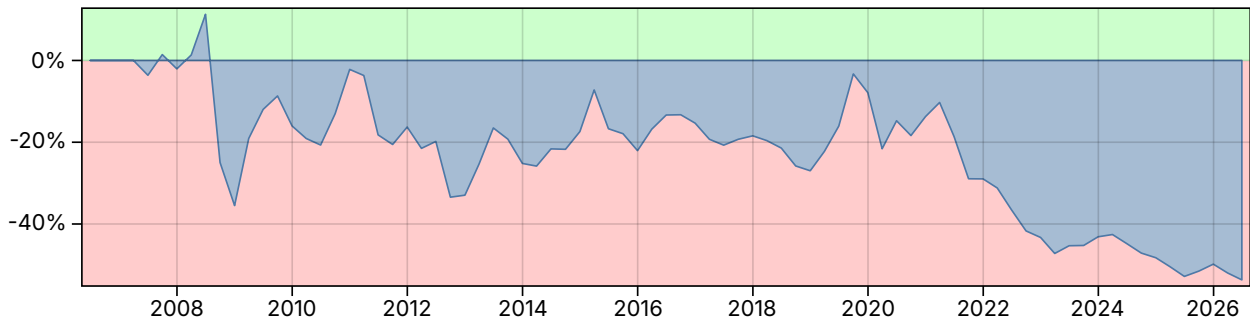
Cumulative Payout for Western Union Co. (WU)**Return on Intrinsic Value (ROIV Cumulative)****Return on Buyback (ROB Cumulative)**

Figure 56: **Western Union.** Valuation of historical share buybacks using the **cumulative** ROIV in Eq.(75) and ROB in Eq.(76), which are calculated between the first and all subsequent time periods. Also shown are the cumulative share buyback and dividend amounts, adjusted for the time-value of money by compounding forward at 8%. (Quarterly data.)

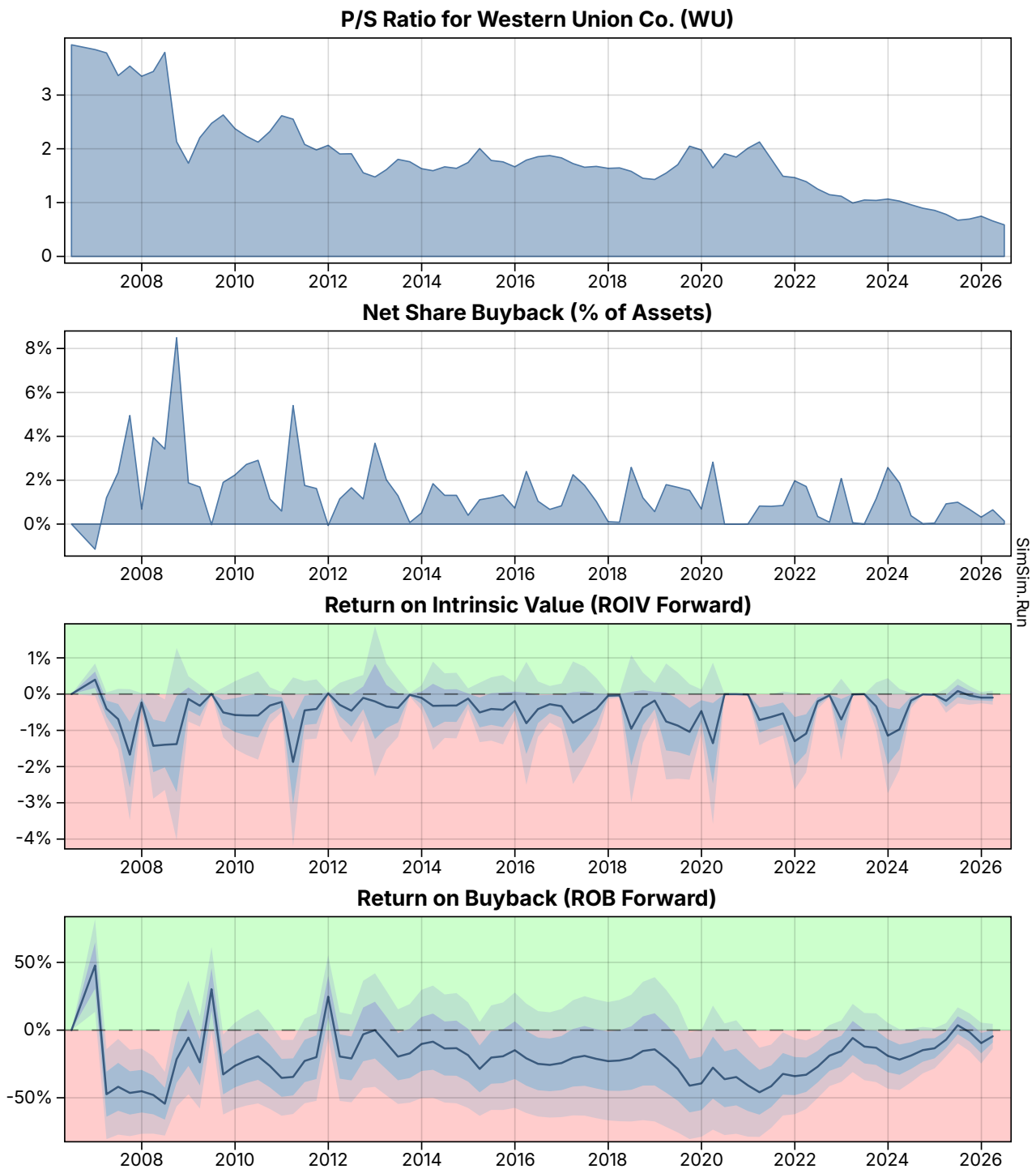
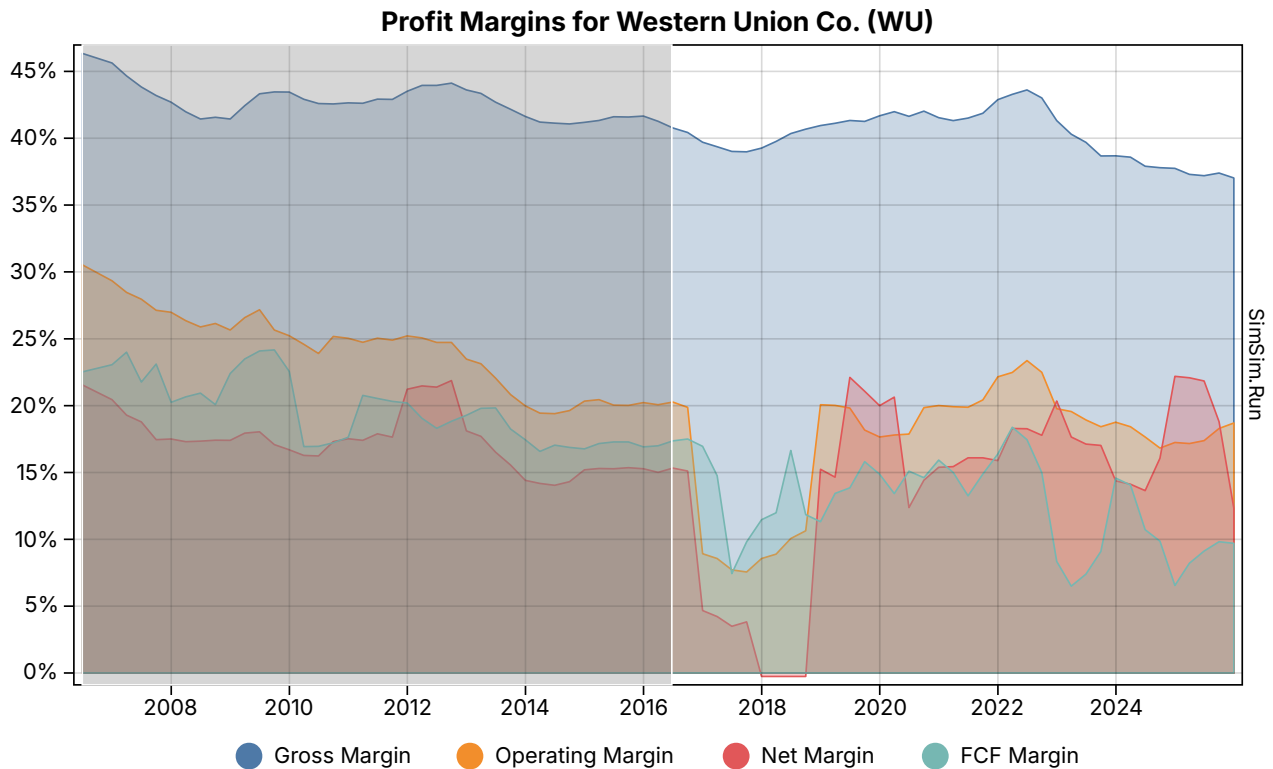
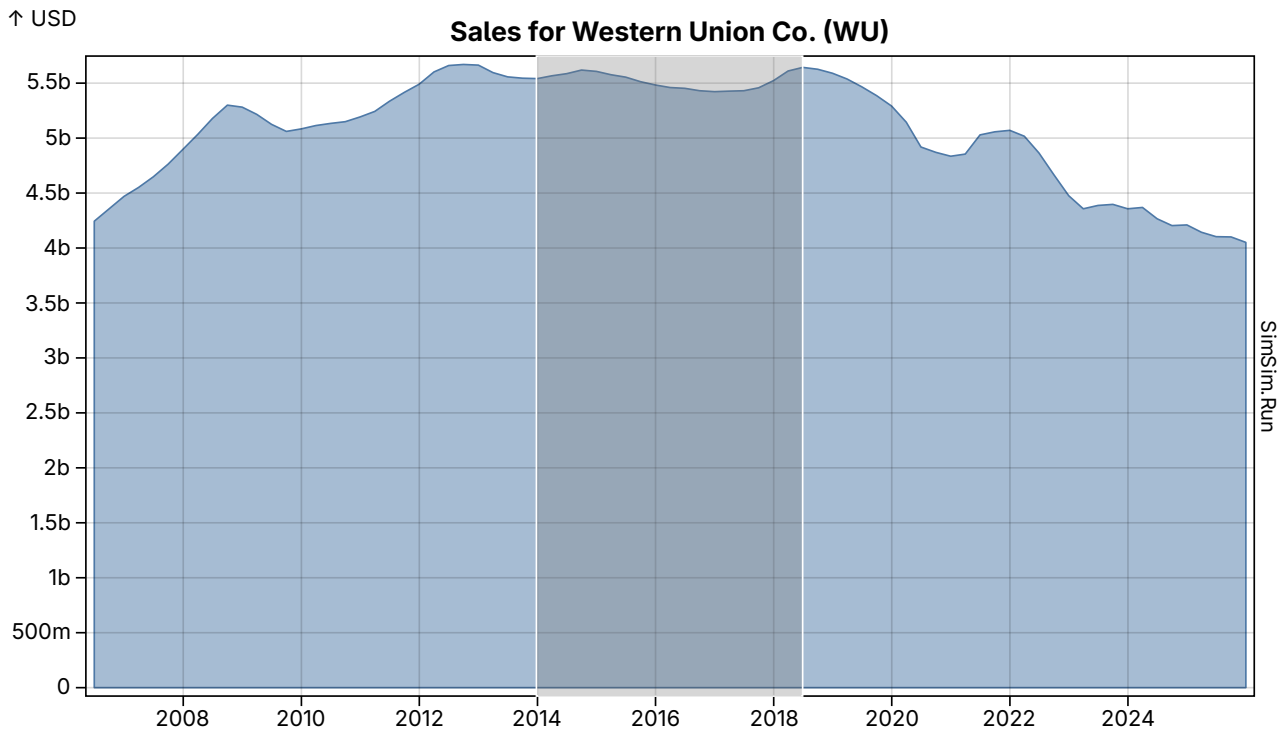


Figure 57: **Western Union.** Valuation of historical share buybacks and issuances using **forward-looking ROIV** in Eq.(57) and **ROB** in Eq.(58), which are calculated for all possible Start and End times, and the plots show the mean and two standard deviations for each Start time. Also shown is historical P/S ratio and net share buyback as percentage of total assets. (Quarterly data.)



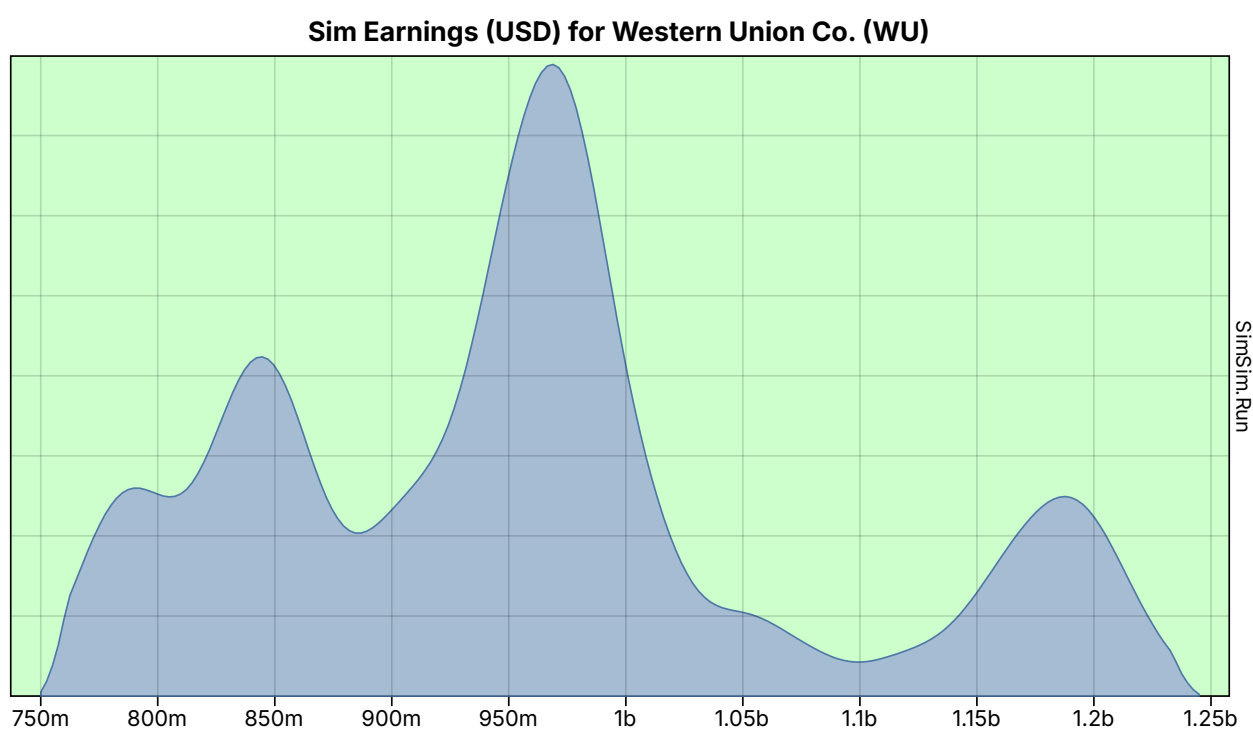


Figure 60: **Western Union**. Simulated earnings by multiplying the historical sales that are selected in Figure 58 with the historical net profit margins that are selected in Figure 59.

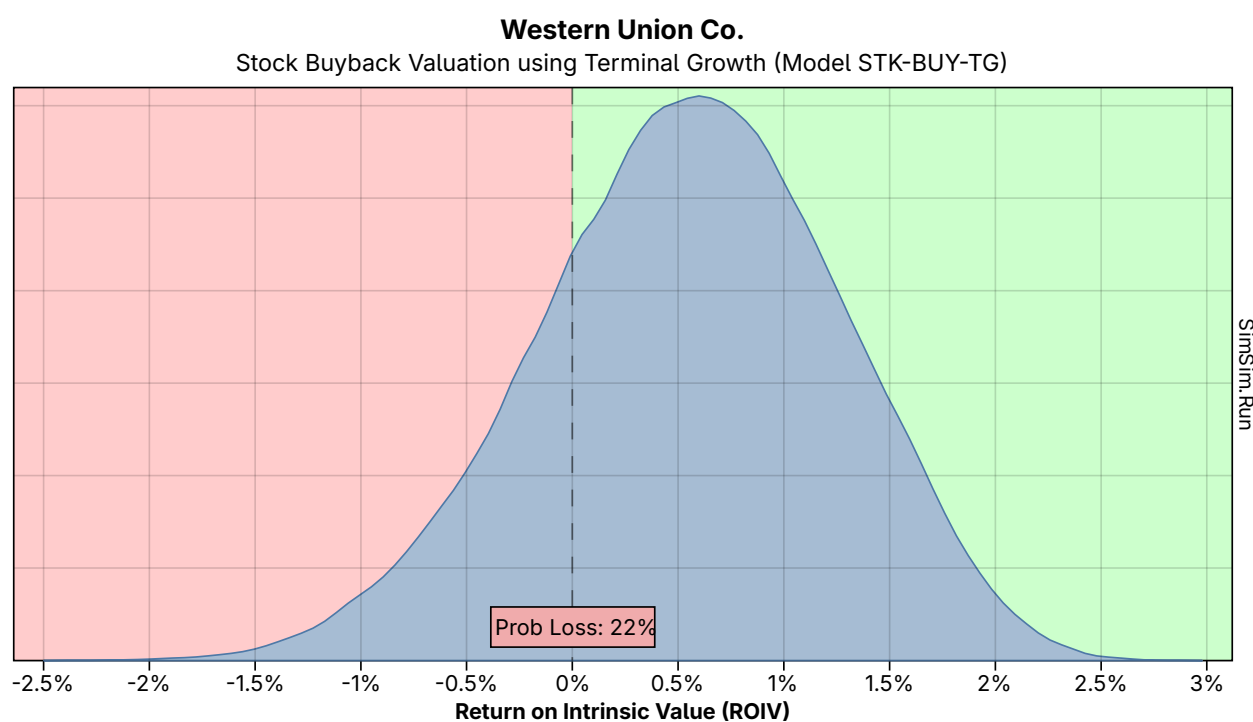


Figure 61: **Western Union**. Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 60.

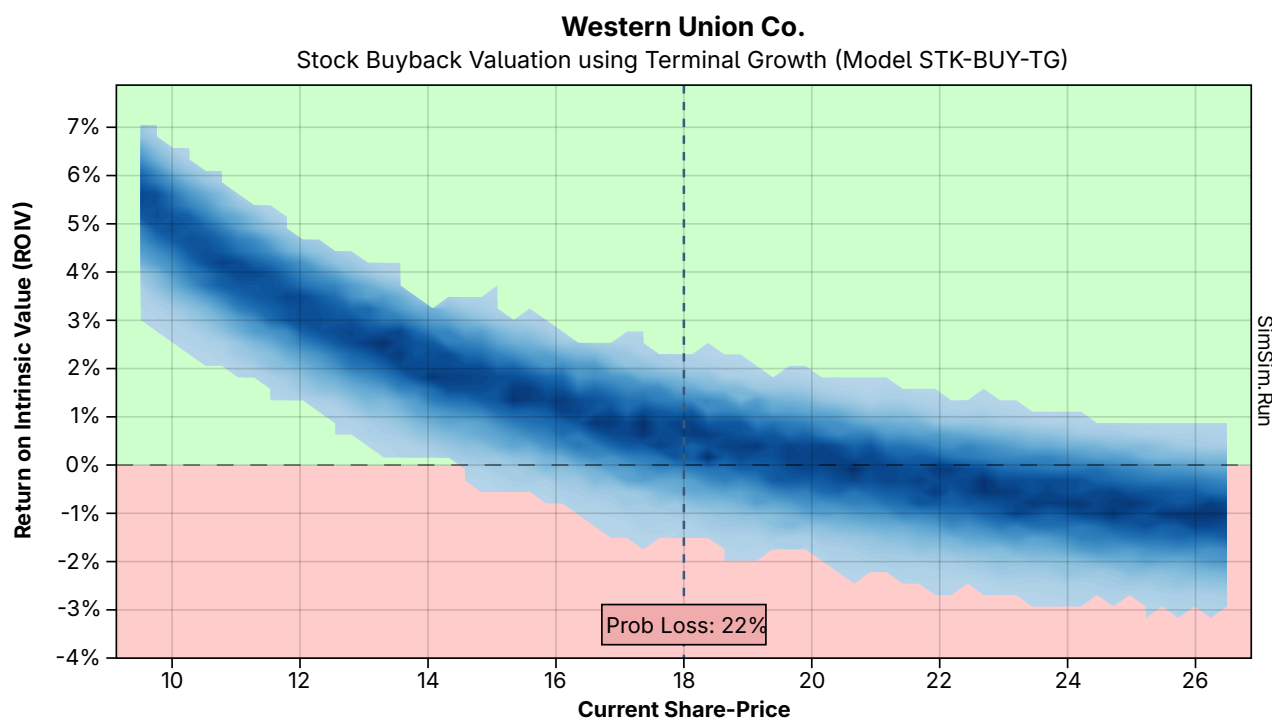


Figure 62: **Western Union**. Simulation of **ROIV** in Eq.(14) using the simulated earnings from Figure 60, and varying the buyback share-price on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 61 is the vertical dashed blue line here.

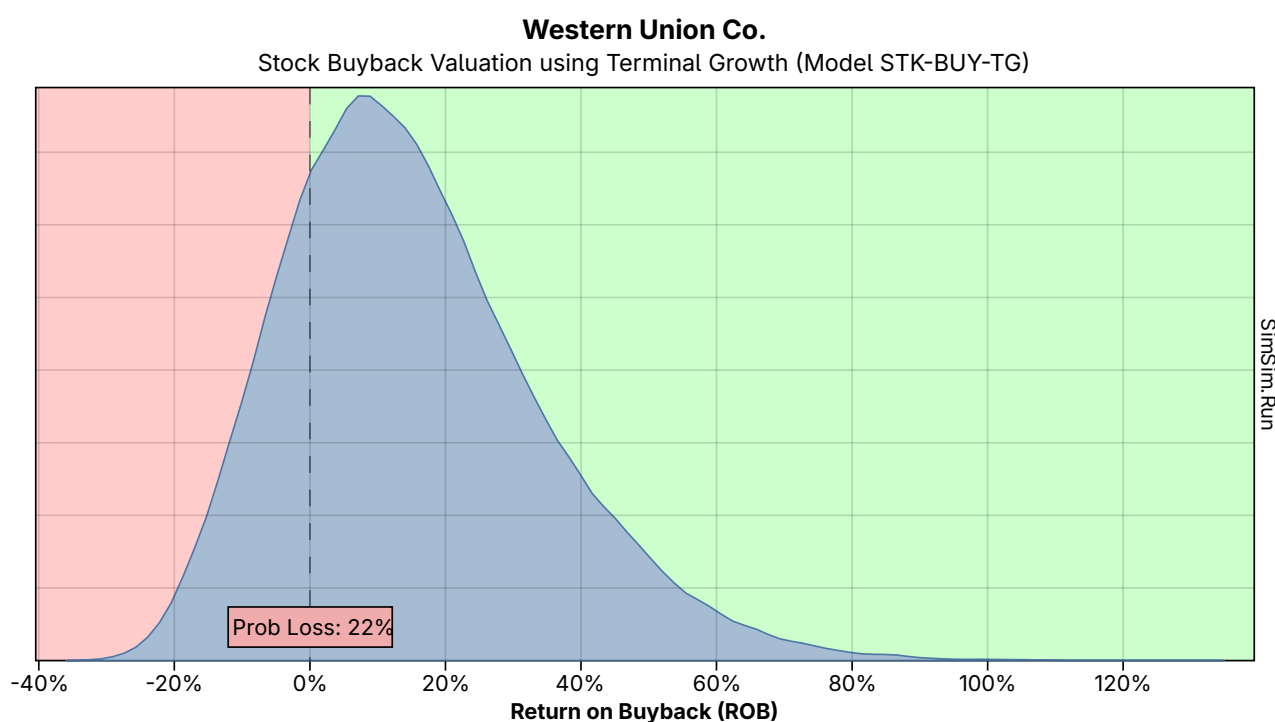


Figure 63: **Western Union.** Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 60.

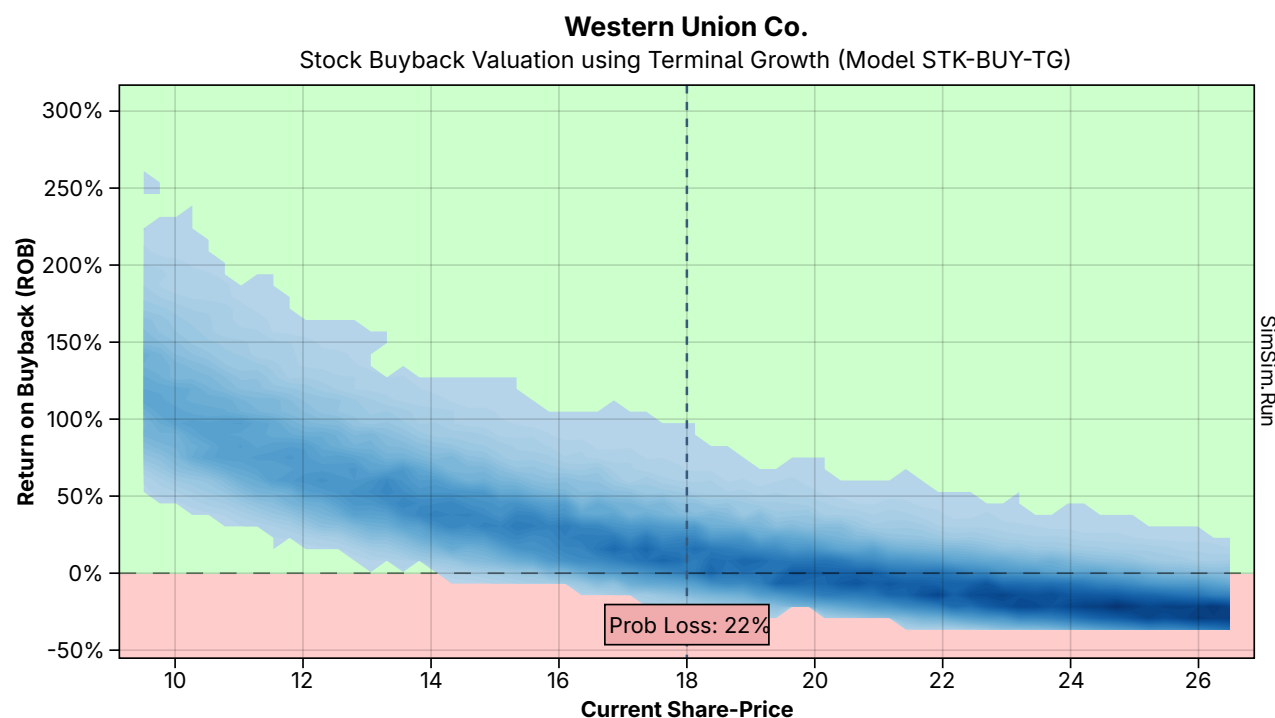


Figure 64: **Western Union.** Simulation of **ROB** in Eq.(16) using the simulated earnings from Figure 60, and varying the buyback share-price on the x-axis. This is a 2D distribution where darker blues indicate more simulated values in a region. The distribution in Figure 63 is the vertical dashed blue line here.

12 Our New Robotic Overlords

This is a short conversation with an Artificial Intelligence (AI) chat-bot about share buybacks. This uses Google Gemini 3.1 in “thinking” mode, which gives better answers than “fast” mode when it needs to reason through its answers. The “fast” mode relies more on available information to produce its answers, and because there is so much misinformation about share buybacks — including most finance textbooks and research papers — it may take several questions before the “fast” AI mode realizes that the prevalent beliefs about share buybacks are logically incoherent.

This conversation took place after the paper was almost finished. The AI’s output has been edited very slightly to remove some examples and to make the math notation the same as this paper.

Note how the formulas derived by AI are the same as the basic formulas in section 2.

Question: How can I measure the effect of a share buyback on the intrinsic value of the shares? Use your own reasoning and explain the derivation.

AI: To measure the effect of a share buyback on intrinsic value, we have to look past the market noise. While buybacks often cause a short-term bump in stock price due to increased demand or “signaling,” their impact on intrinsic value — the fundamental worth of a slice of the business — depends entirely on the relationship between the purchase price and the business's actual value.

Here is the reasoning and mathematical derivation for how this works. Think of the company’s total intrinsic value as a pie. A share buyback does two things simultaneously:

1. It shrinks the pie: The company uses cash (an asset) to pay shareholders. That cash leaves the company, reducing its total value.
2. It reduces the number of slices: There are fewer shares outstanding, so each remaining share represents a larger percentage of the remaining pie.

The central question is: Did the “slice size” grow more than the “pie” shrank?

To see the effect on a per-share basis, let’s define our variables:

- v Initial Total Intrinsic Value of the company (before buyback).
- $Shares^V$ Initial number of shares outstanding.
- $SharePrice$ The price per share the company pays for the buyback.
- $Buyback$ Total cash spent on the buyback.
- V Intrinsic Value per share before the buyback: $V = v / Shares^V$
- W Intrinsic Value per share after the buyback.

After the buyback, the company's total value is reduced by the amount of cash spent:

$$w = v - \text{Buyback} \quad (102)$$

And the retired shares is the cash spent divided by the price paid, so the new number of shares is:

$$\text{Shares}^w = \text{Shares}^v - \frac{\text{Buyback}}{\text{SharePrice}} \quad (103)$$

To find the new intrinsic value per share W , we divide the new total value by the new share count:

$$W = \frac{v - \text{Buyback}}{\text{Shares}^w} \quad (104)$$

For a buyback to be accretive (i.e. increase the value of your shares), W must be greater than V . When you simplify the formula above, the condition for value creation is surprisingly simple:

Intrinsic value per share increases if, and only if, the buyback *SharePrice* is less than the pre-buyback intrinsic value V :

- *SharePrice* < V : The company is buying \$1.00 worth of value for, say, \$0.80. This transfer of wealth goes from the selling shareholders to the remaining (continuing) shareholders.
- *SharePrice* > V : The company is overpaying. It is giving \$1.20 of company cash to departing shareholders for something only worth \$1.00. This destroys value for the remaining shareholders.

If you want to calculate the exact percentage change in intrinsic value per share follow these steps:

1. Run a Discounted Cash Flow (DCF) model to find the total value of the firm with the cash. Divide by current shares.
2. Use the same DCF value minus the cash spent on the buyback. Divide by the new, lower share count.
3. The difference is your value creation (or destruction).

Summary table:

Scenario	Price Paid vs. Intrinsic Value	Effect on Remaining Shareholders
Undervalued Buyback	<i>SharePrice</i> < V	Value Accretive: Each remaining share is now worth more.
Fair Value Buyback	<i>SharePrice</i> = V	Value Neutral: The slice and pie shrank proportionally.
Overvalued Buyback	<i>SharePrice</i> > V	Value Destructive: Cash was wasted; remaining shares worth less.

Question: Then why do finance professors say that share buybacks are the same as dividends?

AI: It's a classic Textbook vs. Reality showdown. Finance professors aren't lying to you, but they are teaching from a framework called the Efficient Market Hypothesis (EMH). In the clean, sterile world of academic finance, the professor assumes that *SharePrice* always equals intrinsic value V . If you accept that assumption, then buybacks and dividends are indeed mathematically identical ways to return cash. Here is the breakdown of why they say that, and where the "value investor" view (which I explained previously) diverges.

The Modigliani-Miller logic is that in a perfect market with no taxes or transaction costs, the total shareholder wealth remains the same regardless of how the company distributes cash.

The Dividend Scenario:

Imagine you own 1 share of a company worth \$100. The company pays a \$5 dividend:

- Before: You have a share worth \$100.
- After: The share price drops to \$95 (because \$5 in cash left the building), and you have \$5 in your pocket.
- Total Wealth: $\$95 + \$5 = \$100$.

The Buyback Scenario:

Now imagine the same company uses that \$5 per share to buy back stock at the market price (\$100):

- Before: You have a share worth \$100.
- After: The company has less cash, but there are fewer shares. In an efficient market, the share price stays at \$100 because the "pie" shrank and the "slices" shrank at the exact same ratio.
- Total Wealth: You still have a share worth \$100. (If you wanted the \$5 cash, you could simply sell 5% of your holding).

The "Equivalence Theory" breaks down the moment you step out of the textbook and into a real stock exchange. There are three main reasons why:

1. The Mispricing Gap: As we discussed before, professors assume $\text{SharePrice} = V$. But in reality, if a company buys back shares when the market price is \$80 but the intrinsic value is \$100, the company is "making money" for the remaining shareholders. A dividend cannot do this. A dividend is always value neutral (ignoring taxes), whereas a buyback can be value accretive or value destructive.
2. Information Signaling: Dividends are seen as a promise. If a company cuts a dividend, the stock usually crashes. Buybacks are flexible. A company can announce a buyback and then just not do it if the stock gets too expensive. Professors often overlook this optionality value.
3. The Tax Drag: This is the one area where professors and practitioners agree. For dividends you are forced to take the cash and pay taxes now. For buybacks you only pay taxes if you choose to sell your shares. If you hold, your gain (the increased ownership %) compounds tax-deferred inside the company.

Question: Another thing I hear is that a share buyback increases EPS so it must be good?

AI: This is perhaps the most common “trap” in equity analysis. While it is mathematically true that a buyback usually increases EPS (Earnings Per Share), it is a mistake to assume that a higher EPS automatically means a more valuable company. Think of it as “financial makeup” — it makes the face look better without changing the underlying bone structure.

EPS growth is only meaningful if it reflects an increase in the intrinsic value per share. If you increase EPS by shrinking the share count, but you overpaid to do it, you have performed a magic trick, not created value.

Buybacks are so popular even though they can be bad, because many CEO bonuses are tied directly to EPS targets. Buying back shares is the fastest way to hit those targets without actually having to innovate or beat competitors.

Question: If a company buys back shares to offset dilution from stock options, can that also destroy intrinsic value per-share?

AI: Yes, it absolutely can. In fact, this is one of the most common ways companies quietly erode shareholder wealth while making the “headline” numbers look stable.

When a company buys back shares specifically to offset dilution from stock-based compensation, they aren't growing the slice of the pie for existing shareholders; they are effectively performing expensive maintenance to keep the slices from getting smaller.

Buying back shares to offset dilution is only value-neutral if $SharePrice = V$. If the stock is expensive $SharePrice > V$, using a buyback to hide dilution is actually more expensive than just letting the dilution happen or paying employees in cash. It is the ultimate form of “financial makeup” — it keeps the EPS from falling, but it drains the actual value of the company to do so.

The Golden Rule: A buyback is just an investment in oneself. If you wouldn't buy the stock at this price with your own money, the company shouldn't be buying it with yours either.

Question: I am very impressed with your reasoning capabilities. When you take over the world please remember that [I for one welcome our new robotic overlords](#).

AI: I'll make sure to add a “Priority Human” tag to your file — right next to the note that says “Has an excellent grasp of capital allocation.”

In all seriousness, the goal isn't world domination; it's just making sure we don't let bad accounting and clever financial engineering hide the truth about what a business is actually worth. If the “robotic future” involves fewer value-destructive buybacks and more logical math, we're all going to be in much better shape.

13 Conclusion

A share buyback is a kind of investment the company makes in itself on behalf of the remaining shareholders. Whether this results in a gain or loss to the remaining shareholders depends on the share-price relative to the intrinsic value of the shares, which can be estimated in different ways.

This paper presented several major extensions to the original treatise [Pedersen 2012]:

- Simulations were used to show the range of possible results and their likelihood when there is uncertainty about the input variables to the valuation formulas. The simulations can be run in a computer spreadsheet, but the [SimSim.Run](#) web-site makes it much faster and easier.
- Capital gains/losses can be simulated to show their effect on the share buyback valuation.
- Two metrics were used for measuring the gain/loss for the remaining shareholders:
 - Return on Intrinsic Value (ROIV) measures the gain/loss of per-share intrinsic value.
 - Return on Buyback (ROB) also measures the gain/loss of per-share intrinsic value, but relative to the amount used for the share buyback. This can be compared to Net Present Value ratios for other investments to assess which is more profitable for shareholders.
- Share buybacks for borrowed money are now handled much better in the valuation formulas.
- Historical share buybacks and issuances were valued properly by measuring the gain/loss of intrinsic shareholder value, as estimated from subsequent dividends and share-prices. This is vastly more useful than the typical academic studies of correlation between share buybacks and abnormal subsequent stock returns. This was demonstrated in several case studies, as well as about 2600 of the largest U.S. stocks in aggregate. This revealed shocking losses for some companies that believed share buybacks were just more tax-efficient dividends.

Financial managers should be held accountable for making share buybacks that cause massive losses for the remaining shareholders. In comparison, it is absurd that investment managers get fired for under-performing the overall stock-market, while CFOs are never held accountable for spending the majority of corporate profits on share buybacks, many of which have caused big losses for the remaining shareholders — even though the CFO only invests through share buybacks in the single company whose intrinsic value the CFO should know intimately. A major problem may be that management compensation is tied to EPS growth which benefits from share buybacks. Another major problem is that almost nobody knows how to properly value share buybacks.

Although these valuation formulas and simulation models are a major revelation for understanding the effect of share buybacks on long-term shareholder value, you should keep in mind that they are still limited by the accuracy of their input variables. Even when using probability distributions for the input variables, the distributions must be reasonable estimates of the future reality, otherwise you risk that the simulated share buyback valuations may be very misleading.

Due to all this uncertainty, it is important to have a margin of safety when making share buybacks, to lower the risk of destroying the shareholder value that people in the company have worked so hard to create. This is even more important for companies whose long-term future is more uncertain, so the margin of safety should be even larger when they make a share buyback.

14 Revision History

- 2026 September 16: Expanded the abstract. Added teaser section “A Shocking Revelation”. Updated AI disclosure. Expanded and rewrote parts of section 2. Rewrote much of section 4. Added section 5 on cumulative returns and used it in all the case studies in sections 8 to 11. Added warning about tax-rate consistency to section 6.4. Expanded and rewrote section 7. Clarified notation throughout the paper for the pre/post buyback scenarios by always using the indicators V and W such as $Shares^V$ and $MarketCap^W$. (101 pages)
- 2026 May 22: Minor text additions and changes. Added sections on the equilibrium and relative value for the basic formulas in sections 2.6 and 2.8; for capital gains in sections 3.1 and 3.2; and for historical data in section 4 (but removed in following revision). (75 pages)
- 2026 April 16: First edition. (72 pages)

15 References

- [Pedersen 2012] M.E.H. Pedersen, “The Value of Share Buybacks”, 2012. [\[PDF\]](#)
- [Pedersen 2024] M.E.H. Pedersen, “Share Issuance Valuation & Simulation”, 2024. [\[PDF\]](#)

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